



Modul : Decision Tree Learning (DTL)

Variable (Attribute) Types

**Source: DataMining Concepts and Techniques
by Jiawei Han, Micheline Kamber, Jian Pei**

Nur ULFA Maulidevi

KK IF - Teknik Informatika- STEI ITB

**Pembelajaran Mesin
(Machine Learning)**



Numeric

Quantitative
(measurable
quantity) ? Integer
or Real Values

Interval Scaled
(equal-size units,
have order)

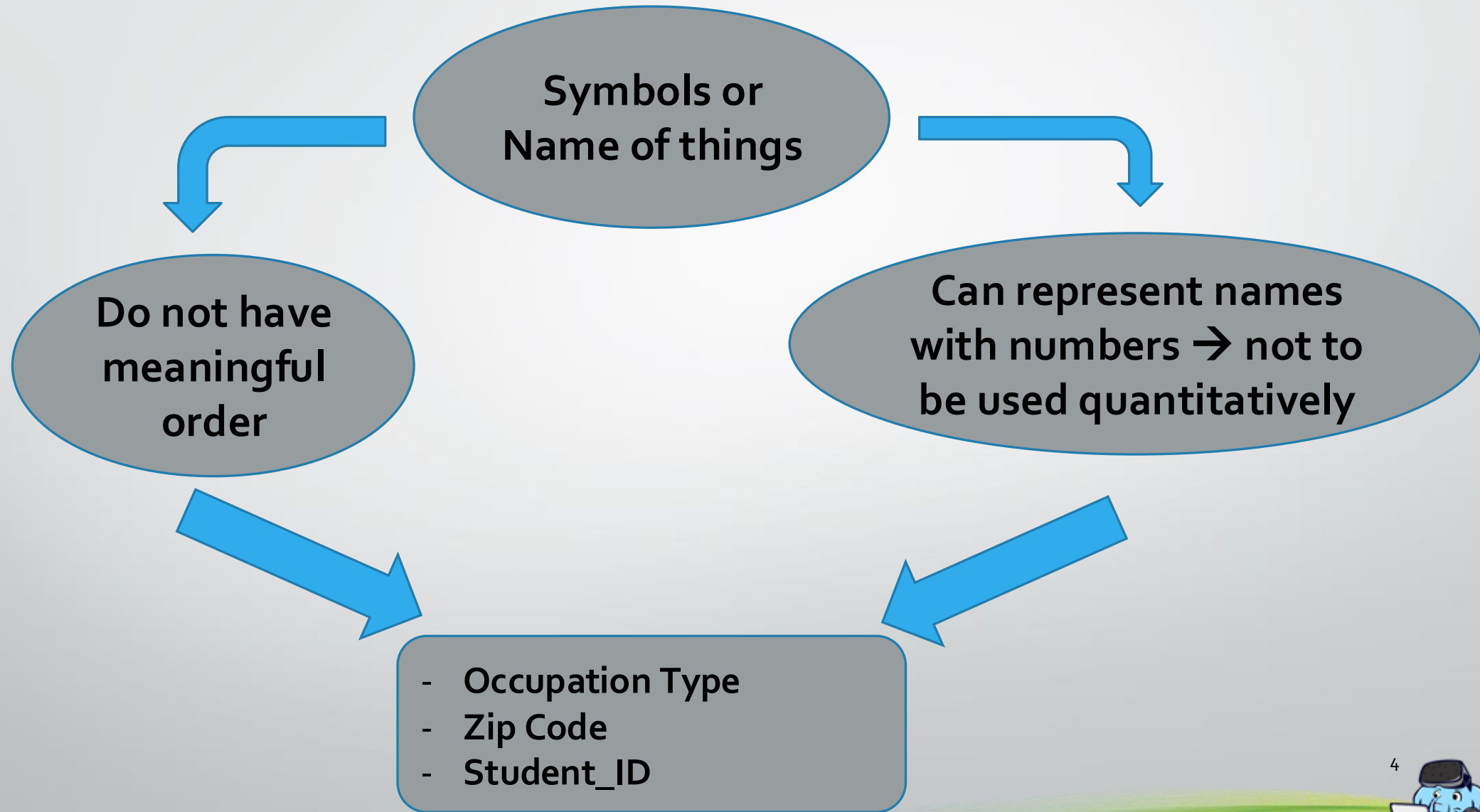
- Temperature
- Calendar dates

Ratio-Scaled
(inherent zero
point, a value can
be multiple of
another value)

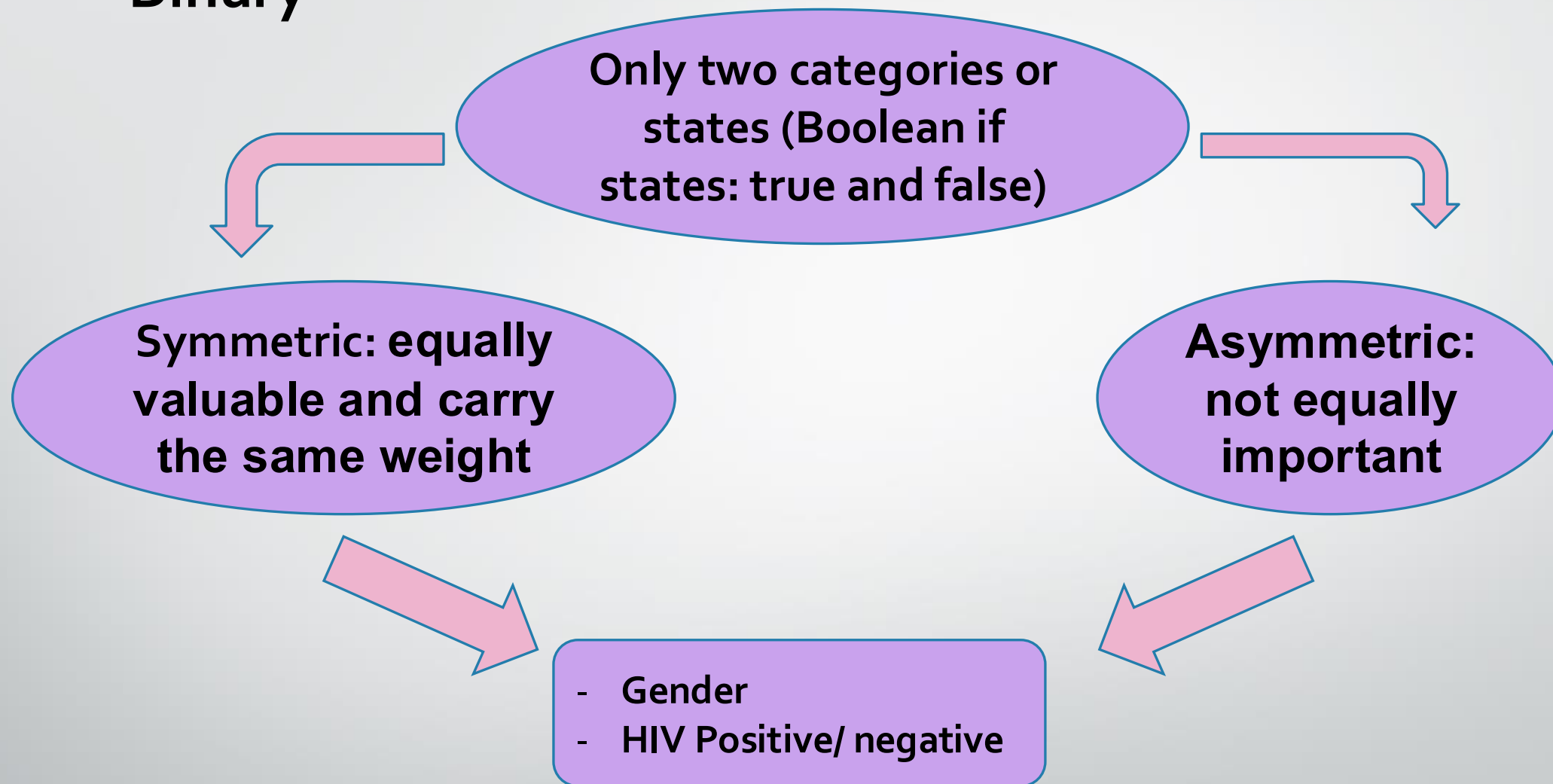
- Years of experience
- Weight
- Number of words



Nominal/ Categorical



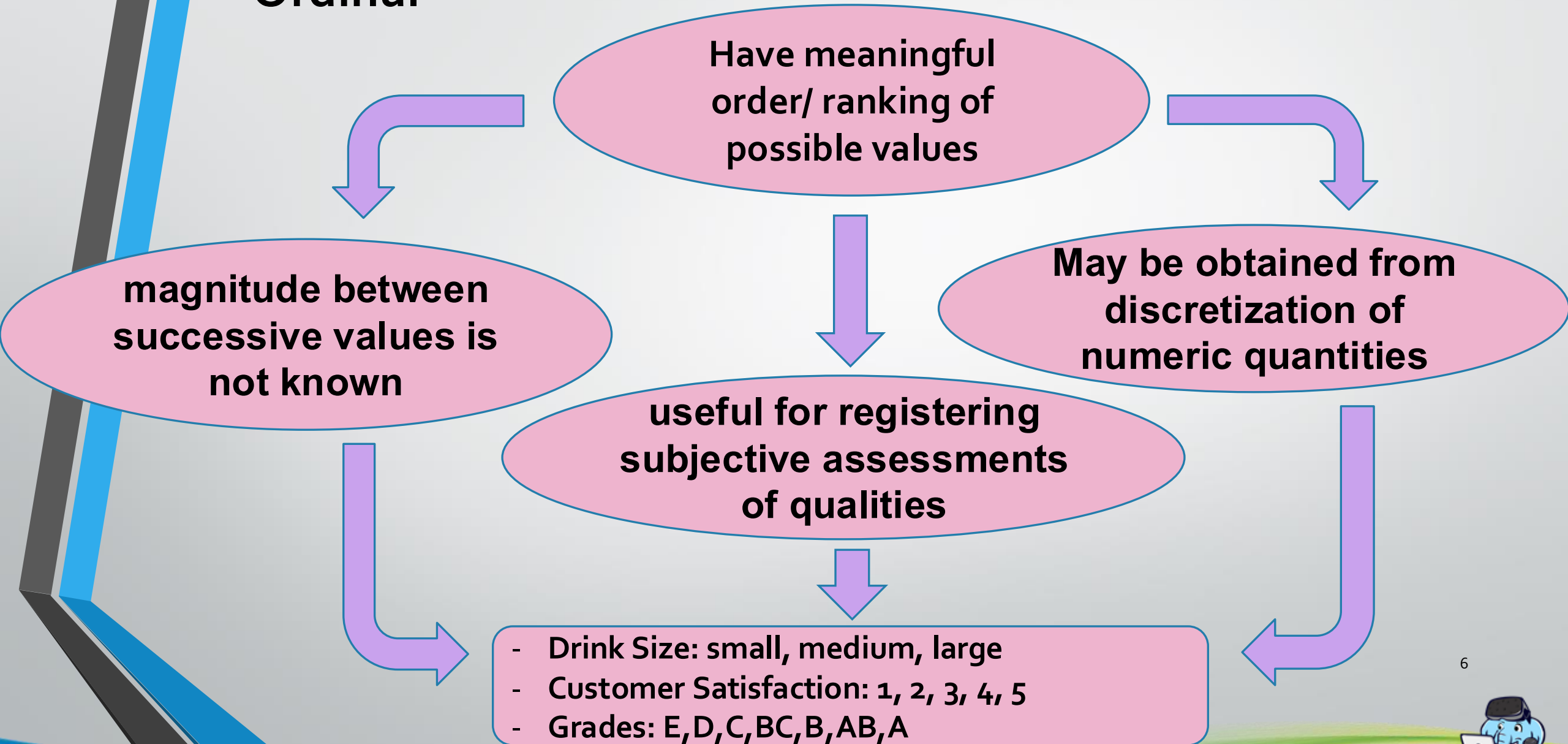
Binary



	red,	green,	blue
1			
2	1,	0,	0
3	0,	1,	0
4	0,	0,	1



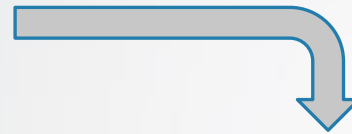
Ordinal



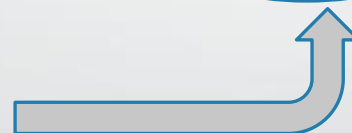
Discrete vs Continuous

has a finite set of values:
Drink size, Age, Medical
test,

has a countably infinite
set of values:
Customer ID, Zip code



Discrete



$\neq$

Continuous



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Modul : Decision Tree Learning (DTL)

What, Why, and When

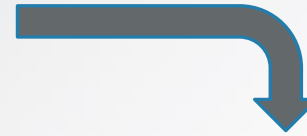
Source: Machine Learning, Tom Mitchell

**Pembelajaran Mesin
(Machine Learning)**

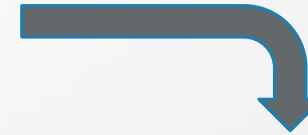


WHAT is DTL

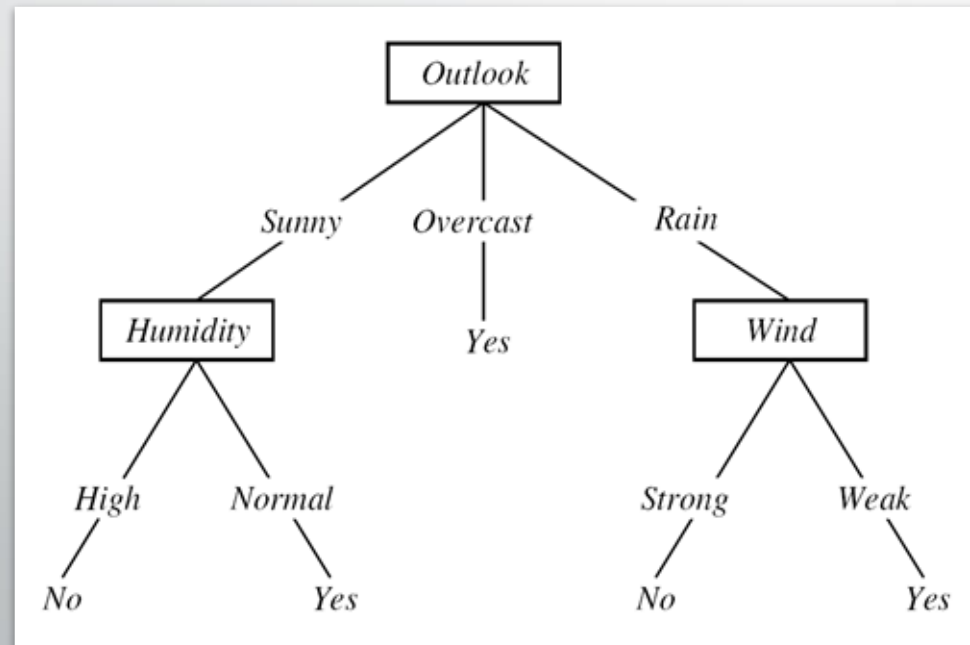
**Method for
approximating
discrete-valued target
functions**



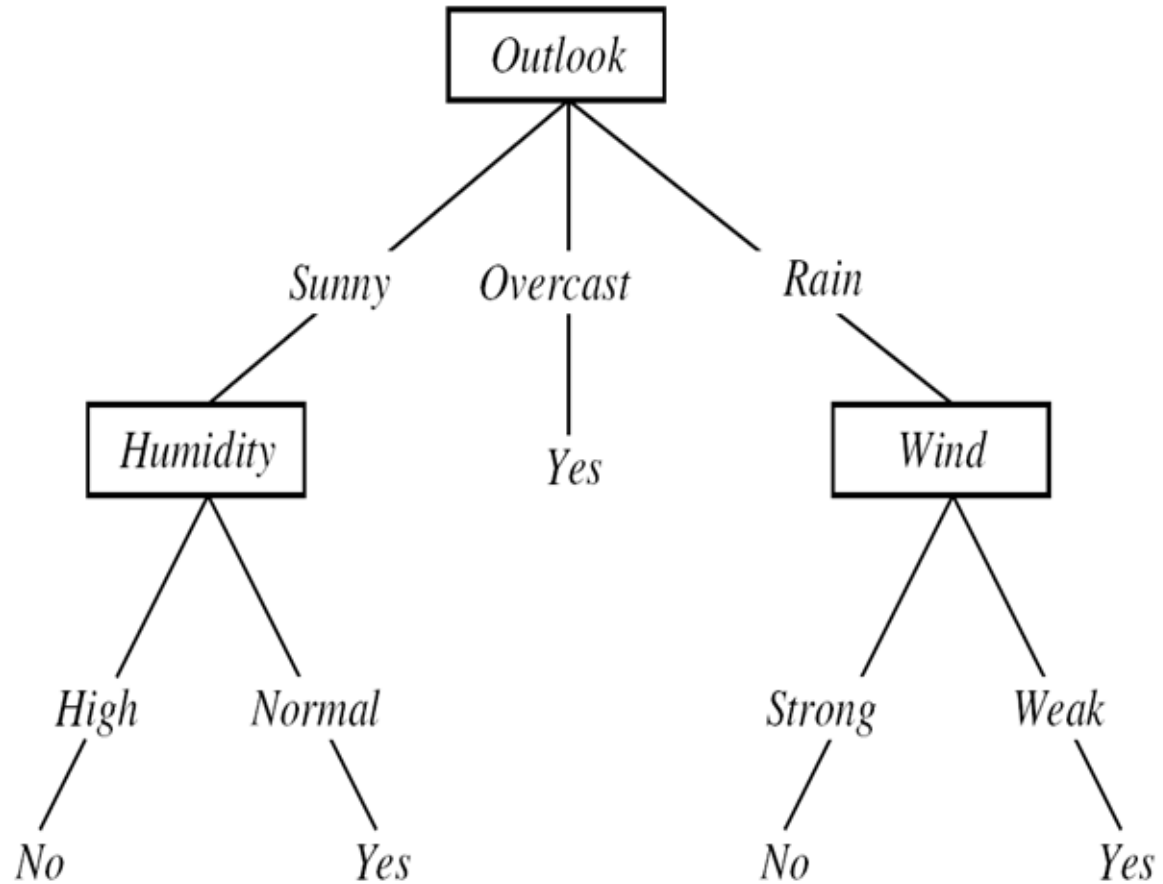
**Represented by
Decision Tree**



**Can be
represented as set
of if-then rules**



DTL Representation



- Each internal node represents test of an attributes
- Branch descending from the node corresponds to one possible value
- Leaf/ terminal nodes represents classification result

Represent a disjunction of conjunctions of constraints on the attribute values of instances

$(\text{outlook}=\text{sunny} \wedge \text{humidity}=\text{normal})$
 $\vee (\text{outlook}=\text{overcast})$
 $\vee (\text{outlook}=\text{rain} \wedge \text{wind}=\text{weak})$



WHY DTL

**Method for
approximating
discrete-valued target
functions**

**Represented by
Decision Tree or
If-Then Rules**

Robust to Noisy Data

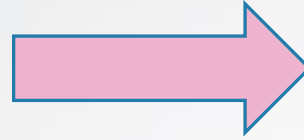
**Capable of learning
disjunctive expression**

- Popular of inductive inference algorithms
- Have been successfully applied to a broad range of tasks: learning to diagnose medical cases, learning to assess credit risk of loan applicants



When we use DTL → Appropriate Problems for DTL

Instances are represented
by attribute-value pairs



For continuous attribute:
discretization

The target function has
discrete output values

Disjunctive descriptions
may be required

The training data may
contain errors

The training data may
contain missing attribute
values



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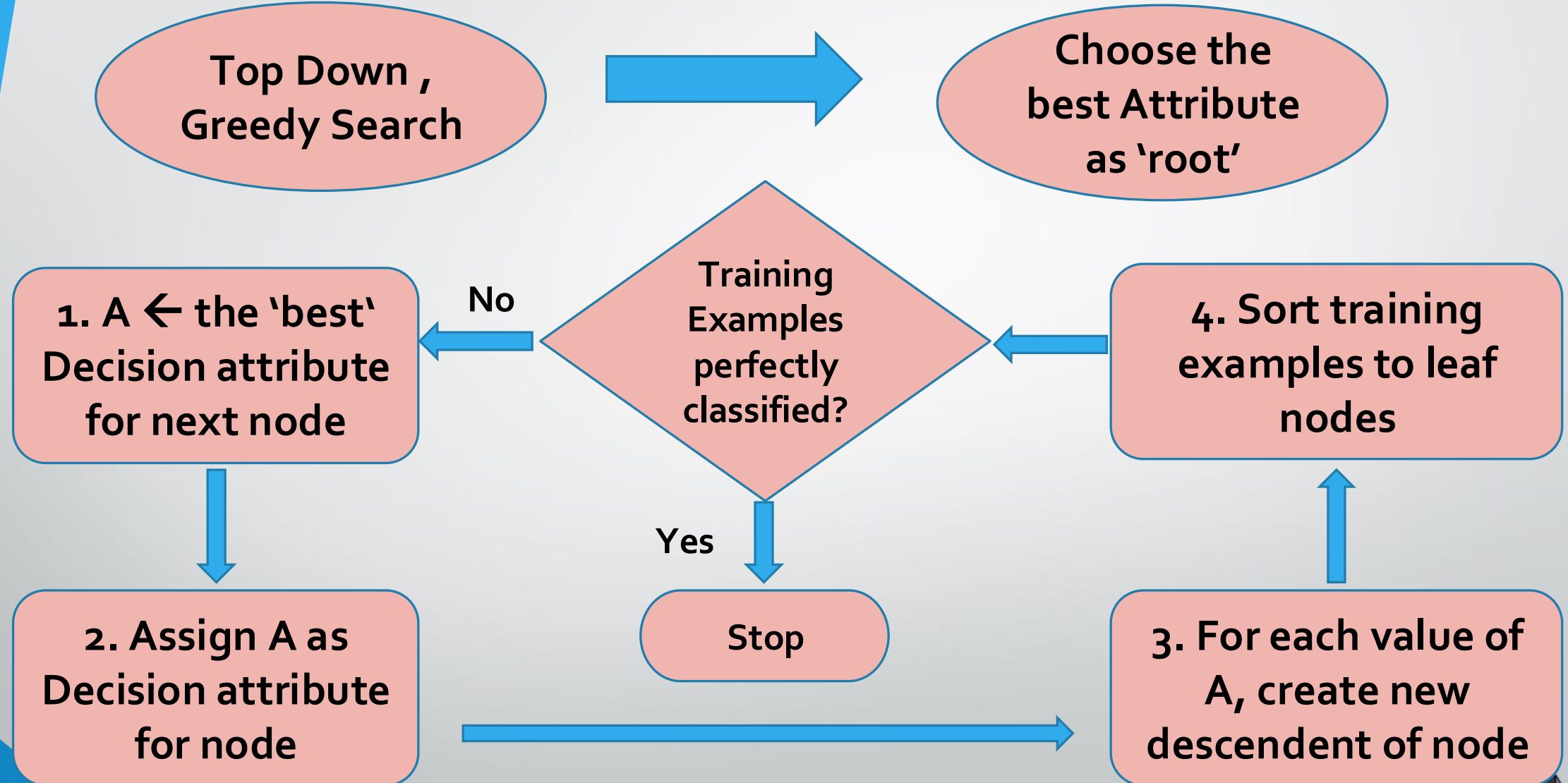
Basic DTL Algorithm (ID3)

Source: Machine Learning, Tom Mitchell

**Pembelajaran Mesin
(Machine Learning)**

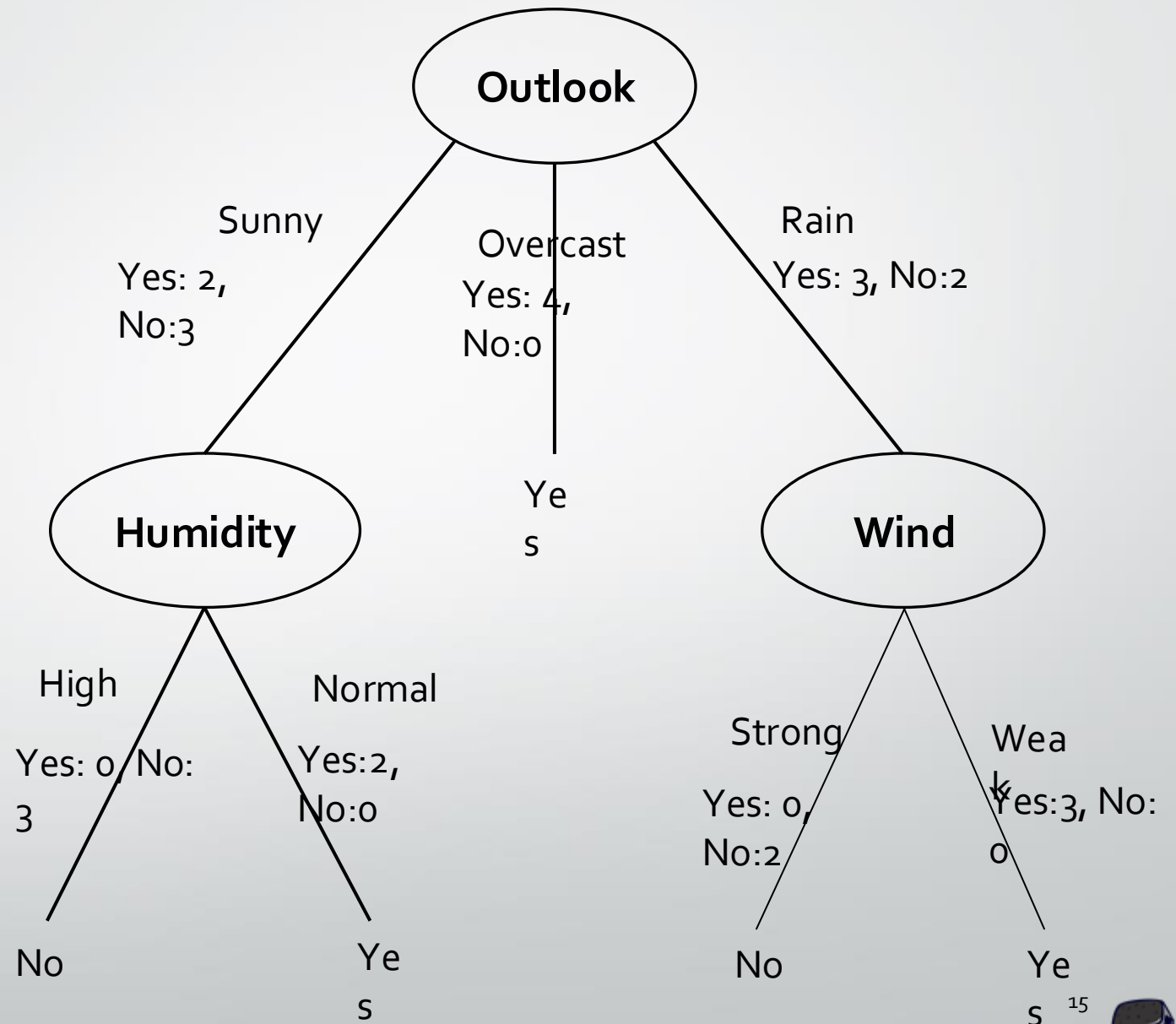


ID₃ Algorithm



Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
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D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



Decision Tree Learning (Russel & Norvig, 2021)

function DECISION-TREE-LEARNING(*examples*, *attributes*, *parent_examples*) **returns**
a tree

if *examples* is empty **then return** PLURALITY-VALUE(*parent_examples*)
else if all *examples* have the same classification **then return** the classification
else if *attributes* is empty **then return** PLURALITY-VALUE(*examples*)
else

$A \leftarrow \operatorname{argmax}_{a \in \text{attributes}} \text{IMPORTANCE}(a, \text{examples})$

tree $\leftarrow$ a new decision tree with root test *A*

for each value v_k of *A* **do**

$\text{exs} \leftarrow \{e : e \in \text{examples} \text{ and } e.A = v_k\}$

subtree $\leftarrow$ DECISION-TREE-LEARNING(*exs*, *attributes* – *A*, *examples*)

add a branch to *tree* with label (*A* = v_k) and subtree *subtree*

return *tree*

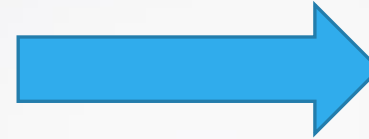
Information gain
Gain ratio

|*A*|=2: binary tree
|*A*| min 2: n-ary tree

The function PLURALITY-VALUE selects the most common output value among a set of examples, breaking ties randomly.

Best Attribute

“Ideally” can classify the training examples into subsets, which has the same class



Information Gain



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Information Gain

Source: Machine Learning, Tom Mitchell

**Pembelajaran Mesin
(Machine Learning)**



Entropy (Information Content)

Measurement in
Information Theory
□ impurity of an
arbitrary collection
of samples

S = set of training examples
Entropy: the
minimum number of bits of
information needed to
encode the classification of
an arbitrary member of S

$$Entropy(S) \equiv \sum_{i=1}^c -p_i \log_2 p_i$$

- S : set of training examples
- c : number of classes
- p_i : proportion of S belonging to class i



Entropy for S with 2 values/ classes

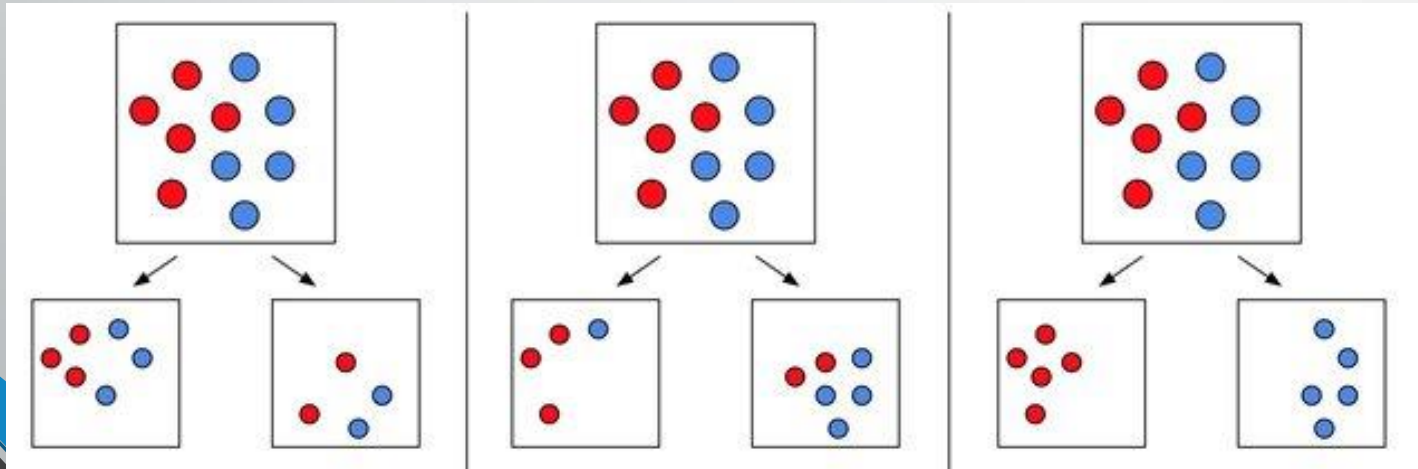
S contains positive examples and negative examples

$$Entropy(S) \equiv -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus}$$

Entropy 0: all examples belong to a class (no surprises, no message need be sent)

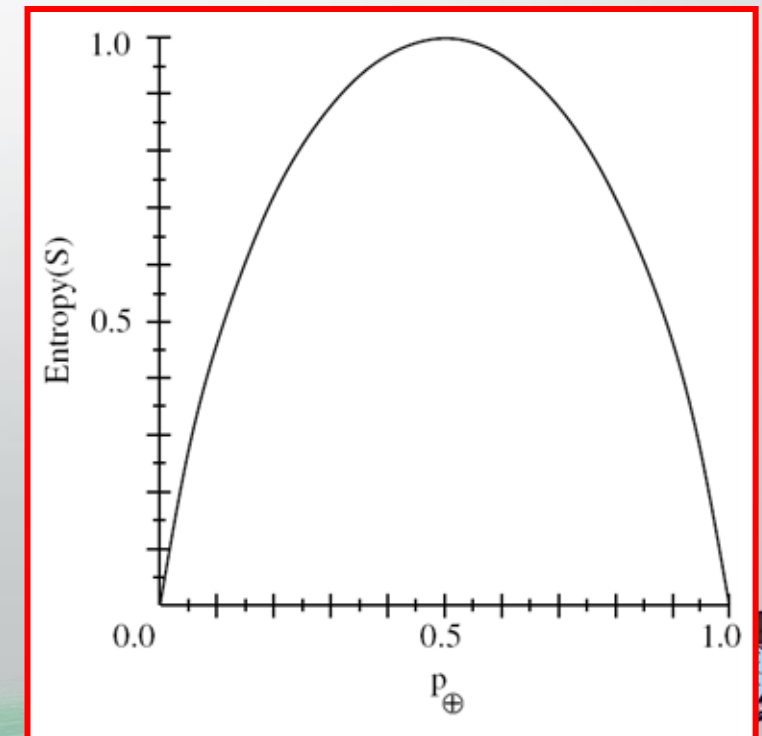
Entropy 1: $p_{+} = p_{-}$
(1 bit is required to indicate the class of the drawn example)

$$I(5/10, 5/10) = 1$$



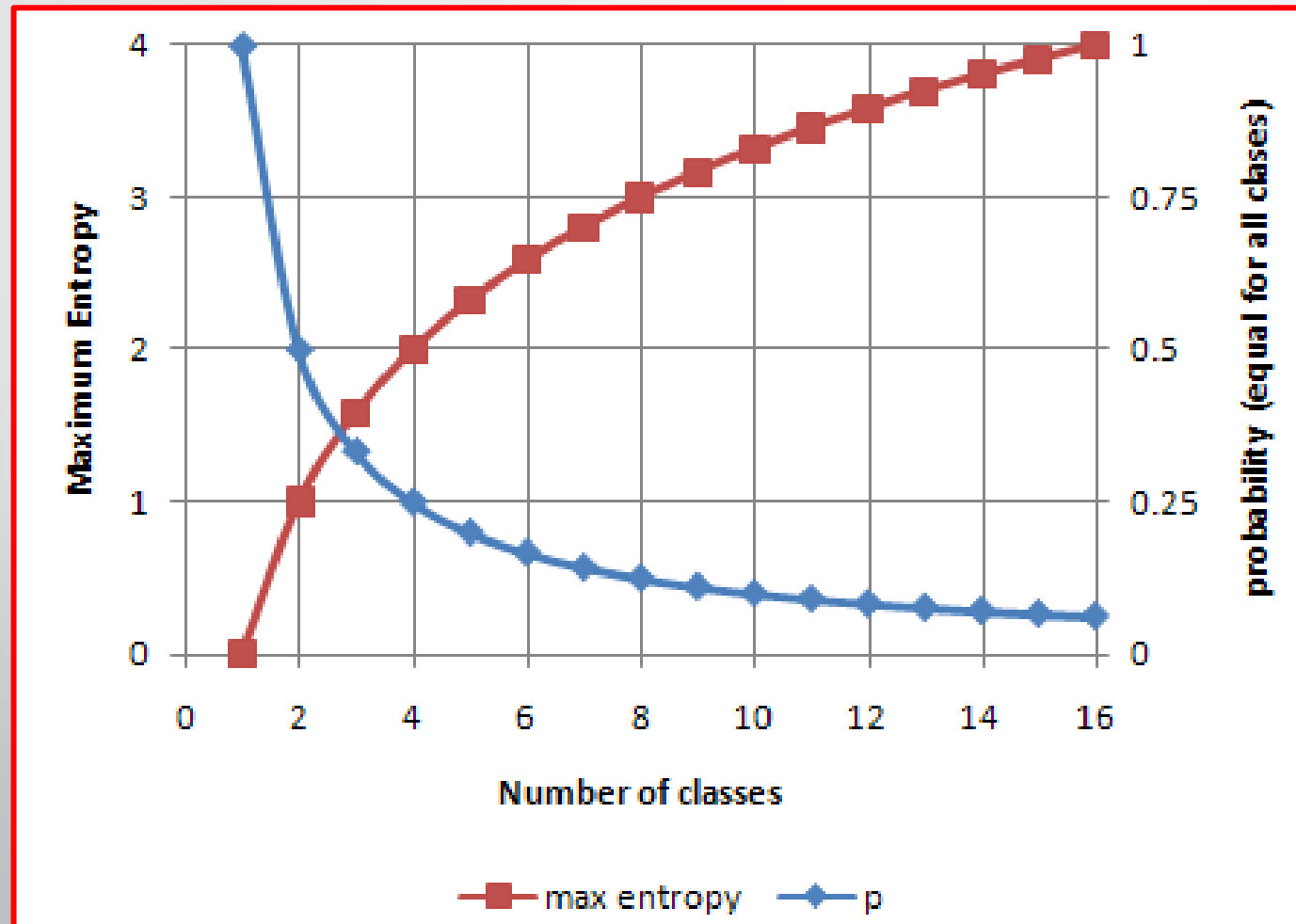
$$I(3/6, 3/6) = 1 \quad I(2/4, 2/4) = 1 \quad I(3/4, 1/4) = 0.81 \quad I(5/5, 0/5) = 0 \quad I(0/5, 5/5) = 0$$

$$I(2/6, 4/6) = 0.92$$



Entropy for S with n values/ classes

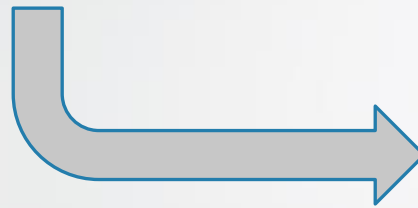
Target attribute has n values: entropy can be as large as $\log_2 n$



Information Gain

Gain(S,A) = expected reduction of entropy due to sorting A

S: set of training examples; A: an attribute



Find A which has maximum Gain(S,A)

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$



Example

Find the 'best' Attribute:
 $\max(\text{Gain}(S, \text{Outlook}),$
 $\text{Gain}(S, \text{Temperature}),$
 $\text{Gain}(S, \text{Humidity}),$
 $\text{Gain}(S, \text{Wind}))$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.940$$

$$\text{Gain}(S, \text{Outlook}) = \text{Entropy}(S) - \sum_{v \in \text{value of Outlook}} \frac{S_v}{S} \text{Entropy}(S_v)$$

$$= 0.940 - \left[\frac{5}{14} \text{Entropy}(\text{Sunny}) + \frac{4}{14} \text{Entropy}(\text{Overcast}) + \frac{5}{14} \text{Entropy}(\text{Rain}) \right]$$

$$= 0.940 - \left[\left(\frac{5}{14} \left(-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right) \right) + \left(\frac{4}{14} \left(-\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} \right) \right) + \left(\frac{5}{14} \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right) \right) \right]$$

$$= 0.246$$



Example

Find the 'best' Attribute:
 $\max(\text{Gain}(S, \text{Outlook}),$
 $\text{Gain}(S, \text{Temperature}),$
 $\text{Gain}(S, \text{Humidity}),$
 $\text{Gain}(S, \text{Wind}))$

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.940$$

$$\begin{aligned} \text{Gain}(S, \text{Humidity}) &= \text{Entropy}(S) - \sum_{v \in \text{value of Humidity}} \frac{S_v}{S} \text{Entropy}(S_v) \\ &= 0.940 - \left[\frac{7}{14} \text{Entropy}(\text{High}) + \frac{7}{14} \text{Entropy}(\text{Normal}) \right] \end{aligned}$$

$$\begin{aligned} &= 0.940 - \left[\left(\frac{7}{14} \left(-\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} \right) \right) + \left(\frac{7}{14} \left(-\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} \right) \right) \right] \\ &= 0.151 \end{aligned}$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
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D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Example

Find the 'best' Attribute:
 $\max(\text{Gain}(S, \text{Outlook}),$
 $\text{Gain}(S, \text{Temperature}),$
 $\text{Gain}(S, \text{Humidity}),$
 $\text{Gain}(S, \text{Wind}))$

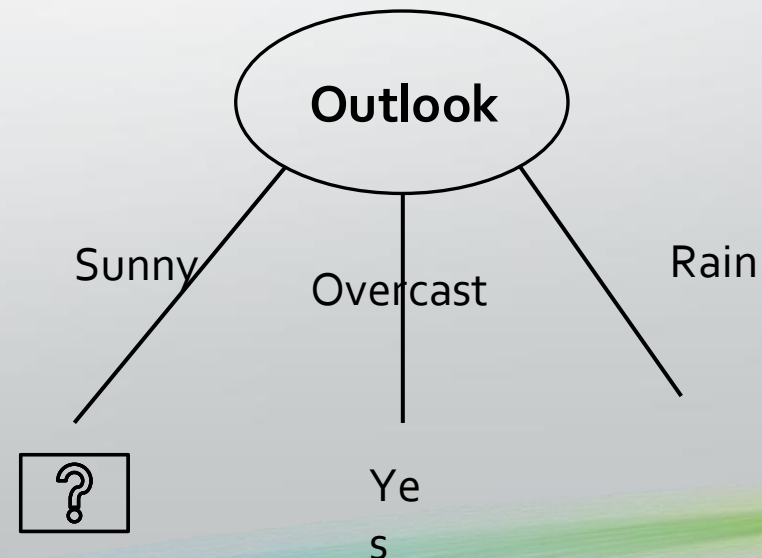
$$\text{Gain}(S, \text{Outlook}) = 0.246$$

$$\text{Gain}(S, \text{Humidity}) = 0.151$$

$$\text{Gain}(S, \text{Temperature}) = 0.029$$

$$\text{Gain}(S, \text{Wind}) = 0.048$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
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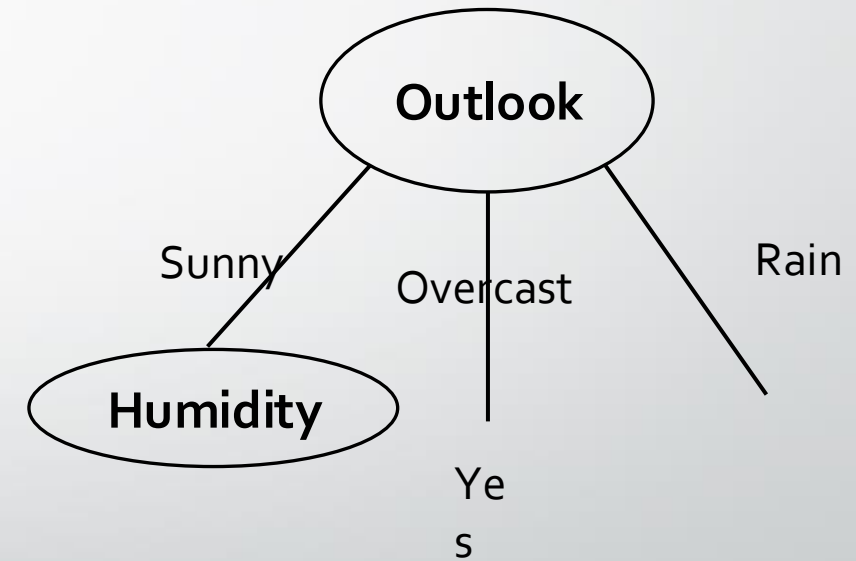
Example

Find the 'best' Attribute:
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$$\text{Entropy}(S) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.971$$

$$\begin{aligned} \text{Gain}(S, \text{Humidity}) &= \text{Entropy}(S) - \sum_{v \in \text{value of Humidity}} \frac{S_v}{S} \text{Entropy}(S_v) \\ &= 0.971 - \frac{3}{5} \text{Entropy}(\text{High}) - \frac{2}{5} \text{Entropy}(\text{Normal}) \\ &= 0.971 - \left[\left(\frac{3}{5} \left(-\frac{0}{3} \log_2 \frac{0}{3} - \frac{3}{3} \log_2 \frac{3}{3} \right) + \left(\frac{2}{5} \left(-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right) \right) \right] \\ &= 0.971 \end{aligned}$$



$$\text{Gain}(S, \text{Humidity}) = 0.971$$

$$\text{Gain}(S, \text{Temperature}) = 0.571$$

$$\text{Gain}(S, \text{Wind}) = 0.020$$



Exercise (Class: Aktivitas)

No	Deadline?	Ada Hangout?	Malas?	Aktivitas
1	Urgent	ya	Ya	Kumpul-kumpul
2	Urgent	Tidak	Ya	Belajar
3	Dekat	ya	Ya	Kumpul-kumpul
4	Tidak ada	Ya	Tidak	Kumpul-kumpul
5	Tidak ada	Tidak	Ya	Jalan-jalan ke mall
6	Tidak ada	Ya	Tidak	Kumpul-kumpul
7	Dekat	Tidak	Tidak	Belajar
8	Dekat	Tidak	Ya	Nonton TV
9	Dekat	Ya	Ya	Kumpul-kumpul
10	Urgent	Tidak	Tidak	Belajar





THANK YOU







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a tree

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else if all *examples* have the same classification **then return** the classification
else if *attributes* is empty **then return** PLURALITY-VALUE(*examples*)
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for each value v_k of *A* **do**

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The function PLURALITY-VALUE selects the most common output value among a set of examples, breaking ties randomly.



Issues in DTL

Overfitting
training
data

Continuous
-valued
attribute

Handling
attributes
with differing
costs

Handling
missing
attribute
value

Alternative
measures for
selecting
attributes



Modul : Issues in Decision Tree Learning (DTL)

Overfitting

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Issues in DTL

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What is Overfit

H : Hypothesis space

A hypothesis: $h \in H$; Alternative hypothesis: $h' \in H$
train: training examples; D : entire distribution of data

$$\text{error}_{\text{train}}(h) < \text{error}_{\text{train}}(h')$$

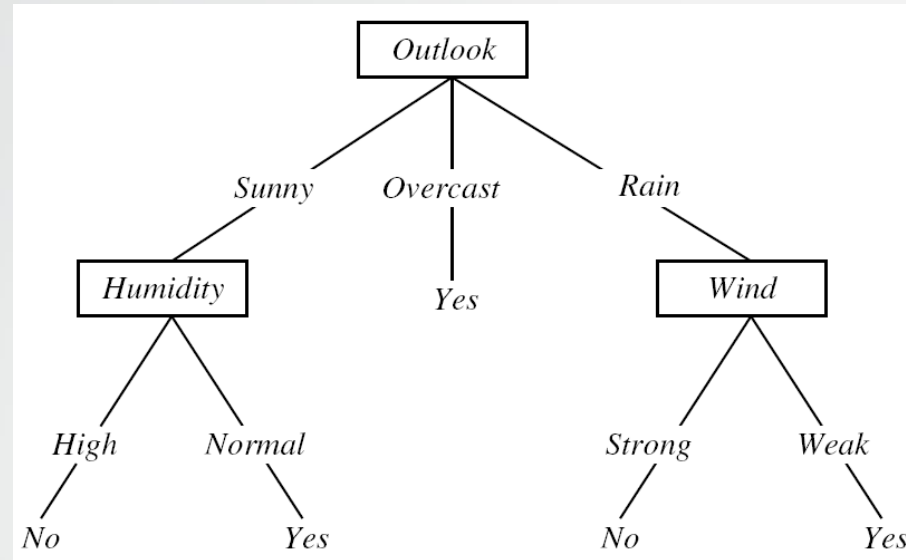
Overfit

$$\text{error}_D(h) > \text{error}_D(h')$$



Illustration

D15 (noisy training examples):
Outlook = Sunny;
Temperature = Hot;
Humidity = Normal;
Wind = Strong;
PlayTennis = No



Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
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D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



Overfitting can happen even training examples is noise-free
(when small numbers of examples are associated with leaf
Nodes) → decrease accuracy 10 – 25% on most problems



Solution Approaches

1. Stop growing the tree earlier, before it reaches the point where it perfectly classifies the training data

2. allow the tree to overfit the data, and then post-prune the tree

Pros

More Direct

More Successfull
in practice

Cons

Difficulty of estimating precisely when to stop growing the tree

Requires more steps (grow until fit, then prune)

what criterion is to be used to determine the correct final tree size



Approaches in Determine the Correct Final Tree Size

1. Use separate examples distinct from training to evaluate the pruning tree

2. Use all available data for training, then conduct a statistical test to check whether expanding (or pruning) a node will produce improvement, example: chi square test

3. Use explicit measure of the complexity for encoding the training examples and decision tree → Minimum Description Length Principle

2/3 Training set

1/3 Validation set

Source:
Machine Learning by Tom Mitchell chapter 6.6

$$h_{MDL} = \operatorname{argmin}_{h \in H} L_{C_1}(h) + L_{C_2}(D|h)$$

$L_{C_1}(h)$: Length (number of bits) of hypothesis encoding
 $L_{C_2}(D|h)$: Length of data D given hypothesis h encoding



Reduced Error Pruning

Consider decision (attribute) node as candidates for pruning → assign the most common classification affiliated with that node



Grow until fit then prune

Split data into training and validation set

Do until further pruning is harmful:

1. Evaluate impact on validation set of pruning each possible node (plus those below it)
2. Greedily remove the one (node) that most improves validation set accuracy

an effective approach provided a large amount of data is available



Rule Post-Pruning

Improvement of ID3
Algorithm: C4.5

Suitable for
limited data

1. Growing the tree from training set, until the training data is fit as well as possible and allowing overfitting to occur.
2. Convert the learned tree into an equivalent set of rules by creating one rule for each path from the root node to a leaf node.
3. Prune (generalize) each rule by removing any preconditions that result in improving its estimated accuracy.
4. Sort the pruned rules by their estimated accuracy, and consider them in this sequence when classifying subsequent instances.



Example

Decision Rules:

**If Outlook = Sunny and Humidity=High
then No**

Pruning:

**If Outlook = Sunny then No
OR**

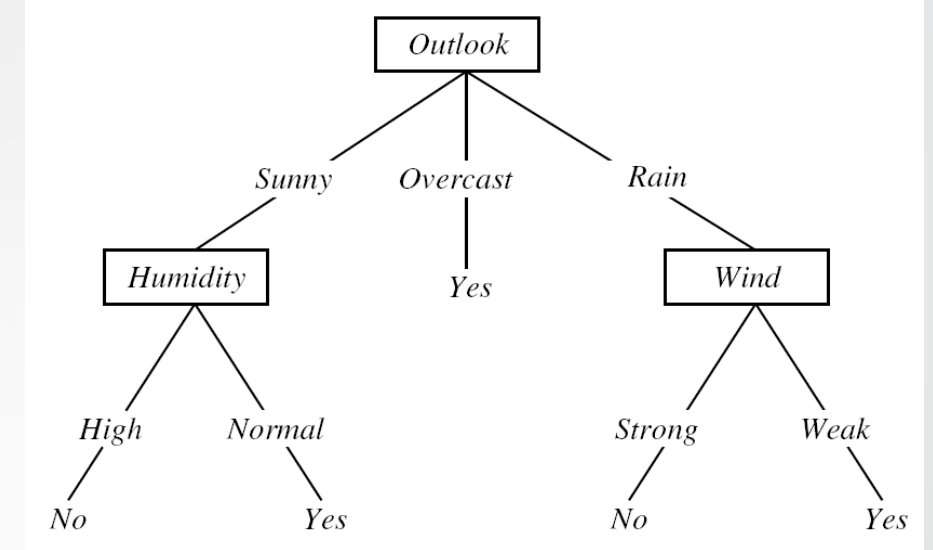
If Humidity=High then No



Increase/ Reduce Accuracy?



**Over validation set/
training set (C4.5)**



Why Decision Tree → Decision Rule ?

1. Distinct path ~ distinct rule:
independent pruning
2. No distinction between
attribute tests
3. Improves readability



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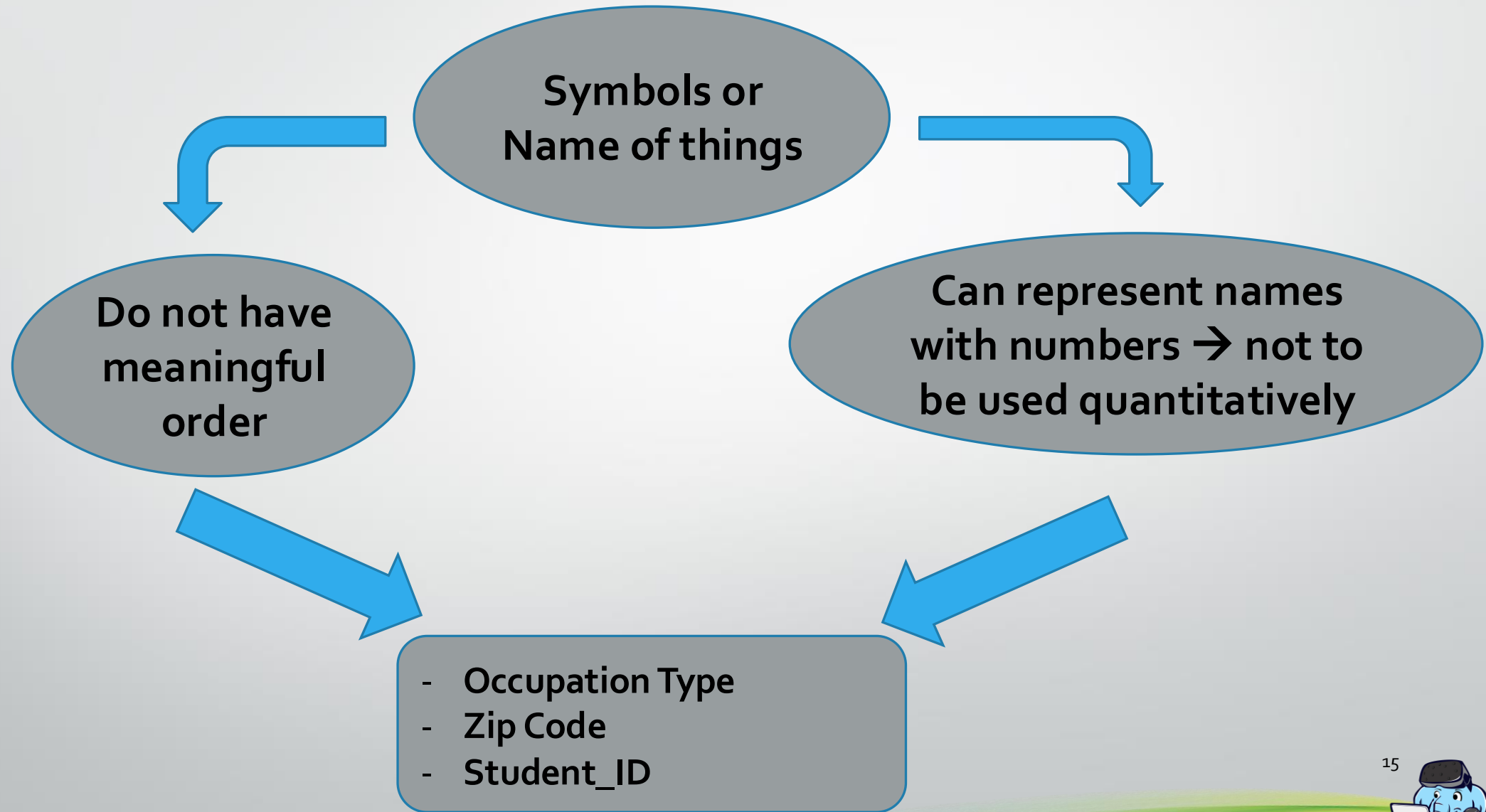
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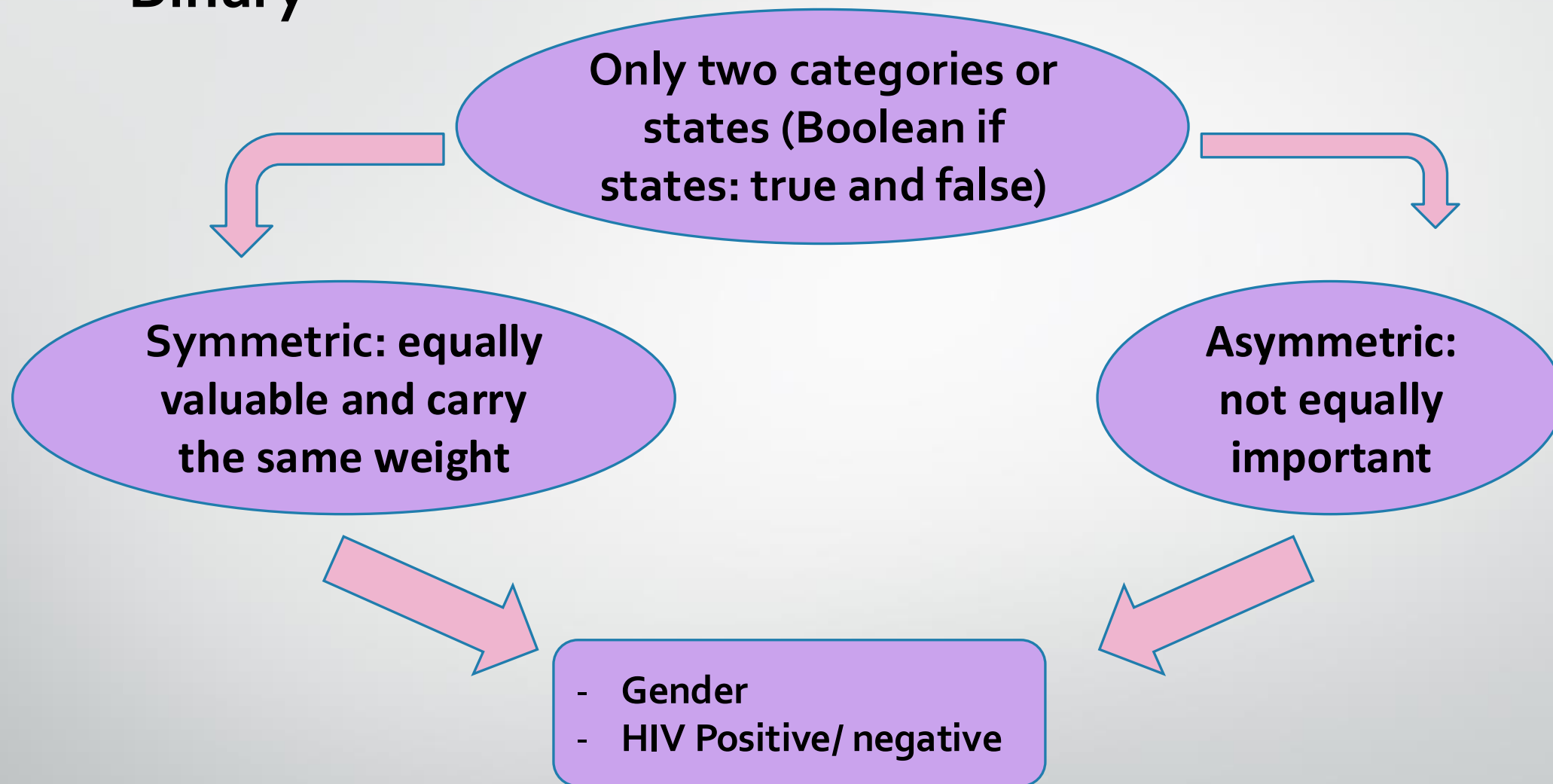
- Years of
experience
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Nominal/ Categorical



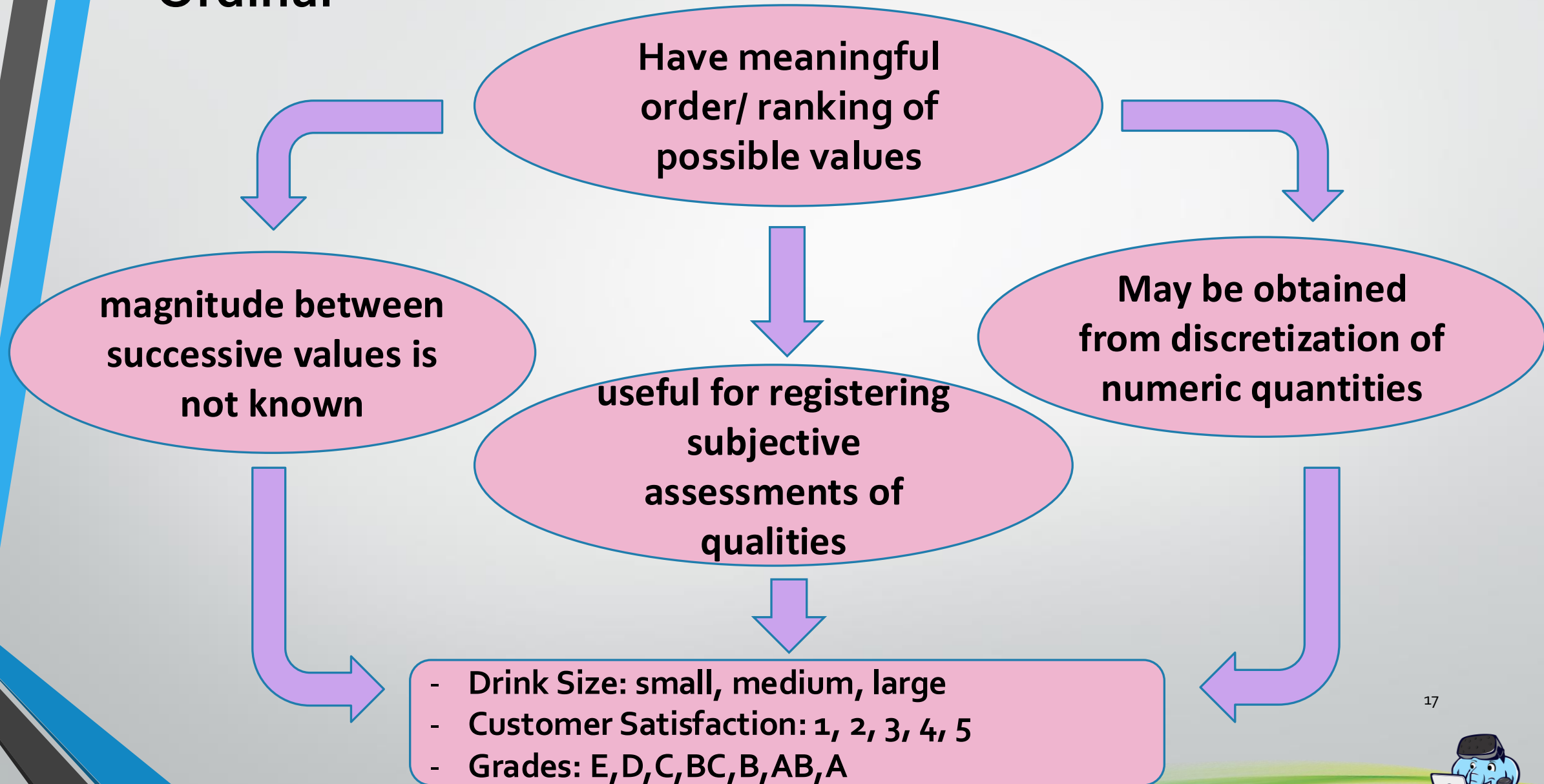
Binary



1	red,	green,	blue
2	1,	0,	0
3	0,	1,	0
4	0,	0,	1



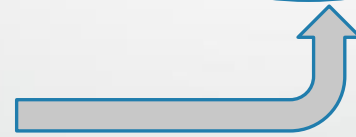
Ordinal



Discrete vs Continuous

has a finite set of values:
Drink size, Age, Medical
test,

has a countably infinite
set of values:
Customer ID, Zip code



Discrete

$\neq$

Continuous



Modul : Issues in Decision Tree Learning (DTL)

Continuous-valued Attribute

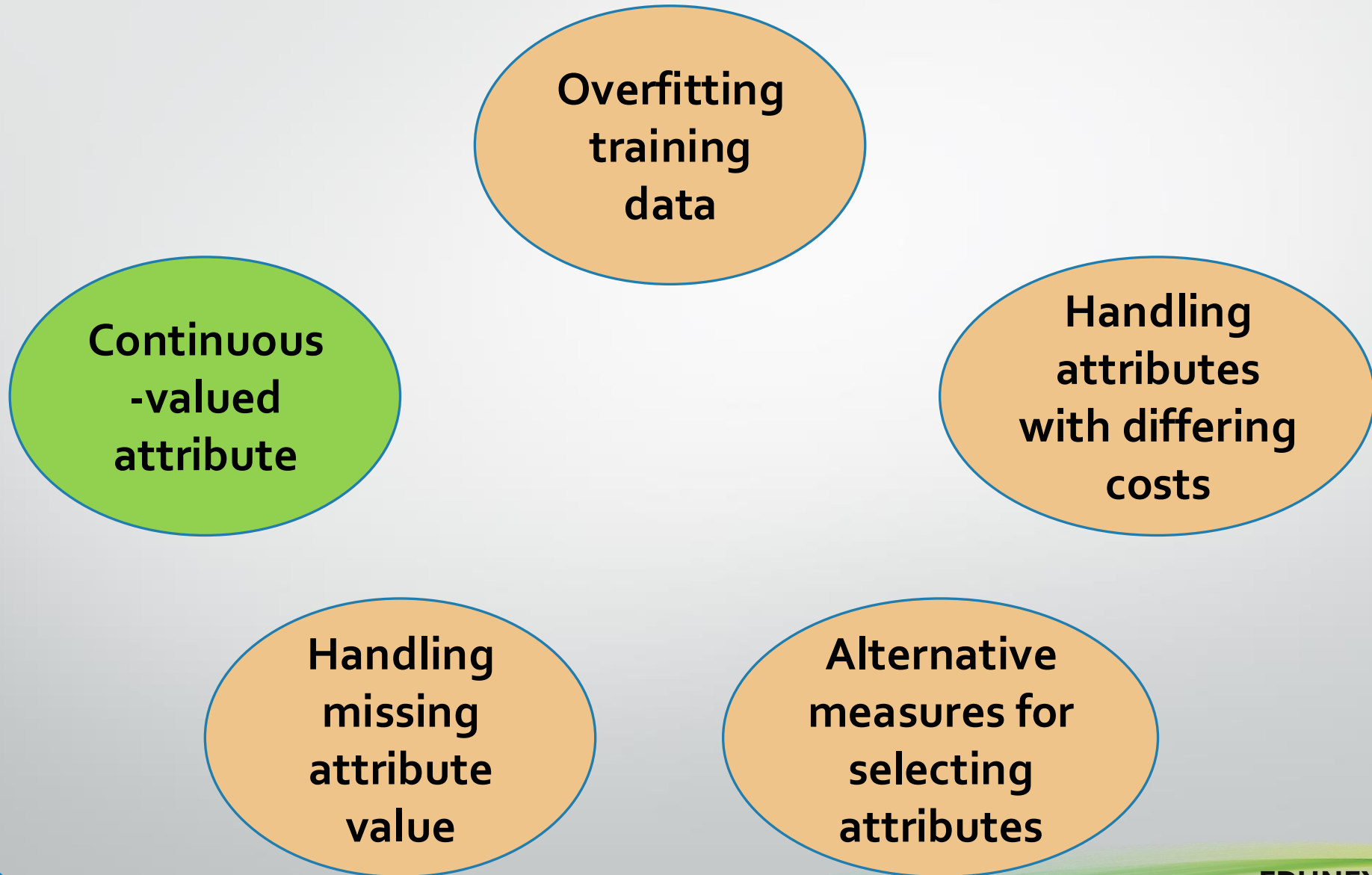
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(Machine Learning)



Issues in DTL



Discretization

Continuous valued attributes →
new discrete valued (boolean)
attribute A_c

True: $A < c$

False: $A < c$

<i>Temperature:</i>	40	48	60	72	80	90
<i>PlayTennis:</i>	No	No	Yes	Yes	Yes	No

Potential optimal breakpoints

$$C = (48+60)/2 = 54$$

Or

$$C = (80+90)/2 = 85$$

What is Best Value
for threshold c ?

Use Information Gain for
each potential breakpoint



Illustration

1. Sort The Continuous-valued attribute

Temperature	40	48	60	72	80	90
Play Tennis	No	No	Yes	Yes	Yes	No



3. Candidates: midway between corresponding values $\rightarrow C : 54$ or $C : 85$

4. Find the greatest Gain from the candidates, and other discrete-valued attributes

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1		72			Yes
D2		40			No
D3		90			No
D4		60			Yes
D5		48			No
D6		80			Yes

2. Identify Adjacent examples that differ in their target class

For $C : 54$

Temperature < 54 : 2 examples $\rightarrow$ yes/0, no/2

Temperature ≥ 54 : 4 examples $\rightarrow$ yes/3, no/1

$$\text{Gain}(S, T_{54}) = \text{Entropi}(S) - [(2/6 * \text{Entropi}(0,2)) + (4/6 * \text{Entropi}(3,1))]$$

For $C : 85$

Temperature < 85 : 5 examples $\rightarrow$ yes/3, no/2

Temperature ≥ 85 : 1 examples $\rightarrow$ yes/0, no/1

$$\text{Gain}(S, T_{85}) = \text{Entropi}(S) - [(5/6 * \text{Entropi}(3,2)) + (1/6 * \text{Entropi}(0,1))]$$

Modul : Issues in Decision Tree Learning (DTL)

Missing Attribute Values

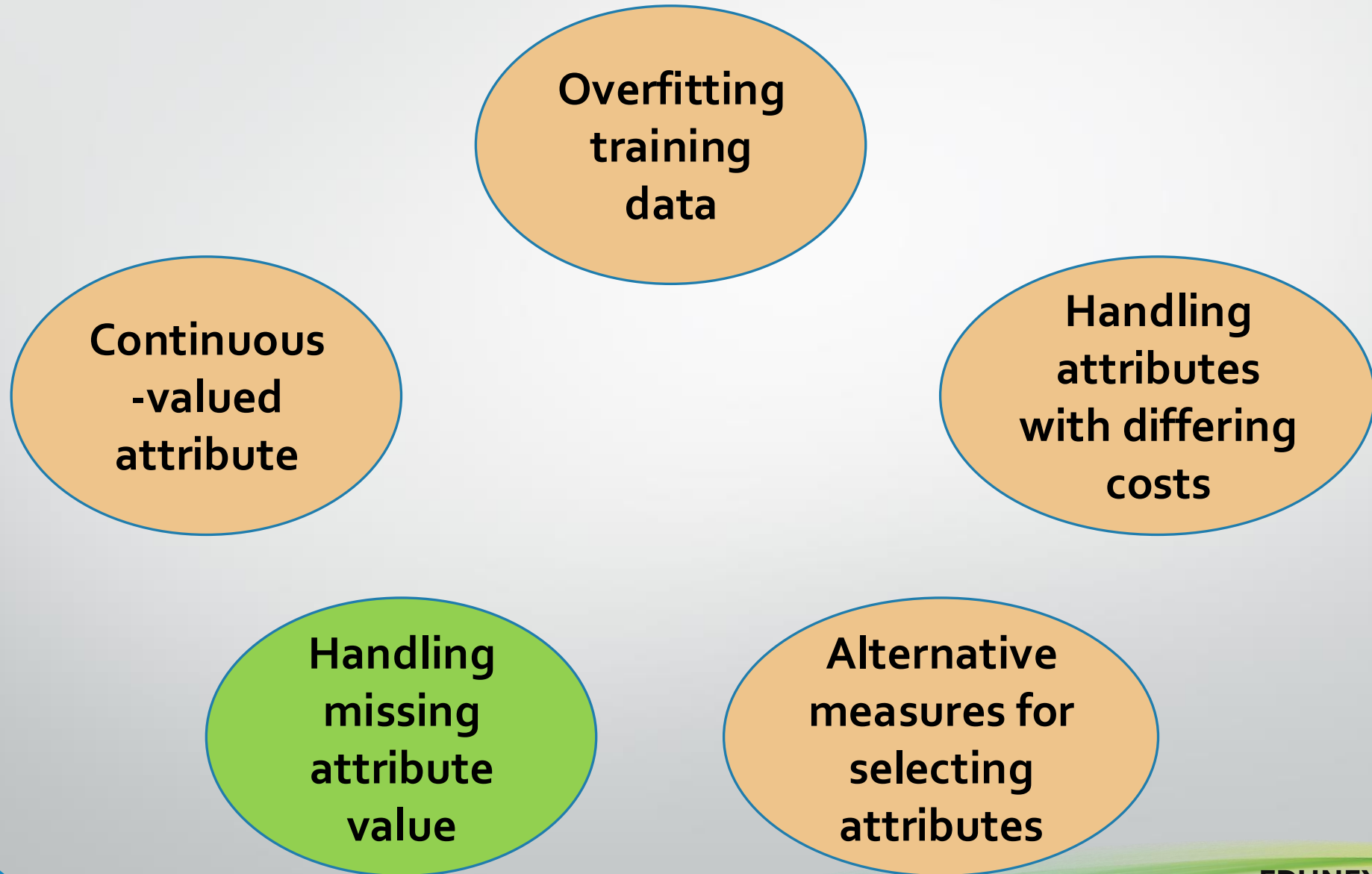
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Issues in DTL



Alternative Strategies

Assign it with the most common value at node n among other examples

Assign it with the most common value at node n that have classification $c(x)$

Assign probability p_i to each possible value v_i of A (used in C4.5)

The probability
Can be used for
classifying
a new
instance
with
missing
value

Gain(S, A) only consider the fraction of training examples with known value
 $\text{Gain}(S, A) = 10/11 * (\text{Entropy}(S) - [\sum \text{proportion} * \text{entropy_of_known_value}])$

$v_1 = 1$, 6 known examples; $v_2 = 0$, 4 known examples, 1 example with missing value of attr A
 $p_i = 6/10$ added to v_1 ; $p_2 = 4/10$ added to $v_2 \rightarrow$ for splitting



Missing Value as Separate Value

Denoted "?" → Null Value
In C4.X

Not
Appropriate
when:



Values are missing due to different reasons

blood sugar value could be missing when it is very high or very low

field IsPregnant missing for a male patient should be treated differently (no) than for a female patient of age 25 (unknown)



Modul : Issues in Decision Tree Learning (DTL)

Alternative Measures for Selecting Attribute

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Issues in DTL

Overfitting
training
data

Continuous
-valued
attribute

Handling
attributes
with differing
costs

Handling
missing
attribute
value

Alternative
measures for
selecting
attributes



Attribute with many values (C4.5)

Gain will always select it → example *Date=2021_Jan_31*

Date will perfectly classify training examples,
but very poor predictor for unseen data



GAIN RATIO

$$GainRatio(S, A) \equiv \frac{Gain(S, A)}{SplitInformation(S, A)}$$

$$SplitInformation(S, A) \equiv - \sum_{i=1}^c \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|}$$

where S_i is subset of S for which A has value v_i

29



Illustration

Date	Atr2	Atr3	Class
2021_Jan_01	v1		No
2021_Jan_02	v1		No
2021_Jan_03	v2		Yes
2021_Jan_04	v2		Yes
2021_Jan_05	v1		Yes
2021_Jan_06	v1		No

$$\begin{aligned} \text{SplitInformation}(S, \text{Date}) &= - \sum_{i=1}^6 \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|} \\ &= - \left(\frac{1}{6} \log_2 \frac{1}{6} + \frac{1}{6} \log_2 \frac{1}{6} + \frac{1}{6} \log_2 \frac{1}{6} + \frac{1}{6} \log_2 \frac{1}{6} + \frac{1}{6} \log_2 \frac{1}{6} + \frac{1}{6} \log_2 \frac{1}{6} \right) \end{aligned}$$

$$\begin{aligned} \text{SplitInformation}(S, \text{Atr2}) &= - \sum_{i=1}^2 \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|} \\ &= - \left(\frac{4}{6} \log_2 \frac{4}{6} + \frac{2}{6} \log_2 \frac{2}{6} \right) \end{aligned}$$

What if SplitInformation is very small or zero ($|S_i| \approx |S|$)
 → GainRatio undefined or very large



Heuristic: Apply GainRatio test only for Attribute with above average Gain



Modul : Issues in Decision Tree Learning (DTL)

Attributes with Differing Costs

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Pembelajaran Mesin
(Machine Learning)



Issues in DTL

**Overfitting
training
data**

**Continuous
-valued
attribute**

**Handling
attributes
with differing
costs**

**Handling
missing
attribute
value**

**Alternative
measures for
selecting
attributes**



Attribute with Different Cost

Attributes: Temperature, BiopsyResult, Pulse, BloodTestResults

Have Different Cost (monetary and patient comfort)



Use low cost attribute where possible, high cost only when required to produce reliable classification

Cost is considered in calculating Gain of each attribute



Approaches

Tan and Schlimmer (1990)
and Tan (1993):

$$\frac{Gain^2(S, A)}{Cost(A)}$$

Nunez (1988):

$$\frac{2^{Gain(S, A)} - 1}{(Cost(A) + 1)^w}$$

Where $w \in [0, 1]$ determine
importance of cost



Exercise

Outlook	Temp	Humidity	Windy	Class
sunny	75	70	TRUE	Play
sunny	80	90	TRUE	Don't Play
sunny	85	85	FALSE	Don't Play
sunny	72	95	FALSE	Don't Play
sunny	69	70	FALSE	Play
?	72	90	TRUE	Play
overcast	83	78	FALSE	Play
overcast	64	65	TRUE	Play
overcast	81	75	FALSE	Play
rain	71	80	TRUE	Don't Play
rain	65	70	TRUE	Don't Play
rain	75	80	FALSE	Play
rain	68	80	FALSE	Play
rain	70	96	FALSE	Play

1. What is GainRatio for Outlook?
2. What are Examples (instances) for Outlook = sunny and what is the weight for instance with outlook=? related with outlook=sunny
3. Based on the result of question (2), define the threshold in "Humidity" discretization, and the leaf node for each branch.



THANK YOU







IF3170 Artificial Intelligence

Regression

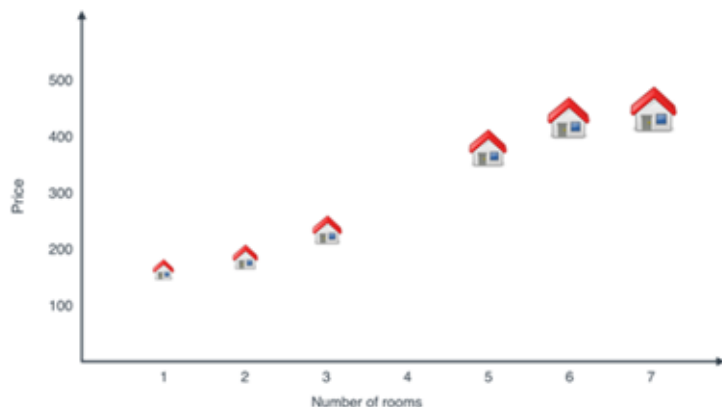
Masayu Leylia Khodra

Capaian Pembelajaran

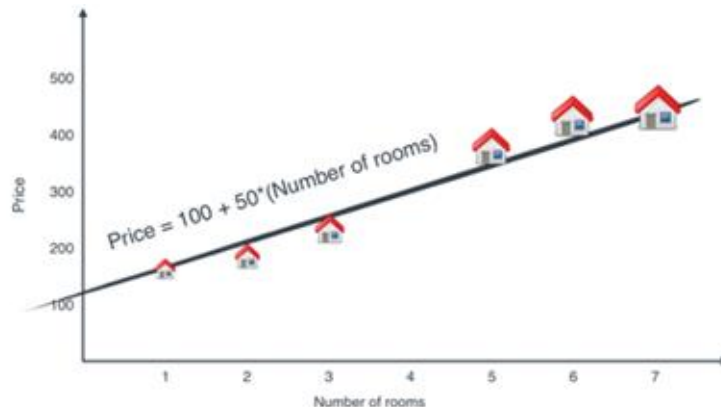
- Mahasiswa dapat menjelaskan dan menerapkan penggunaan konsep supervised learning
- Mahasiswa dapat menjelaskan dan menerapkan teknik linear regression untuk persoalan pembelajaran mesin yang sesuai.

Regression: Introduction

As an example, we are a real estate agent, and in charge of selling a new house. We don't know the price, and we want to infer it by comparing it with other houses. We look at features of the house which could influence the house, i.e. number of rooms. At the end of the day, what we want is **a formula on the feature** which gives us the price of the house, or at least an estimate for it.



train



Linear Relationship

- The price of the house is **dependent variable** or **response**, and number of rooms is **independent variable** or **regressor** or **predictor**.
- A reasonable form of a relationship between the response Y and the regressor x is the linear relationship: $Y = \beta_0 + \beta_1 x$

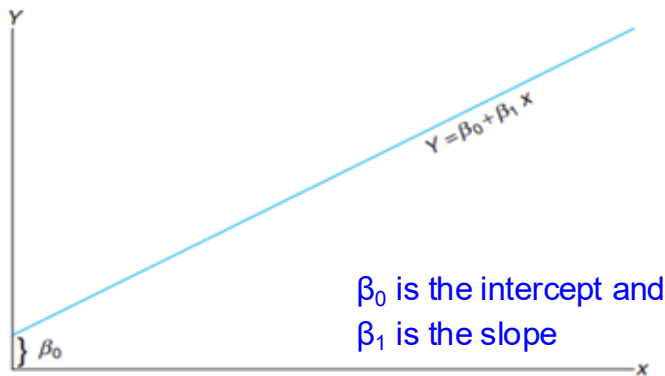
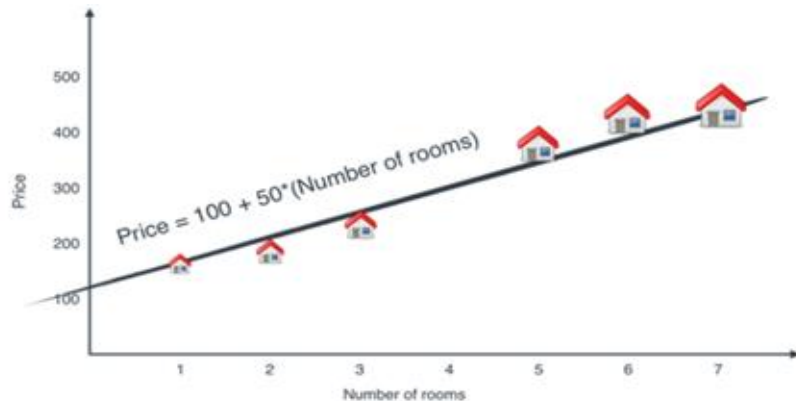
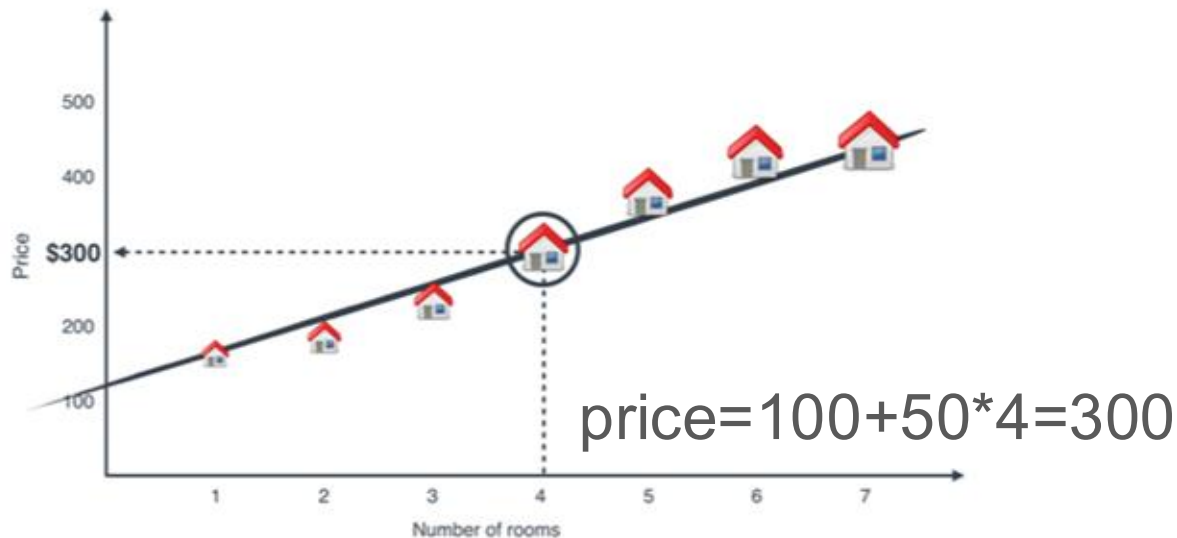


Figure 11.1: A linear relationship; β_0 : intercept; β_1 : slope.

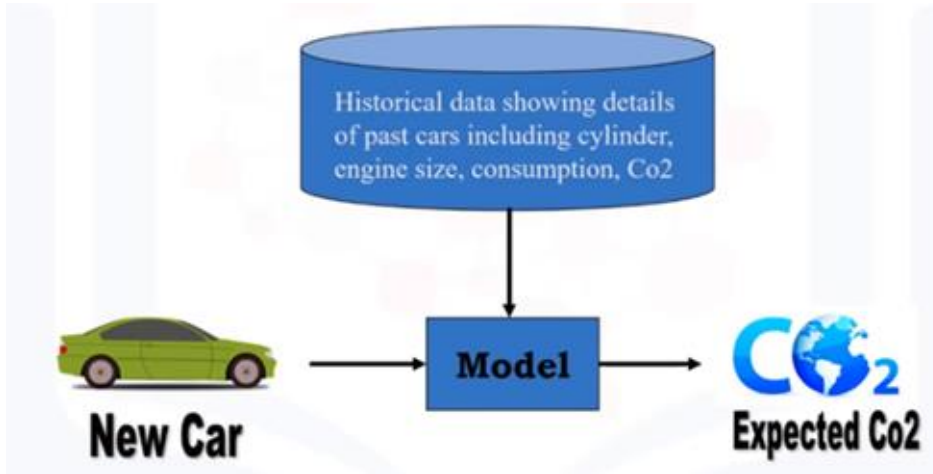


Regression: Predict Response Y based on Regressor x

Figure 6. Our task is now to predict the price of the house with 4 rooms. Using the model (line), we deduce that the predicted price of this house is \$300.



Regression Modeling: Example

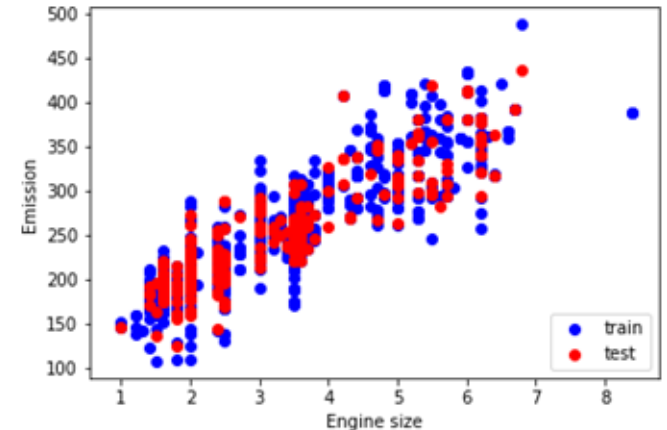


```
1 df.shape
```

(1067, 13)

```
1 df[['ENGINE SIZE', 'CO2EMISSIONS']].sample(5)
```

	ENGINE SIZE	CO2EMISSIONS
697	2.0	196
872	3.8	253
614	2.4	200
351	5.4	382
426	3.5	258

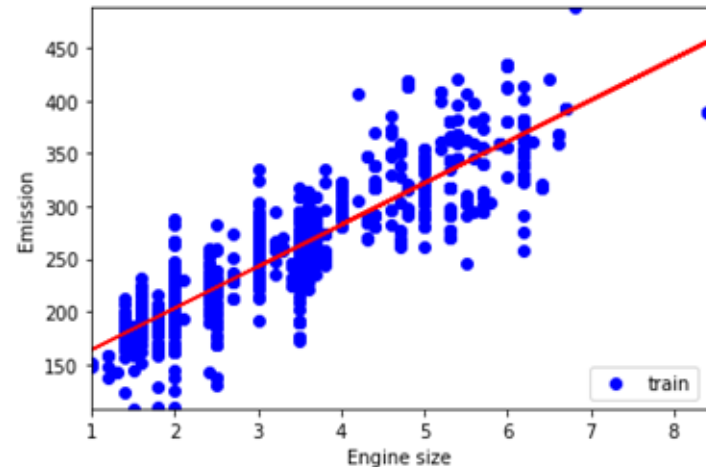


Simple Linear Regression: Training

```
1 from sklearn import linear_model
2 regr = linear_model.LinearRegression()
3 train_x = np.asanyarray(train[['ENGINE SIZE']])
4 train_y = np.asanyarray(train[['CO2EMISSIONS']])
5 regr.fit(train_x, train_y)
6 # The coefficients
7 print('Coefficients: ', regr.coef_)
8 print('Intercept: ', regr.intercept_)
```

Coefficients: $[[39.43522758]]$

Intercept: $[124.41641136]$



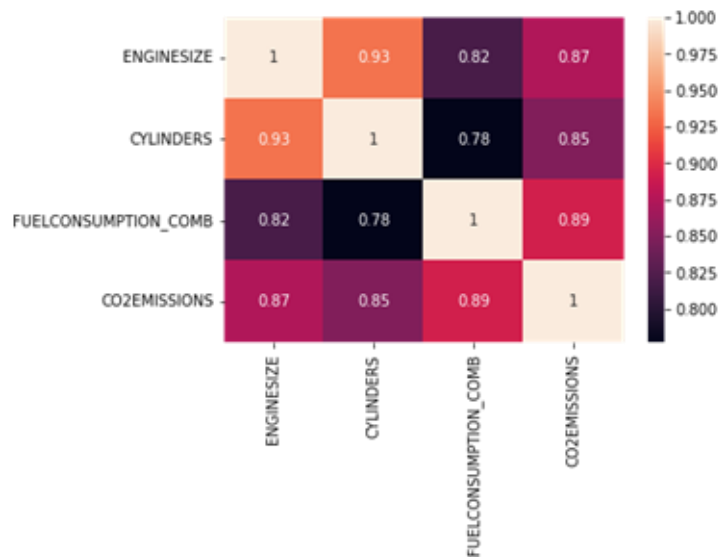
$$y = 125.49 + 39.14x$$

Multivariate Linear Regression: Training & Testing

```
1 from sklearn import linear_model
2 regr2 = linear_model.LinearRegression()
3 train_x = np.asanyarray(train[['ENGINE_SIZE', 'CYLINDERS', 'FUELCONSUMPTION_COMB']])
4 train_y = np.asanyarray(train[['CO2EMISSIONS']])
5 regr2.fit(train_x, train_y)
6 print('Coefficients: ', regr2.coef_)
7 print('Intercept: ', regr2.intercept_)
```

Coefficients: [[11.36694334 7.26349823 9.58219844]]
Intercept: [64.98474043]

	ENGINE_SIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244



Estimating Parameters: Least Square Estimator

- Objective function: minimize sum of the squares of the residuals/errors (SSE)

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

$$\frac{\partial(SSE)}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i)$$

$$nb_0 + b_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$b_0 = \frac{\sum_{i=1}^n y_i - b_1 \sum_{i=1}^n x_i}{n} = \bar{y} - b_1 \bar{x}$$

$$\frac{\partial(SSE)}{\partial b_1} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) x_i$$

$$b_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sigma_{xy} / \sigma_x^2$$

Let X be a random variable with probability distribution $f(x)$ and mean μ . The variance of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x), \quad \text{if } X \text{ is discrete, and}$$

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx, \quad \text{if } X \text{ is continuous.}$$

The positive square root of the variance, σ , is called the **standard deviation** of X .

Let X and Y be random variables with joint probability distribution $f(x, y)$. The covariance of X and Y is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y)$$

if X and Y are discrete, and

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) dx dy$$

if X and Y are continuous.

Let X and Y be random variables with covariance σ_{XY} and standard deviations σ_X and σ_Y , respectively. The correlation coefficient of X and Y is

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

Implement From Scratch: Simple Linear Regression

From Scratch

```
b1=covariance(X,Y)/variance(X)
```

```
b0=mean(Y)-b1*mean(X)
```

```
1 # Coefficients: [[39.43522758]]
2 # Intercept: [124.41641136]
3 print('cov(X,Y)=',train.CO2EMISSIONS.cov(train.ENGINESIZE))
4 print('var(X)=',train.ENGINESIZE.var())
5 b1=train.CO2EMISSIONS.cov(train.ENGINESIZE)/train.ENGINESIZE.var()
6 b0=train.CO2EMISSIONS.mean()-b1*train.ENGINESIZE.mean()
7 print('Coefficient=',b1)
8 print('Intercept=',b0)
```

```
cov(X,Y)= 78.95604822766053
var(X)= 2.0021704722919877
Coefficient= 39.43522757943557
Intercept= 124.41641136306387
```

```
1 from sklearn import linear_model
2 regr = linear_model.LinearRegression()
3 train_x = np.asanyarray(train[['ENGINE_SIZE']])
4 train_y = np.asanyarray(train[['CO2EMISSIONS']])
5 regr.fit (train_x, train_y)
6 # The coefficients
7 print ('Coefficients: ', regr.coef_)
8 print ('Intercept: ',regr.intercept_)
```

```
Coefficients: [[39.43522758]]
Intercept: [124.41641136]
```

Linear Regression: Supervised Learning

Learning a (possibly incorrect) general function from specific input-output pairs

I/O pairs
Training set
<data, label>

Least Square
Estimation

Hypothesis h
 $h \approx f$

$$y = 64.98 + 11.37x_1 + 7.26x_2 + 9.58x_3$$

	ENGINE SIZE	CYLINDERS	FUEL CONSUMPTION_COMB	CO2 EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244

0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244

Unseen
data

Predict

label

	ENGINE SIZE	CYLINDERS	FUEL CONSUMPTION_COMB
0	2.0	4	8.5
1	2.4	4	9.6
2	1.5	4	5.9

0	2.0	4	8.5
1	2.4	4	9.6
2	1.5	4	5.9

```
1 test_y[0:3]
array([[196],
       [221],
       [136]], )
```

$$y = 64.98 + 11.37x_1 + 7.26x_2 + 9.58x_3$$

```
1 #test_y_ = regr2.predict(test_x)
2 test_y_[0:3]
array([[198.22130673],
       [213.30850235],
       [167.62411912]])
```

Linear Regression: Testing

```
1 from sklearn.metrics import r2_score
2
3 test_x = np.asanyarray(test[['ENGINE_SIZE']])
4 test_y = np.asanyarray(test[['CO2EMISSIONS']])
5 test_y_ = regr.predict(test_x)
6
7 print("Mean absolute error: %.2f" % np.mean(np.absolute(test_y_ - test_y)))
8 print("Residual sum of squares (MSE): %.2f" % np.mean((test_y_ - test_y) ** 2))
9 print("R2-score: %.2f" % r2_score(test_y_ , test_y) )
```

Mean absolute error: 22.50

Residual sum of squares (MSE): 840.97

R2-score: 0.73

$$y = 64.98 + 11.37x_1 + 7.26x_2 + 9.58x_3$$

```
1 from sklearn.metrics import r2_score
2
3 test_x = np.asanyarray(test[['ENGINE_SIZE', 'CYLINDERS', 'FUELCONSUMPTION_COMB']])
4 test_y_ = regr2.predict(test_x)
5
6 print("Mean absolute error: %.2f" % np.mean(np.absolute(test_y_ - test_y)))
7 print("Residual sum of squares (MSE): %.2f" % np.mean((test_y_ - test_y) ** 2))
8 print("R2-score: %.2f" % r2_score(test_y_ , test_y) )
```

Mean absolute error: 15.20

Residual sum of squares (MSE): 443.90

R2-score: 0.87

$$y = 125.49 + 39.14x$$

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Coefficient of Determination, R^2

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

Linear Regression - LSE: Summary

Target function is estimated by **hypothesis** as a linear function of X :

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \dots + \theta_n x_n = \sum \theta_i x_i, i \text{ in } [0..n]$$
$$\rightarrow h_{\theta}(x) = \theta^T \cdot x$$

Cost function $J(\theta)$ assigns a large cost to bad predictions $h(x)$ and small cost to good predictions $h(x)$. For LSE, we used **sum of squares errors (SSE)**.

Find good θ (parameters or weights): $\theta^* = \operatorname{argmin} J(\theta)$

Analytical solution: find θ that solve $\nabla J(\theta) = 0$.

If $\nabla J(\theta) = 0$ is unable to solve, use iterative numerical methods. The simplest method is gradient descent.

Pertanyaan ?



Logistic Regression

What & Why

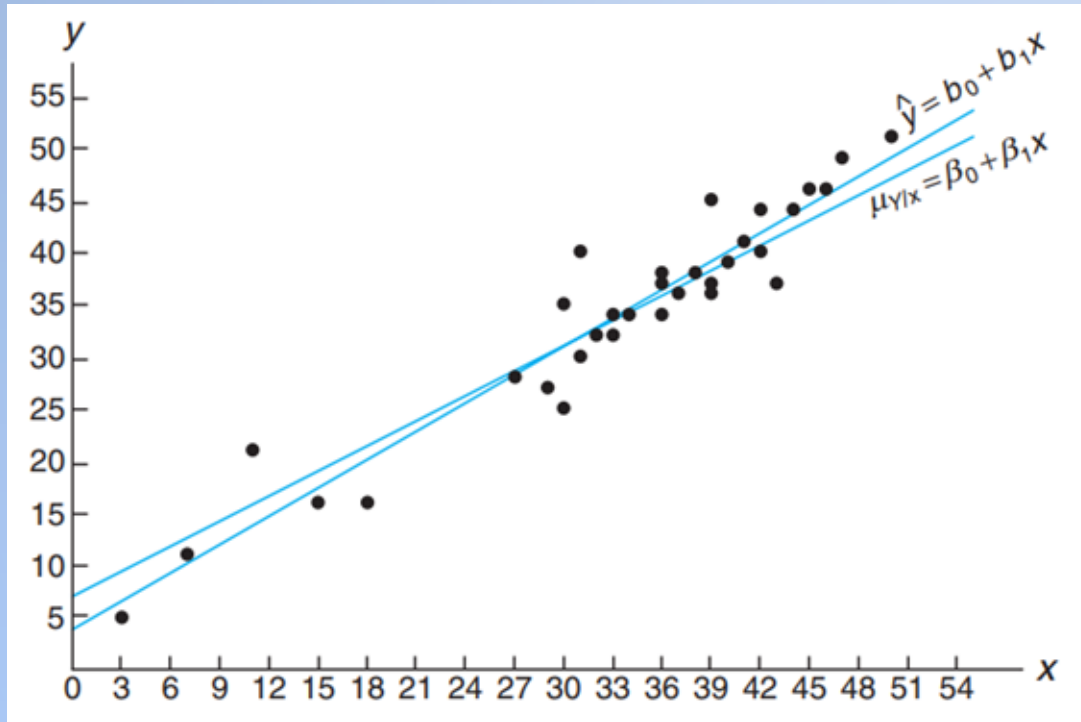
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(masayu@informatika.org)

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ITB

Pembelajaran Mesin
(*Machine Learning*)



REGRESSION ANALYSIS



Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (2012). Probability and Statistics for engineering and sciences. *Pearson Education*, 430-435.

Finding the best relationship between Y and x : not deterministic, random error

Quantifying the strength of that relationship. Random error with $E(\text{error})=0$ and homoscedasticity. Least Squared Estimation (LSE)

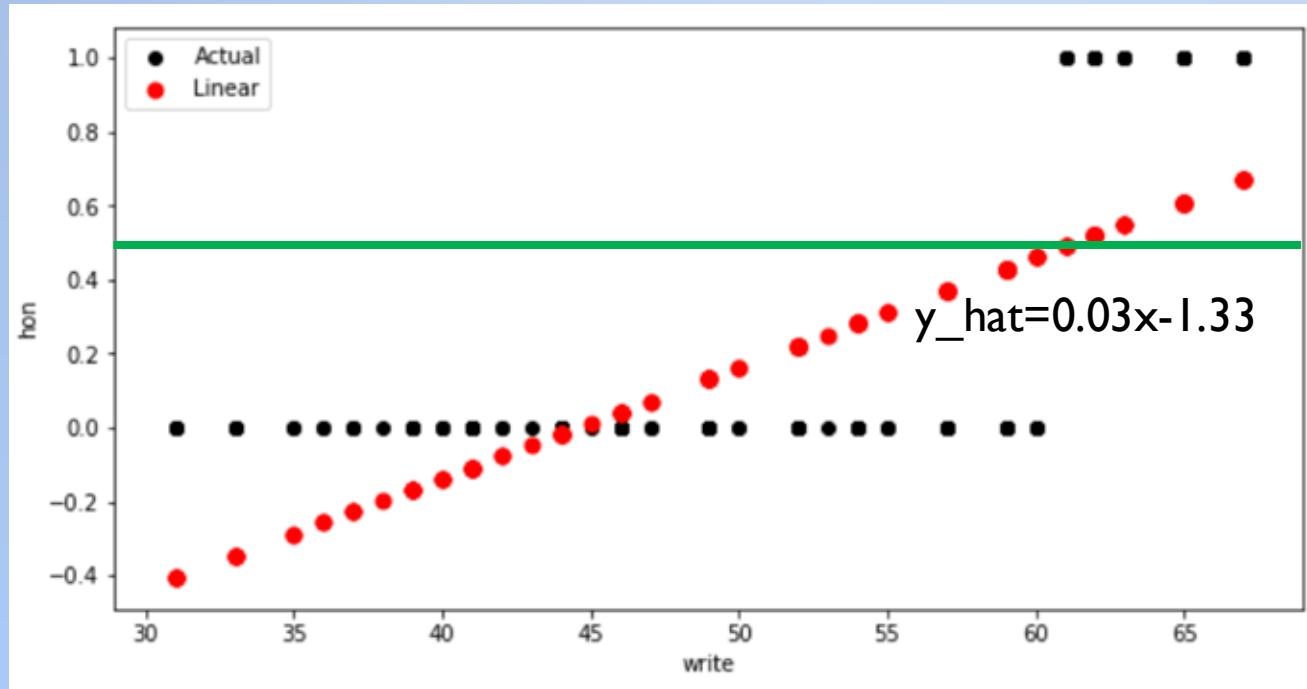
Predicting of the response value y given values of the regressor x .



CLASSIFICATION USING LINEAR REGRESSION

LINEAR REGRESSION + THRESHOLD

Example: Dataset has 200 observations (5 attributes), and the target attribute is **hon**, indicating if a student is in an honors class or not.



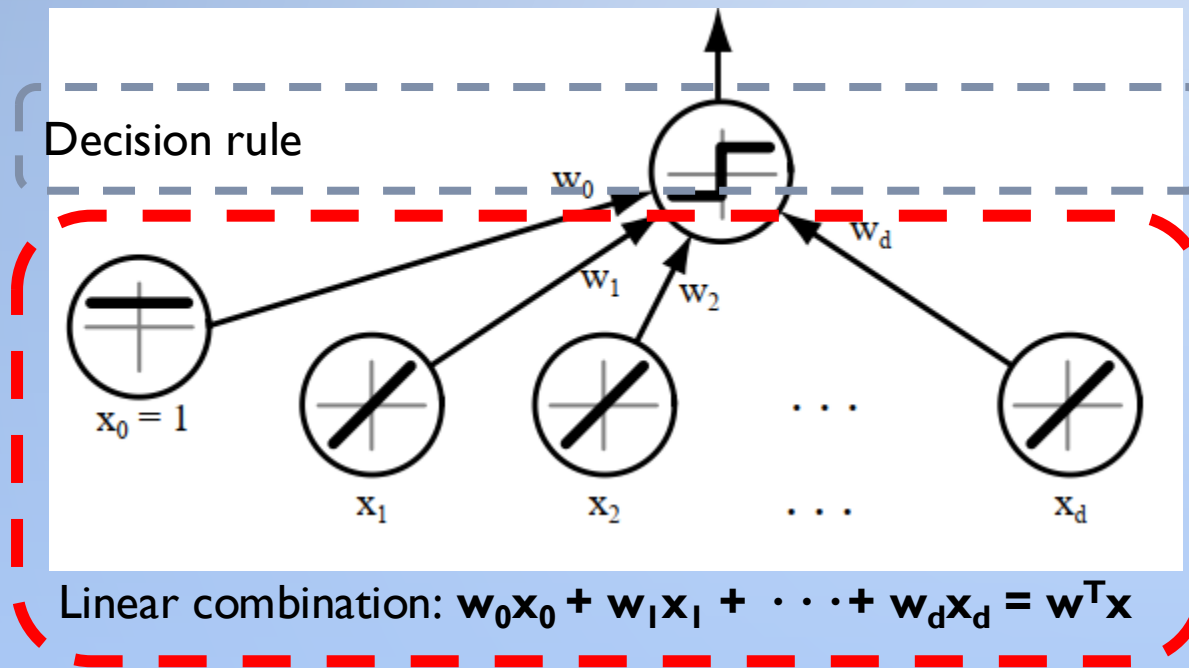
We are able to predict the value along the Y-axis. If Y is greater than 0.5 (above the green line), predict that the student is in honor class otherwise not in honor class.

Linear Model or
Linear Discriminant Function
or Linear Classifier



LINEAR DISCRIMINANT FUNCTION

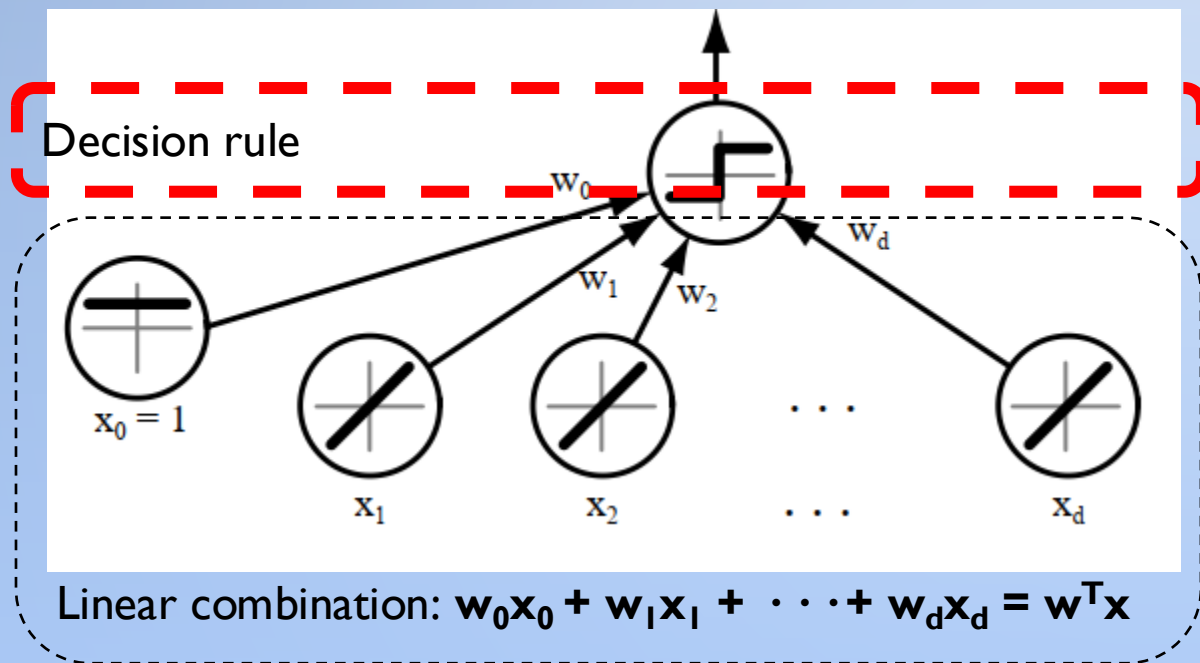
LINEAR COMBINATION + DECISION RULE



Discriminant function:
a linear combination of the
components of $\mathbf{x}$
$$g(\mathbf{x}) = w_0x_0 + w_1x_1 + \dots + w_dx_d = \mathbf{w}^T \mathbf{x}$$



HYPERPLANE DECISION SURFACE



Discriminant function:
 $g(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$

Decision rule:
 Decide ω_1 if $g(\mathbf{x}) > 0$ and ω_2 if $g(\mathbf{x}) < 0$
 $g(\mathbf{x}) = 0$: decision surface
 ω_1 and ω_2 are target classes



HYPERPLANE DECISION SURFACE

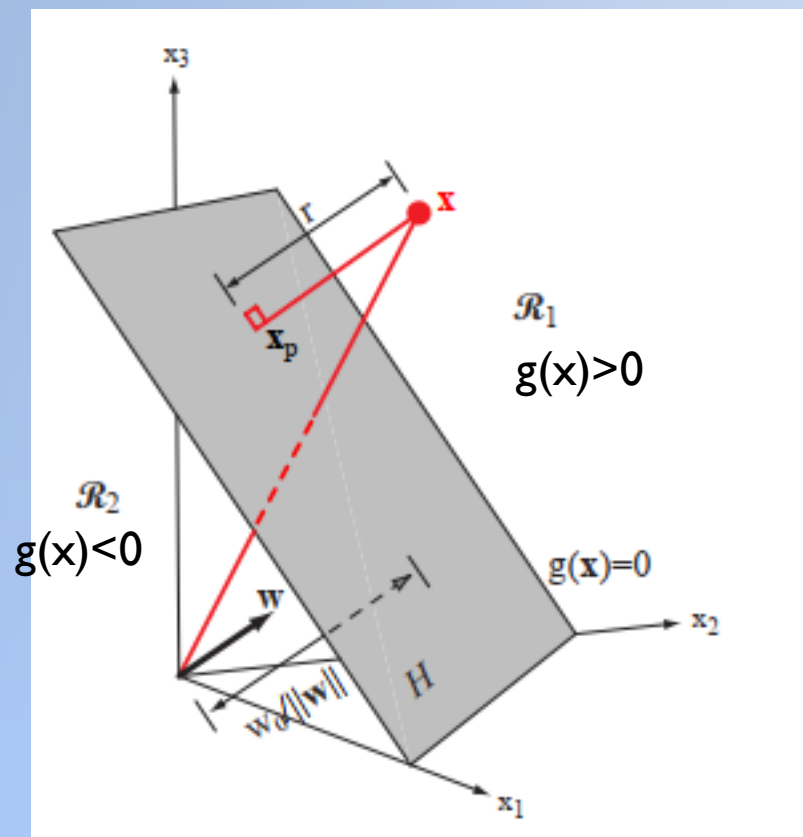


Figure 5.2: The linear decision boundary H , where $g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 = 0$, separates the feature space into two half-spaces $\mathcal{R}_1$ (where $g(\mathbf{x}) > 0$) and $\mathcal{R}_2$ (where $g(\mathbf{x}) < 0$).

Discriminant function:

$$g(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$$

Decision rule:
 Decide ω_1 if $g(\mathbf{x}) > 0$ and ω_2 if $g(\mathbf{x}) < 0$
 $g(\mathbf{x}) = 0$: decision surface
 ω_1 and ω_2 are target classes



Linear Regression

Logistic Regression

Estimate the probability of classes

Least squares

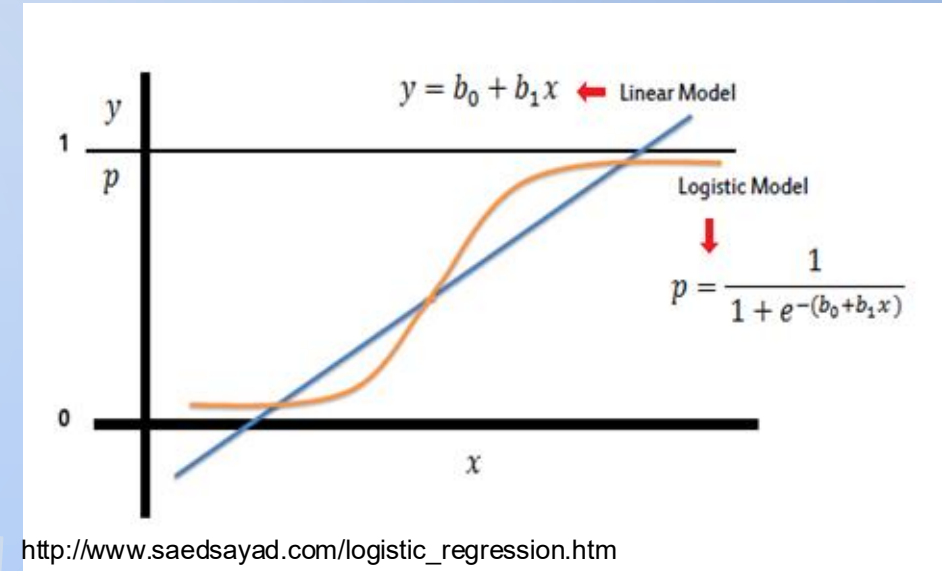
Maximum likelihood

Constant variance

Non-constant variance

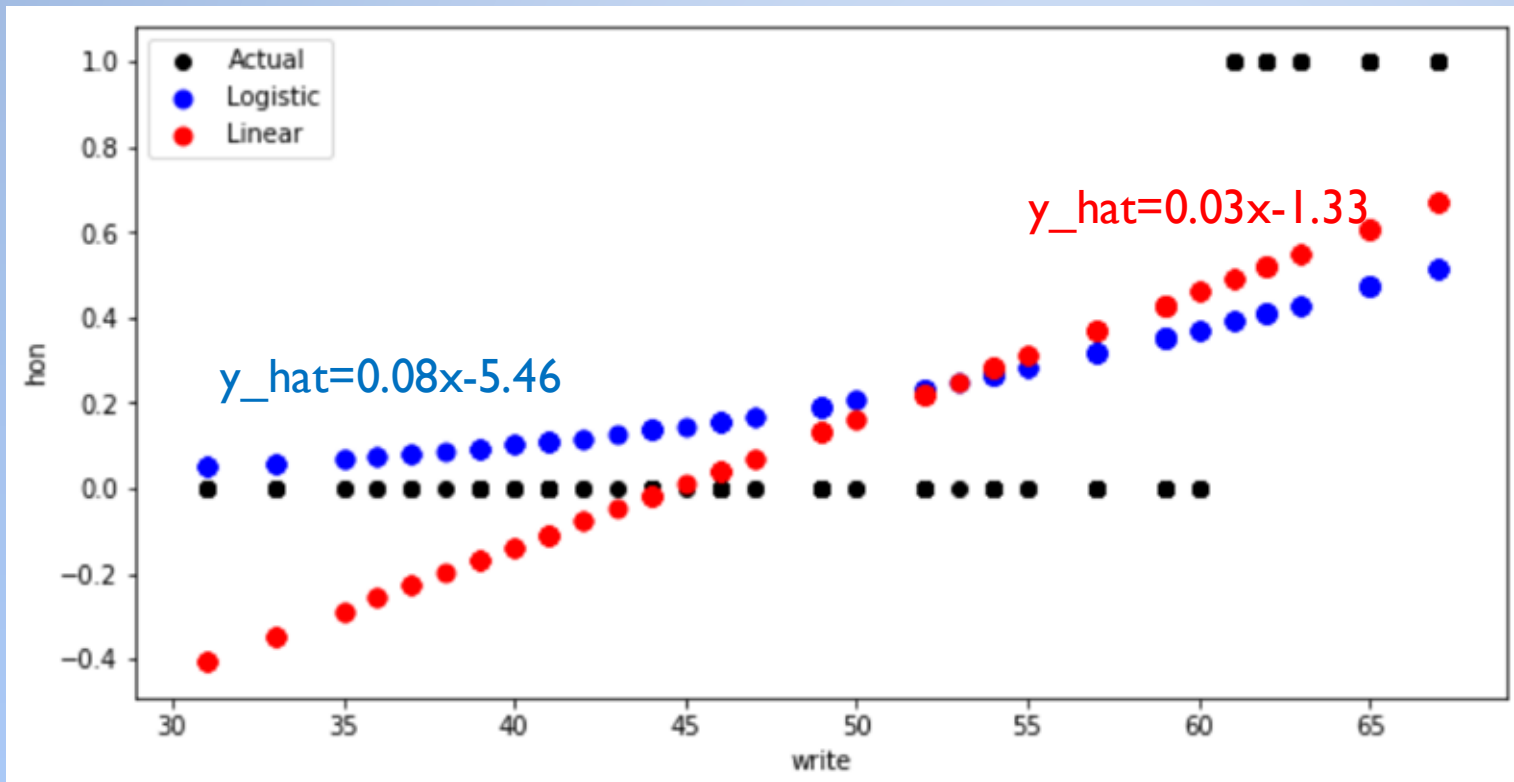
Response variable is normal

Response variable is binary



LOGISTIC REGRESSION

$$\hat{y} = \log\left(\frac{p}{1-p}\right) = \mathbf{b}^T \mathbf{x} = b_0 + b_1 x_1 + \dots + b_d x_d$$



Walpole et al. (2012):

Odds of success = $p/(1-p)$

Logistic regression estimates the probability of classes:

$$p = P(y = 1 | \mathbf{x}, \mathbf{b}) = \frac{1}{1 + e^{-\mathbf{b}^T \mathbf{x}}}$$

Coefficients in logistic regression are in terms of the log odds, that is, the coefficient 0.08 implies that a one unit change in “write” results in a 0.08 unit change in the log of the odds.

<https://stats.idre.ucla.edu/stata/faq/how-do-i-interpret-odds-ratios-in-logistic-regression/>



SUMMARY: LOGISTIC REGRESSION

☒ Linear model

$$\hat{y} = \log\left(\frac{p}{1-p}\right) = \mathbf{b}^T \mathbf{x} = b_0 \cdot 1 + b_1 x_1 + \dots + b_d x_d$$

☒ Binary classification

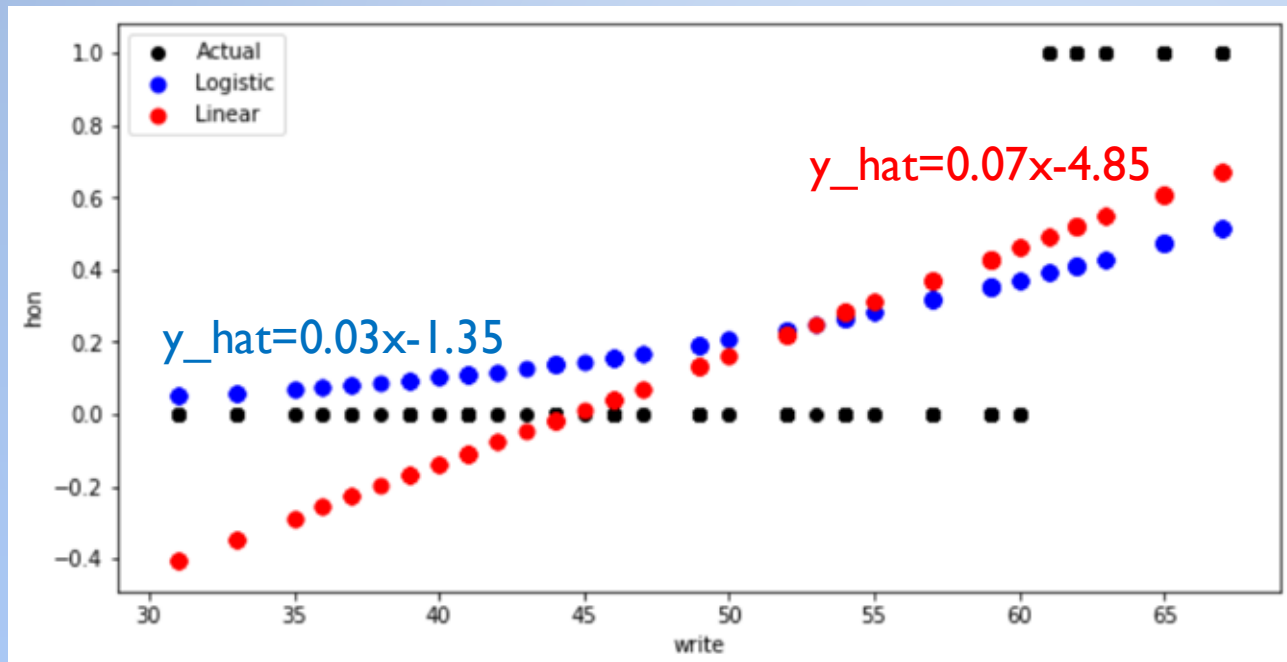
$$p = P(y = 1 | \mathbf{x}, \mathbf{b}) = \frac{1}{1 + e^{-\mathbf{b}^T \mathbf{x}}}$$

☒ Soft classification





EXERCISE



Given these linear regression and logistic regression models, determine whether a student that has write score 65 is in honors class.



12 REFERENCES

- Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (2012). Probability and Statistics for engineering and sciences. *Pearson Education*, 430-435. Chapter 11 & 12.12, 9.14
- RO Duda, PE Hart, and DG Stork, Pattern Classification, 2nd edition, John Wiley & Sons, 2001. Chapter 5
- Charles Elkan (2014). Maximum Likelihood, Logistic Regression, and Stochastic Gradient Training. <https://cseweb.ucsd.edu/~elkan/250B/logreg.pdf>
- Russell, S., & Norvig, P. (2010). Artificial intelligence: a modern approach. 3rd edition. Chapter 18.6.4.

Logistic Regression

Stochastic Gradient Ascent

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Pembelajaran Mesin
(*Machine Learning*)



14 LOGISTIC REGRESSION

$$\hat{y} = \log\left(\frac{p}{1-p}\right) = \mathbf{b}^T \mathbf{x} = b_0 \cdot 1 + b_1 x_1 + \dots + b_d x_d$$

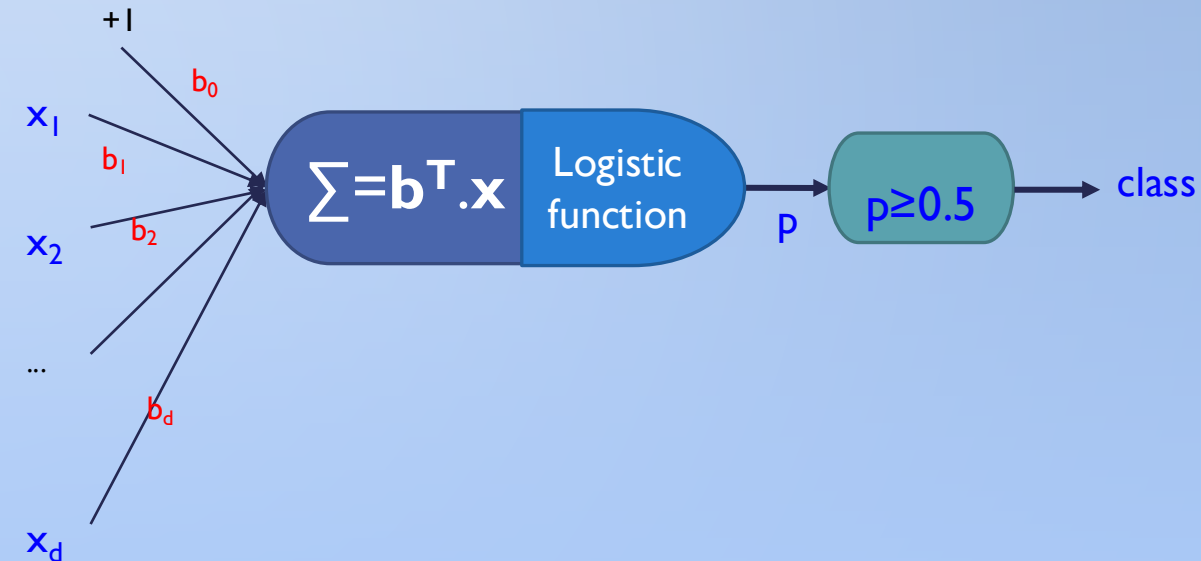
$$p = P(y = 1 | \mathbf{x}, \mathbf{b}) = \frac{1}{1 + e^{-\mathbf{b}^T \mathbf{x}}}$$

Input $\mathbf{x} = (1, x_1, x_2, \dots, x_d)$

Model $\mathbf{b} = (b_0, b_1, b_2, \dots, b_d)$

$$\Sigma = \mathbf{b}^T \mathbf{x} = b_0 \cdot 1 + b_1 x_1 + \dots + b_d x_d$$

Output = $\sigma(\Sigma)$



15 MAXIMUM LIKELIHOOD ESTIMATOR FOR LOGISTIC REGRESSION

ESTIMATOR THAT RESULTS IN A MAXIMUM VALUE FOR ITS JOINT PROBABILITY OR MAXIMIZES THE LIKELIHOOD OF THE SAMPLE

Formal
definition

(Elkan, 2014):

Given the training set $\{<x_1, y_1> .. <x_n, y_n>\}$, learn logistic regression classifier by maximizing the log joint conditional likelihood, that is the sum of log conditional likelihood (LCL) for each training example. x_{ij} is the value of the j th feature of the i th training example.

$$LCL = \sum_{i=1}^n \log L(\theta; y_i | x_i) = \sum_{i=1; y_i=1}^n \log p_i + \sum_{i=1; y_i=0}^n \log(1 - p_i)$$

$$\rightarrow \frac{\partial LCL}{\partial b_j} = \sum_i (y_i - p_i) x_{ij}$$

Stochastic gradient **ascent** is optimization method that changes the coefficient values (as random approximation to true derivative) to **increase** the log likelihood based on a randomly chosen example at a time.

Stochastic gradient update of b_j
is: learning rate

$$b_j = b_j + \eta (y_i - p_i) x_{ij}$$

<https://cseweb.ucsd.edu/~elkan/250B/logreg.pdf>



16 STOCHASTIC GRADIENT ASCENT FOR LOGISTIC REGRESSION

INPUT: TRAINING DATA $D=\{<X_1,Y_1>...<X_N,Y_N>\}$; MAX-ITER T ; LEARNING RATE η

Initialize **b**

For $t=1, \dots, T$:

For each example $<x_i,y_i>$: #randomly chosen example

p_i =prediction for x_i using the current coefficients **b**

For each non-zero feature of x_i : $b_j = b_j + \eta(y_i - p_i)x_{ij}$

Return **b**

One
iteration =
one epoch

$D: \{<[52,41],0>, <[62,58],1>\}; T=1; \eta=0.1$

b=[0,0,0] #b0=0; b1=0; b2=0

$t=1$:

$<[62,58],1>$:

$p_i = 1 / (1 + e^{-0}) = 0.5$

$b_0 = 0 + 0.1(1 - 0.5) * 1 = 0.05$

$b_1 = 0 + 0.1(1 - 0.5) * 62 = 3.1$

$b_2 = 0 + 0.1(1 - 0.5) * 58 = 2.9$

$<[52,41],0>$:

$p_i = 1 / (1 + e^{-(0.05 + 3.1 * 52 + 2.9 * 41)})$
 $= 1 / (1 + e^{-280.15}) = 1$

$b_0 = 0.05 + 0.1(0 - 1) * 1 = -0.05$

$b_1 = 3.1 + 0.1(0 - 1) * 52 = -2.1$

$b_2 = 2.9 + 0.1(0 - 1) * 41 = -1.2$

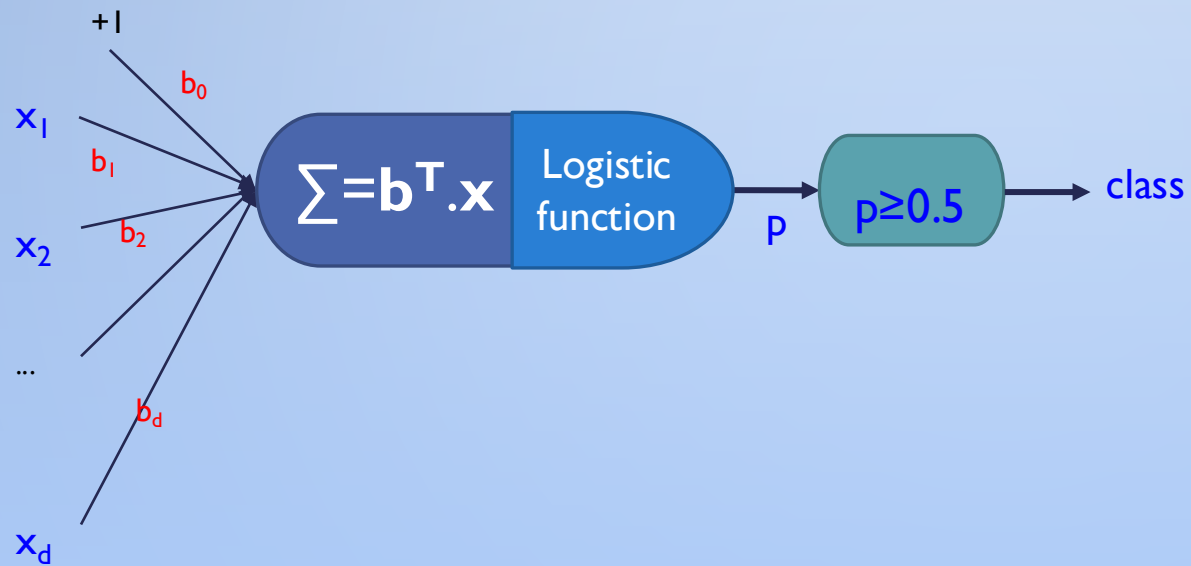


17 PREDICTION

- $b_0 = -0.05; b_1 = -2.1; b_2 = -1.2$
- $D: \{ \langle [52, 41], 0 \rangle, \langle [62, 58], 1 \rangle \}$
 - $\mathbf{x}_1 = [52, 41]: p_1 = 1 / (1 + e^{-(0.05 - 2.1 \cdot 52 - 1.2 \cdot 41)}) = 1 / (1 + e^{158.45}) = 1.53 \cdot 10^{-69} \Rightarrow \text{class} = 0 \text{ (} p_1 < 0.5 \text{)}$
 - $\mathbf{x}_2 = [62, 58]: p_2 = 1 / (1 + e^{-(0.05 - 2.1 \cdot 62 - 1.2 \cdot 58)}) = 1 / (1 + e^{199.85}) = 1.61 \cdot 10^{-87} \Rightarrow \text{class} = 0 \text{ (} p_2 < 0.5 \text{)}$
- Akurasi training = $\frac{1}{2} = 50\%$



18 SUMMARY: LOGISTIC REGRESSION



Model: $\mathbf{b} \in \mathbb{R}^{d+1}$

Maximum Likelihood estimator

Stochastic gradient ascent



19 REFERENCES

- Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (2012). Probability and Statistics for engineering and sciences. *Pearson Education*, 430-435. Chapter 11 & 12.12, 9.14
- RO Duda, PE Hart, and DG Stork, Pattern Classification, 2nd edition, John Wiley & Sons, 2001. Chapter 5
- Charles Elkan (2014). Maximum Likelihood, Logistic Regression, and Stochastic Gradient Training. <https://cseweb.ucsd.edu/~elkan/250B/logreg.pdf>
- Russell, S., & Norvig, P. (2010). Artificial intelligence: a modern approach. 3rd edition. Chapter 18.6.4.





Support Vector Machines

IF-3170 Inteligensi Artifisial
Teknik Informatika ITB



Modul: Supervised Learning

01 SVM: What & Why?

IF3170 - Inteligensi Artifisial

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Outline

Sejarah SVM

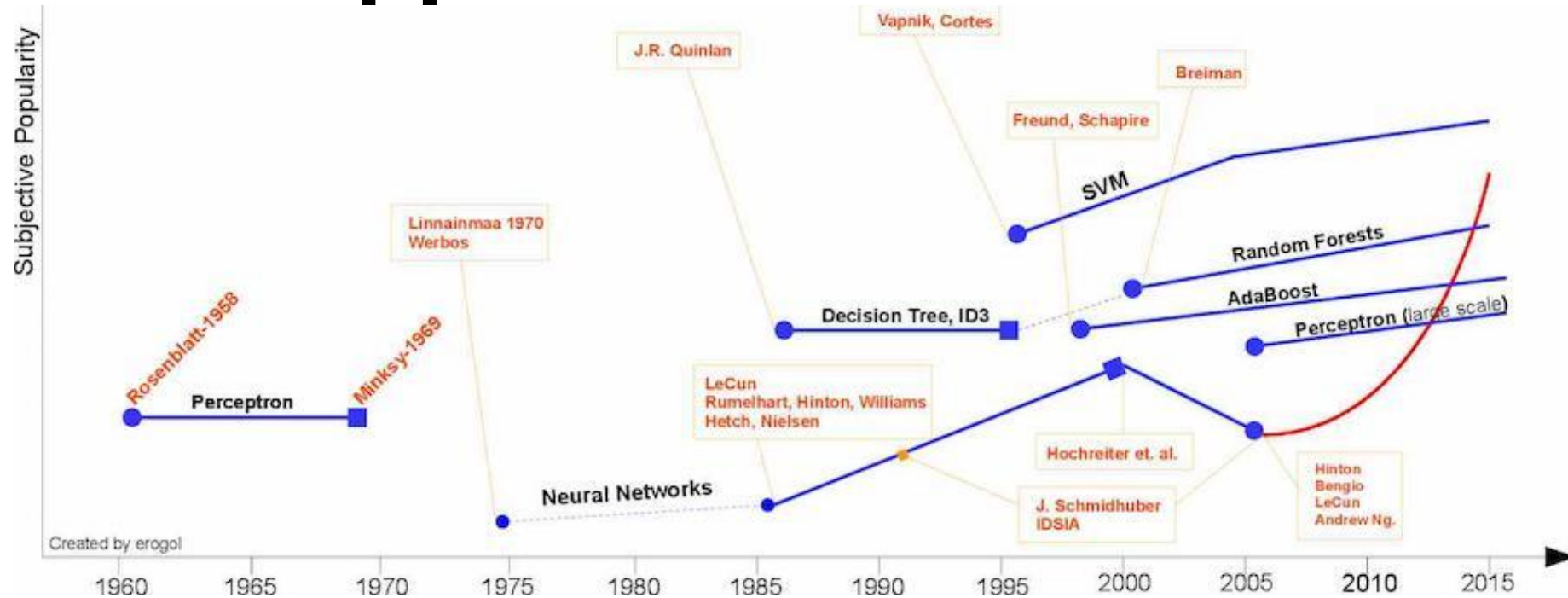
Bidang Pemisah
Terbaik

Tujuan SVM

Klasifikasi Biner –
Linear Separability

Hyperplane Classifier

Support Vector Machine



Eren Golge: <https://www.kdnuggets.com/2014/10/deep-learning-make-machine-learning-algorithms-obsolete.html>

- SVM diperkenalkan tahun 1992 oleh Vapnik, Boser, & Guyon
- Kinerja baik di berbagai aplikasi seperti *bioinformatics*, klasifikasi teks, pengenalan tulisan tangan dan lain-lain.

SVM

- 1980an
 - DTL dan NN memungkinkan pembelajaran nonlinear yang efisien
 - Kurang didukung dasar teoritis dan memungkinkan terjadinya local minima
- 1990an
 - Algoritma pembelajaran yang efisien untuk fungsi non linier berbasis teori komputasi

SVM Introduction

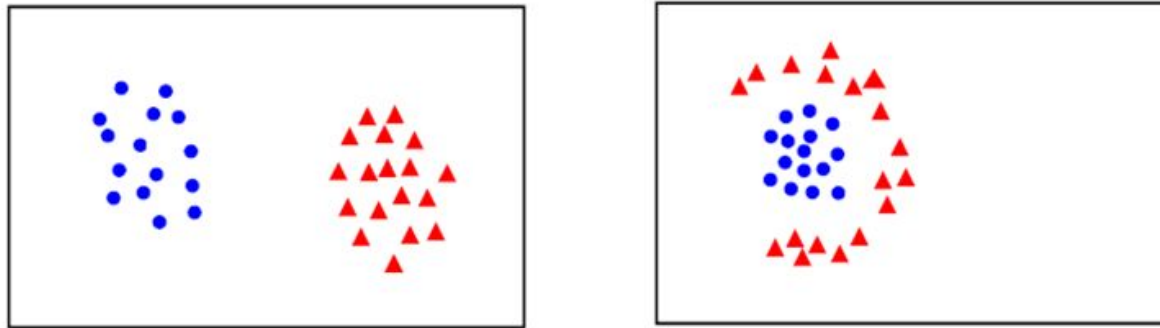
- Boser, B. E., Guyon, I. M., & Vapnik, V. N. (1992, July). A training algorithm for optimal margin classifiers. In *Proceedings of the fifth annual workshop on Computational learning theory* (pp. 144-152). ACM.
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine learning*, 20(3), 273-297.

Klasifikasi Biner

Given training data $(\mathbf{x}_i, y_i)$ for $i = 1 \dots N$, with $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$, learn a classifier $f(\mathbf{x})$ such that

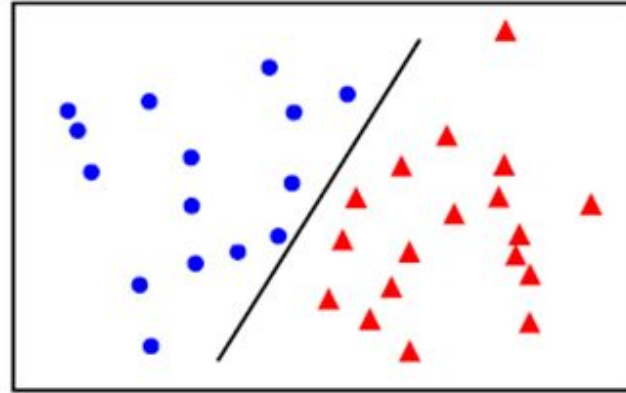
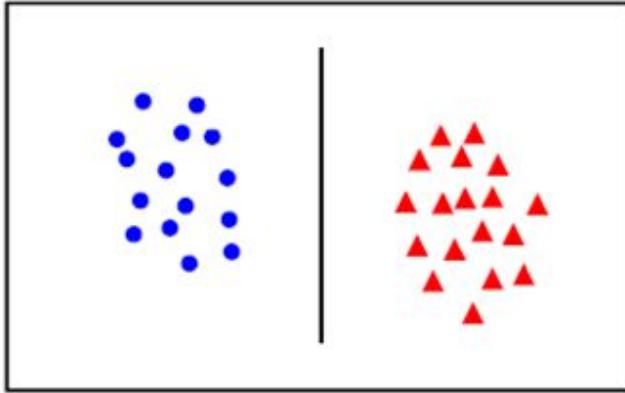
$$f(\mathbf{x}_i) \begin{cases} \geq 0 & y_i = +1 \\ < 0 & y_i = -1 \end{cases}$$

i.e. $y_i f(\mathbf{x}_i) > 0$ for a correct classification.

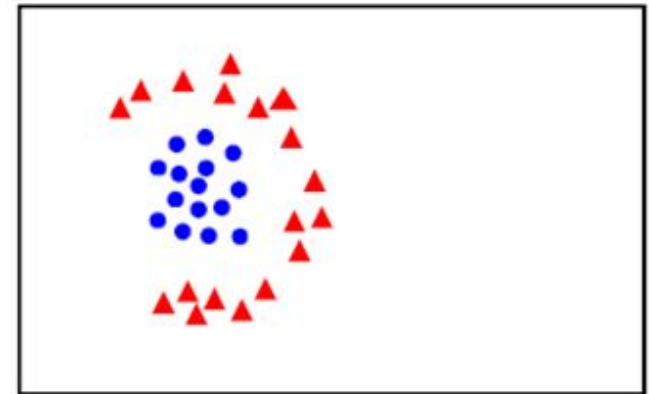
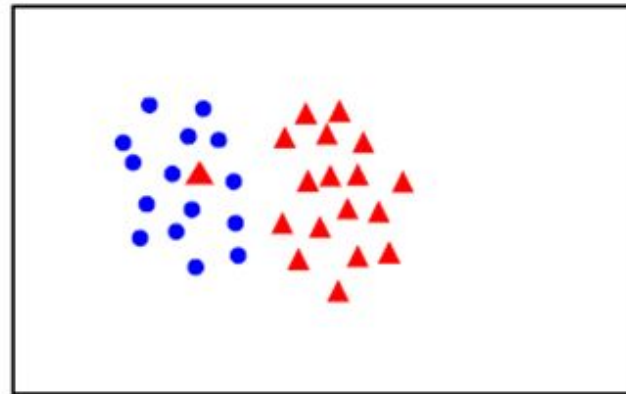


Linear Separability

linearly
separable



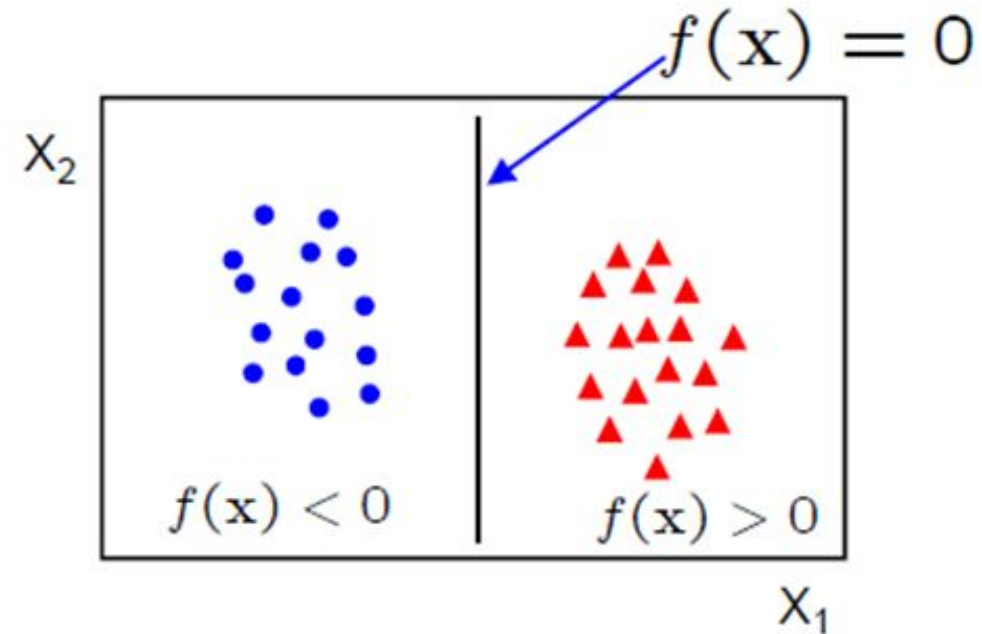
not
linearly
separable



Linear Classifier

A linear classifier has the form

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$$

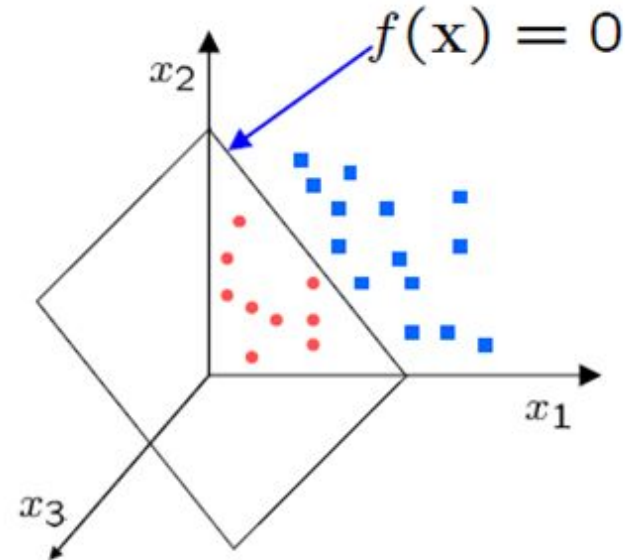


- in 2D the discriminant is a line
- $\mathbf{W}$ is the **normal** to the line, and b the **bias**
- $\mathbf{W}$ is known as the **weight vector**

Linear Classifier

A linear classifier has the form

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$$



- in 3D the discriminant is a plane, and in nD it is a hyperplane

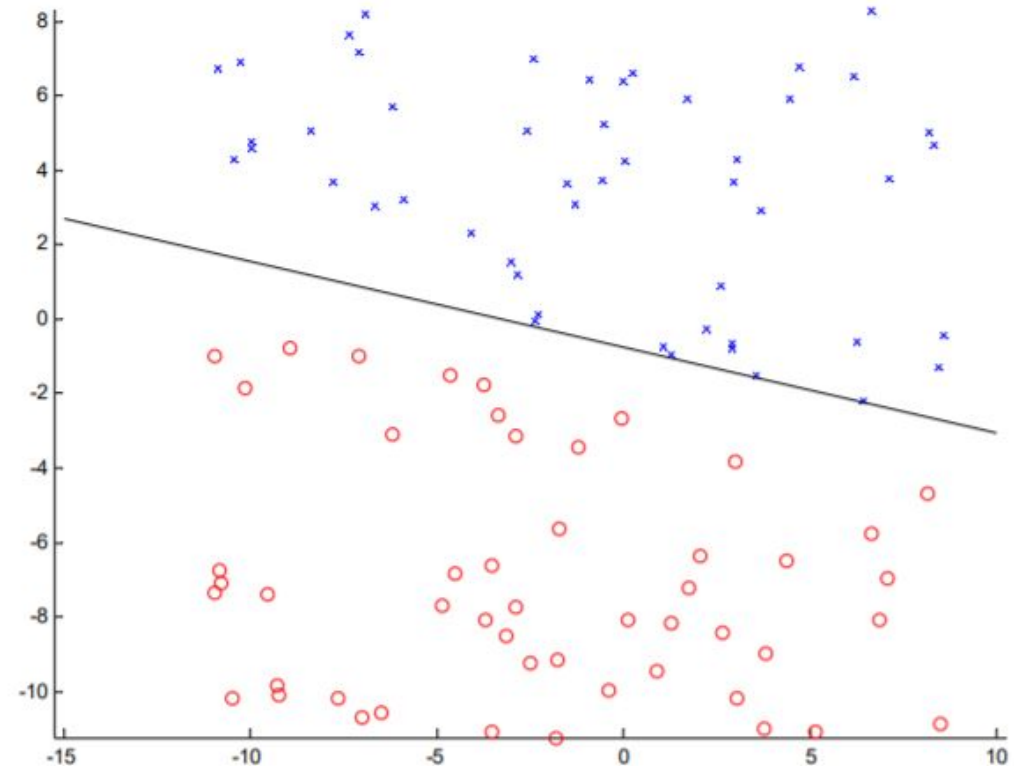
For a K-NN classifier it was necessary to 'carry' the training data

For a linear classifier, the training data is used to learn $\mathbf{w}$ and then discarded

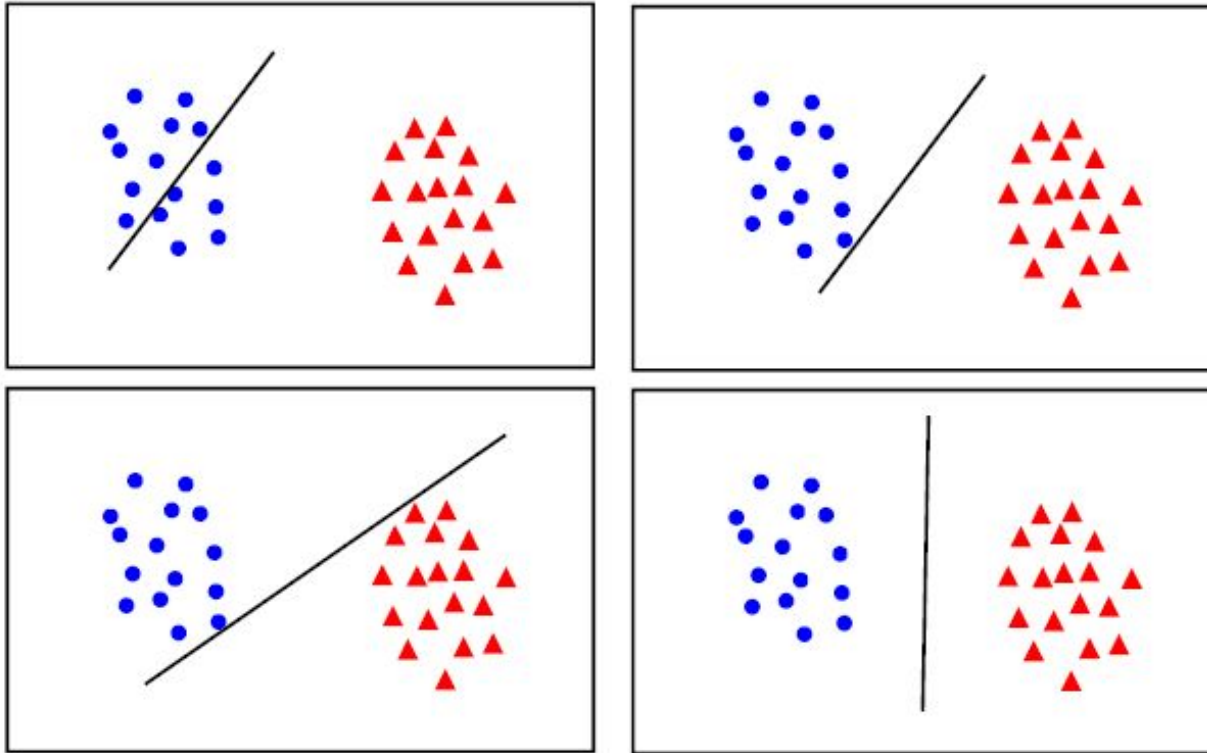
Only $\mathbf{w}$ is needed for classifying new data

Perceptron Weakness

- Perceptron biggest weakness is that it will not find the same hyperplane every time.
 - Not all separating hyperplanes are equals.
 - If the Perceptron gives you a hyperplane that is very close to all the data points from one class, you have a right to believe that it will generalize poorly when given new data.
- After an accurate hyperplane is found, the training process will stop and it is considered to have converged.

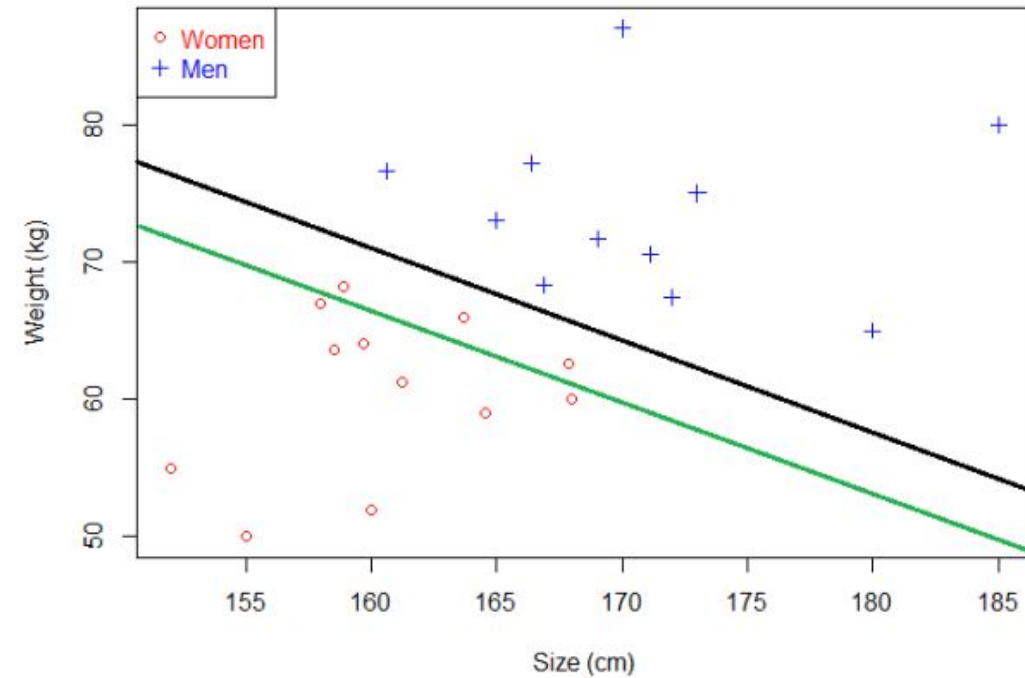
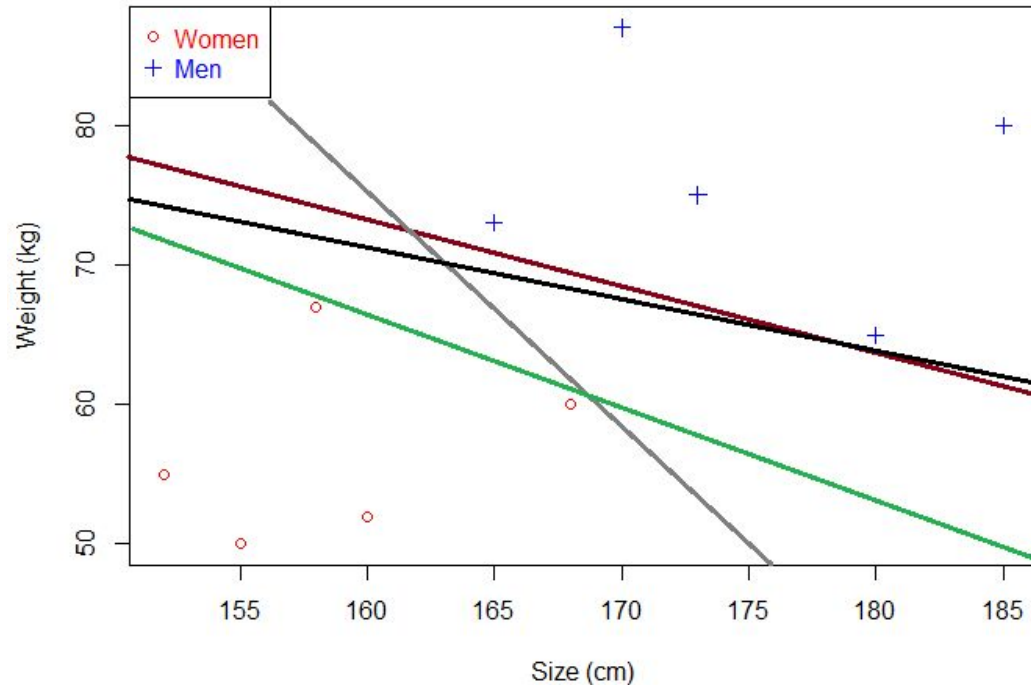


Bidang Pemisah Terbaik?



• Mengapa?

Bidang Pemisah Terbaik (lanj)



Kiri: semua bidang pemisah valid karena memisahkan kedua kelas pada training data.
Kanan: real-life data. Bidang pemisah hitam lebih baik daripada hijau.

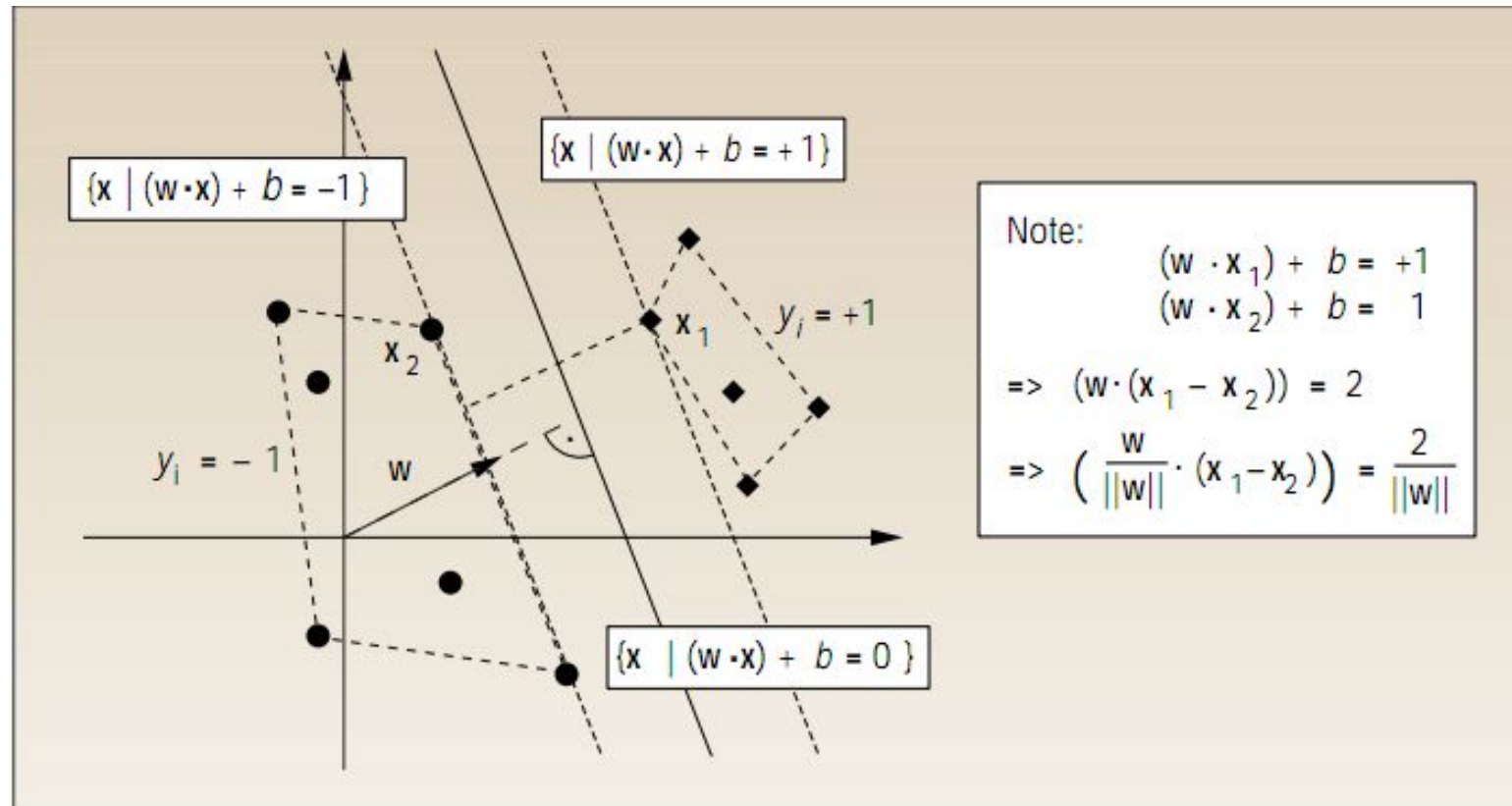
SVM Objective

- Objective of the SVM is to find the optimal separating hyperplane which maximizes the margin of the training data. There will never be any data point inside the margin.
- Menggunakan optimasi kuadratik untuk menghindari 'local minimum' isu yang ada pada NN (Greedy)
- Menggunakan fungsi kernel untuk memisahkan non-linear region

Hyperplane Classifier

- Hipotesis:
- $x_1, x_2 \in \text{training data}$

$$f(x) = \text{sign}(\vec{w} \cdot \vec{x} + b); \vec{w}, \vec{x} \in \mathbb{R}^N; b \in \mathbb{R}$$



Vector Direction

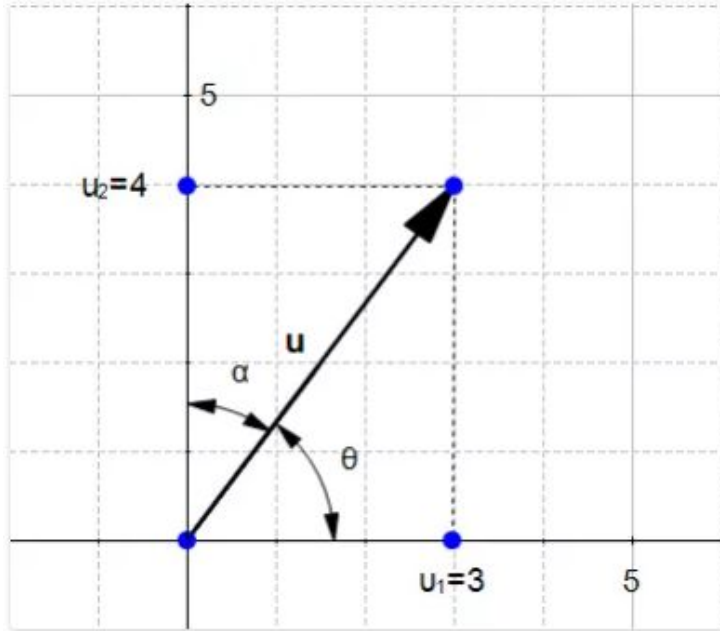


Figure 4 - direction of a vector

$u(u_1, u_2)$ with $u_1=3$ and $u_2=4$

$$\cos(\theta) = \frac{u_1}{\|u\|}$$

$$\cos(\alpha) = \frac{u_2}{\|u\|}$$

- Naive definition 1: The direction of the vector u is defined by the angle θ with respect to the horizontal axis, and with the angle α with respect to the vertical axis.
- Naive definition 2: The direction of the vector u is defined by the cosine of the angle θ and the cosine of the angle α .



02 SVM for Linearly Separable Data

IF3170 Artificial Intelligence

Modul: Supervised Learning

02 SVM for Linearly Separable Data

IF3170 - Inteligensi Artifisial

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Outline

Support Vectors

Optimal
Hyperplane

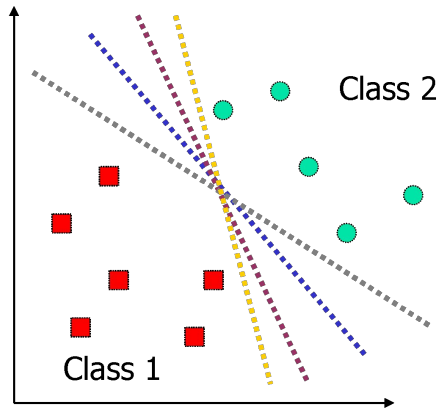
Quadratic
Optimization
Problem

Bidang Pemisah
Terbaik

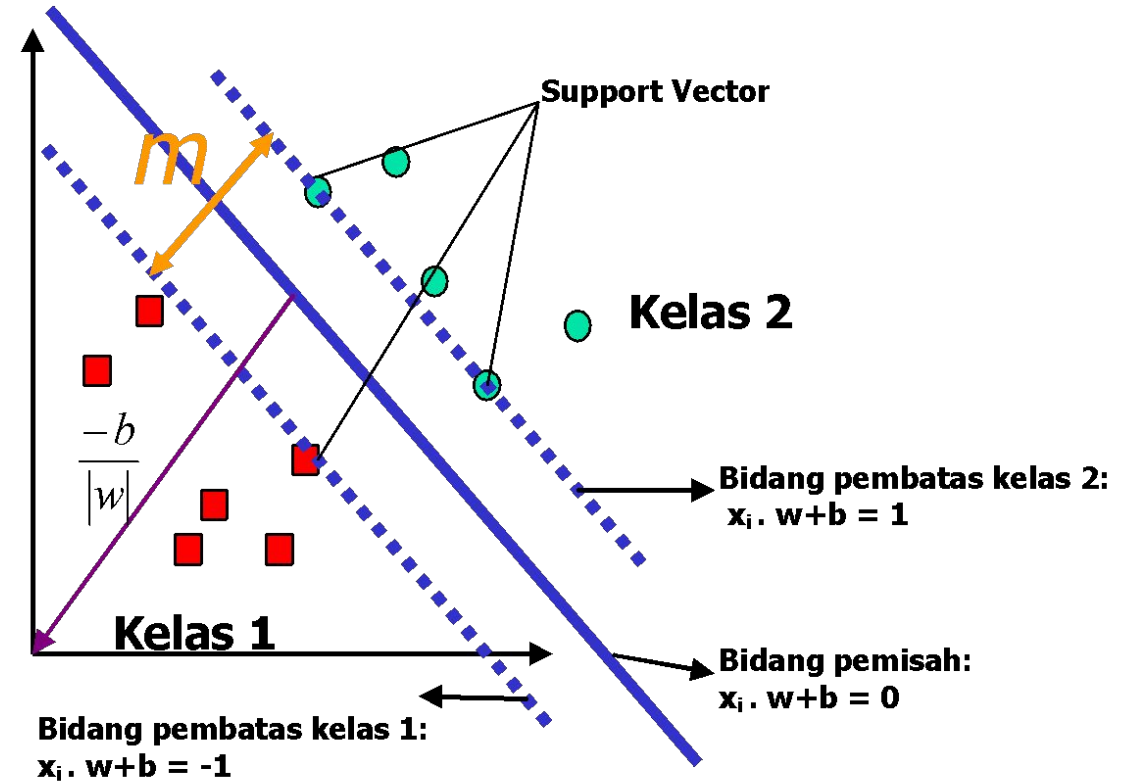
Calculation Example



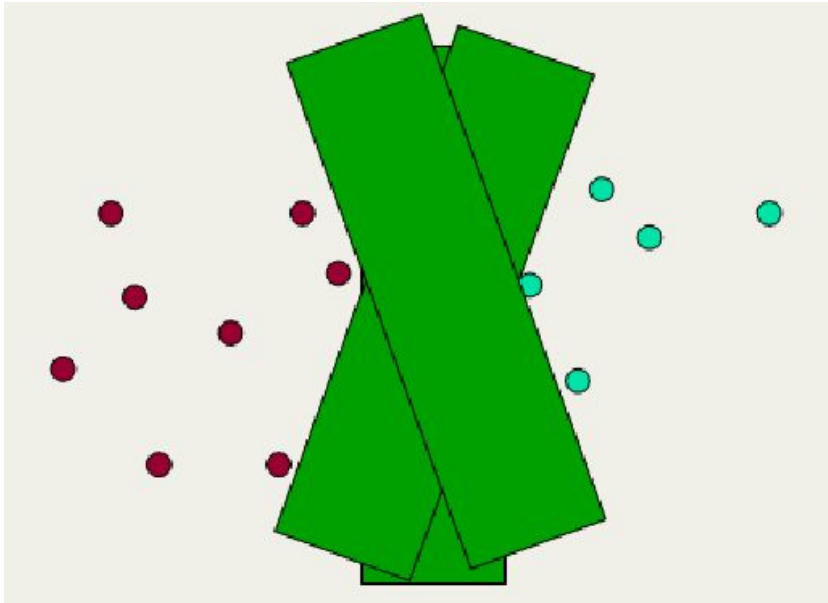
Mencari Bidang Pemisah Terbaik (1)



Mencari bidang pemisah dengan margin (m) terbesar



Mencari Bidang Pemisah Terbaik (2)



- 2 kelas dapat dipisahkan oleh sepasang bidang pembatas yang sejajar.
 - Bidang pembatas pertama membatasi kelas pertama
 - Bidang pembatas kedua membatasi kelas kedua
- **Support Vector** : Vector pada training data yang *men-support* bidang pemisah

Mengapa mencari margin terbesar?

- Bidang pemisah terbaik dengan margin terbesar memiliki generalisasi yang lebih baik
 - Titik yang dekat dengan bidang pemisah merepresentasikan ketidakyakinkan klasifikasi : 50% peluang pengambilan keputusan oleh classifier
- Kapasitas memori untuk menyimpan model menjadi lebih sedikit
 - Hanya support vector yang mempengaruhi pengambilan keputusan

Support Vectors

- The **data points** that lie closest to the decision surface (or hyperplane)
- They are the data points **most difficult to classify**
- They have direct bearing on the optimum location of the decision surface

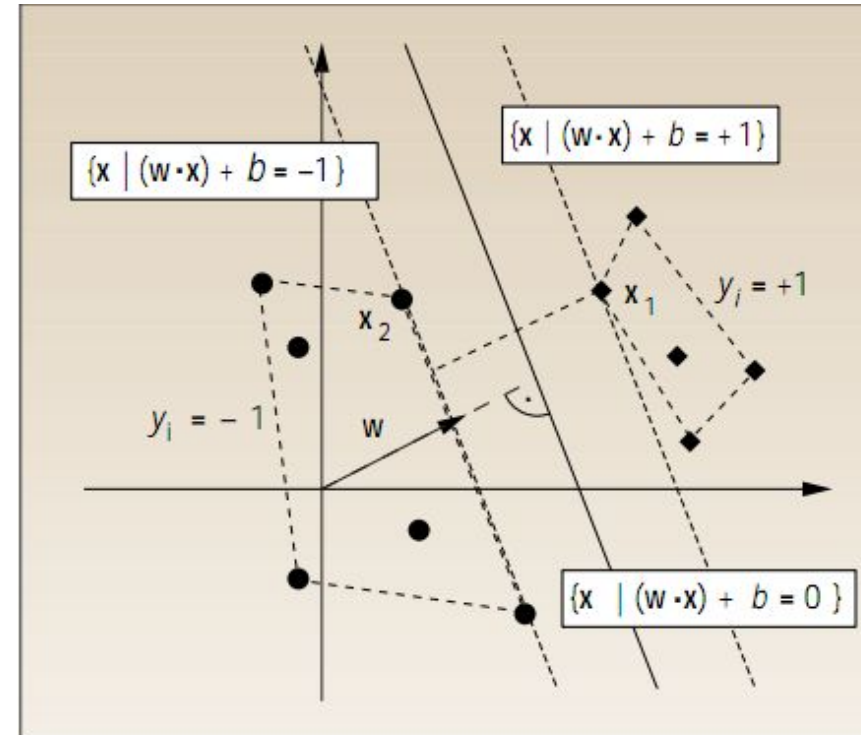
Support Vectors

- Support vectors memenuhi:

$$| \vec{w} \cdot \vec{x}_i + b | = 1$$

- Untuk itu, semua training data berlabel “+” dan “-” memenuhi juga:

$$y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1$$



Optimal Hyperplane

- Optimal hyperplane:
 - *maximum-margin hyperplane*
 - atau hyperplane with maximal margin separation between two classes.
- Maximal margin ($2/||w||$) $\approx$ minimize $||w||$ dengan batasan konsistensi training data tetap terjaga (semua klasifikasi benar).

Minimize:
$$V(\vec{w}, b) = \frac{1}{2} \vec{w} \cdot \vec{w}$$

subject to:

$$y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1$$

Recall the distance from a point (x_0, y_0) to a line: $Ax + By + c = 0$ is: $|Ax_0 + By_0 + c| / \sqrt{A^2 + B^2}$, so,
 The distance between H_0 and H_1 is then:
 $|w \cdot x + b| / ||w|| = 1 / ||w||$, so
 The total distance between H_1 and H_2 is thus: $2 / ||w||$

Optimal Hyperplane (2)

- Pencarian bidang pemisah terbaik dapat dirumuskan menjadi:

$$\begin{array}{ll} \min & \frac{1}{2}|w|^2 \\ \text{s.t} & y_i(x_i \cdot w + b) - 1 \geq 0 \end{array}$$

- Want to look for solution point p where

$$\nabla f(p) = \nabla \lambda g(p)$$

$$g(x) = 0$$
- Or, combining these two as the *Langrangian* L & requiring derivative of L be zero:

$$L(x, a) = f(x) - ag(x)$$

$$\nabla(x, a) = 0$$

- Supaya lebih mudah diselesaikan ubah ke formula Lagrangian

$$\begin{array}{ll} \min_{w, b} & L_p(w, b, \alpha) \equiv \frac{1}{2}|w|^2 - \sum_{i=1}^n \alpha_i y_i (x_i \cdot w + b) + \sum_{i=1}^n \alpha_i \\ & \alpha \geq 0 \end{array}$$

- Minimumkan L_p terhadap b dan w (saddle point), diperoleh:

$$\frac{\partial}{\partial b} L_p(w, b, \alpha) = 0 \quad \rightarrow \quad \sum_{i=1}^n \alpha_i y_i = 0$$

$$\frac{\partial}{\partial w} L_p(w, b, \alpha) = 0 \quad \rightarrow$$

$$\frac{\partial L_p}{\partial w} = w - \sum_{i=1}^l a_i y_i x_i = 0$$

$$w = \sum_{i=1}^n \alpha_i y_i x_i$$

Langrangian Dual Problem

The Lagrangian Dual Problem: instead of minimizing over $\mathbf{w}$, b , subject to constraints involving a 's, we can maximize over a (the dual variable) subject to the relations obtained previously for $\mathbf{w}$ and b

Our solution must satisfy these two relations:

$$\mathbf{w} = \sum_{i=1}^l a_i y_i \mathbf{x}_i, \quad \sum_{i=1}^l a_i y_i = 0$$

By substituting for $\mathbf{w}$ and b back in the original eqn we can get rid of the dependence on $\mathbf{w}$ and b .

Note first that we already now have our answer for what the weights $\mathbf{w}$ must be: they are a linear combination of the training inputs and the training outputs, x_i and y_i and the values of a . We will now solve for the a 's by differentiating the dual problem wrt a , and setting it to zero. Most of the a 's will turn out to have the value zero. The non-zero a 's will correspond to the support vectors

Substitute the Primal Problem

Primal problem:

$$\min L_P = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^l a_i y_i (\mathbf{x}_i \cdot \mathbf{w} + b) + \sum_{i=1}^l a_i$$

$$\text{s.t. } \forall i \ a_i \geq 0$$

$$\mathbf{w} = \sum_{i=1}^l a_i y_i \mathbf{x}_i, \quad \sum_{i=1}^l a_i y_i = 0$$

Dual problem:

$$\max L_D(a_i) = \sum_{i=1}^l a_i - \frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l a_i a_j y_i y_j (\mathbf{x}_i \cdot \mathbf{x}_j)$$

$$\text{s.t. } \sum_{i=1}^l a_i y_i = 0 \ \& \ a_i \geq 0$$

(note that we have removed the dependence on $\mathbf{w}$ and b)

Quadratic Optimization Problem

- Minimize:
$$W(\vec{\alpha}) = -\sum_{i=1}^n \alpha_i + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i y_j \alpha_i \alpha_j (\vec{x}_i \cdot \vec{x}_j)$$

subject to:
$$\sum_{i=1}^n y_i \alpha_i = 0; \forall i \in [1..n]: \alpha_i \geq 0$$

- Support vectors: $\alpha_i > 0$

- Hyperplane:
$$\vec{w} \cdot \vec{x} = \left(\sum_{i=1}^n \alpha_i y_i \vec{x}_i \right) \cdot \vec{x} = \sum_{i=1}^n \alpha_i y_i (\vec{x}_i \cdot \vec{x})$$

$$b = y_{sv} - \vec{w} \cdot \vec{x}_{sv}$$

Optimization Problem Example

Suppose we have two 2D data points: $[(x_{11}, x_{12}), (x_{21}, x_{22})]$ with labels $[y_1, y_2]$

- Minimize: $W(a) = -(a_1 + a_2) + 0.5 \cdot [y_1 \cdot y_1 \cdot a_1 \cdot a_1 \cdot (x_{11} \cdot x_{11} + x_{12} \cdot x_{12}) + y_1 \cdot y_2 \cdot a_1 \cdot a_2 \cdot (x_{11} \cdot x_{21} + x_{12} \cdot x_{22}) + y_2 \cdot y_1 \cdot a_2 \cdot a_1 \cdot (x_{21} \cdot x_{11} + x_{22} \cdot x_{12}) + y_2 \cdot y_2 \cdot a_2 \cdot a_2 \cdot (x_{21} \cdot x_{21} + x_{22} \cdot x_{22})]$

subject to: $(y_1 \cdot a_1 + y_2 \cdot a_2) = 0$

- Support vectors: $a_1 > 0, a_2 > 0$
- Hyperplane:

$$w \cdot x = a_1 \cdot y_1 \cdot ((x_{11}, x_{12}) \cdot x) + a_2 \cdot y_2 \cdot ((x_{21}, x_{22}) \cdot x)$$

$$b = y_{sv} - w \cdot x_{sv}$$

$$W(\vec{\alpha}) = -\sum_{i=1}^n \alpha_i + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i y_j \alpha_i \alpha_j (\vec{x}_i \cdot \vec{x}_j)$$

$$\sum_{i=1}^n y_i \alpha_i = 0; \forall i \in [1..n]: \alpha_i \geq 0$$

$$\vec{w} \cdot \vec{x} = \left(\sum_{i=1}^n \alpha_i y_i \vec{x}_i \right) \cdot \vec{x} = \sum_{i=1}^n \alpha_i y_i (\vec{x}_i \cdot \vec{x})$$

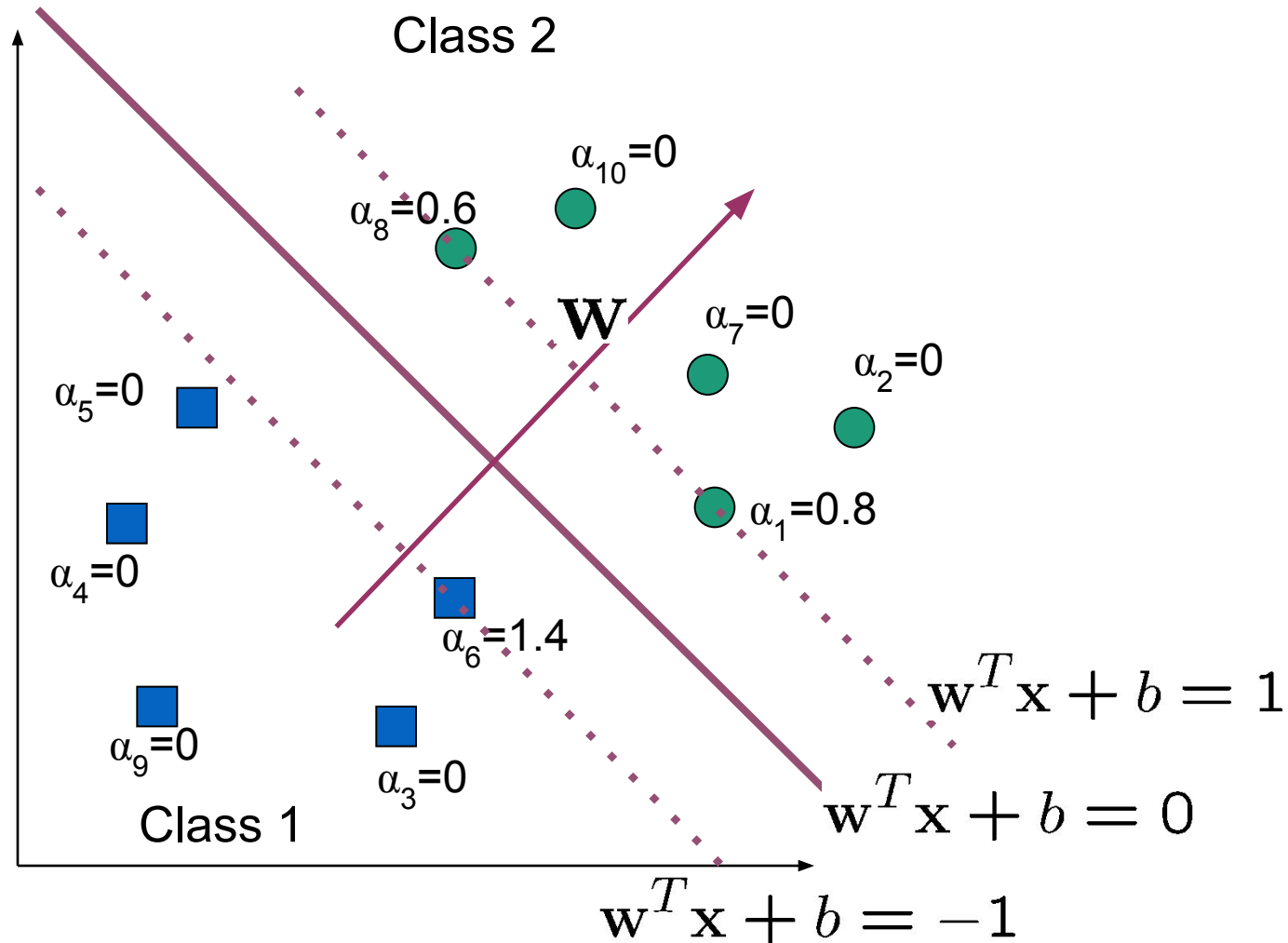
$$b = y_{sv} - \vec{w} \cdot \vec{x}_{sv}$$

Quadratic Programming

- Quadratic programming (QP) is the problem of optimizing a quadratic objective function and is one of the simplest form of non-linear programming.
- The objective function can contain bilinear or up to second order polynomial terms, and the constraints are linear and can be both equalities and inequalities.

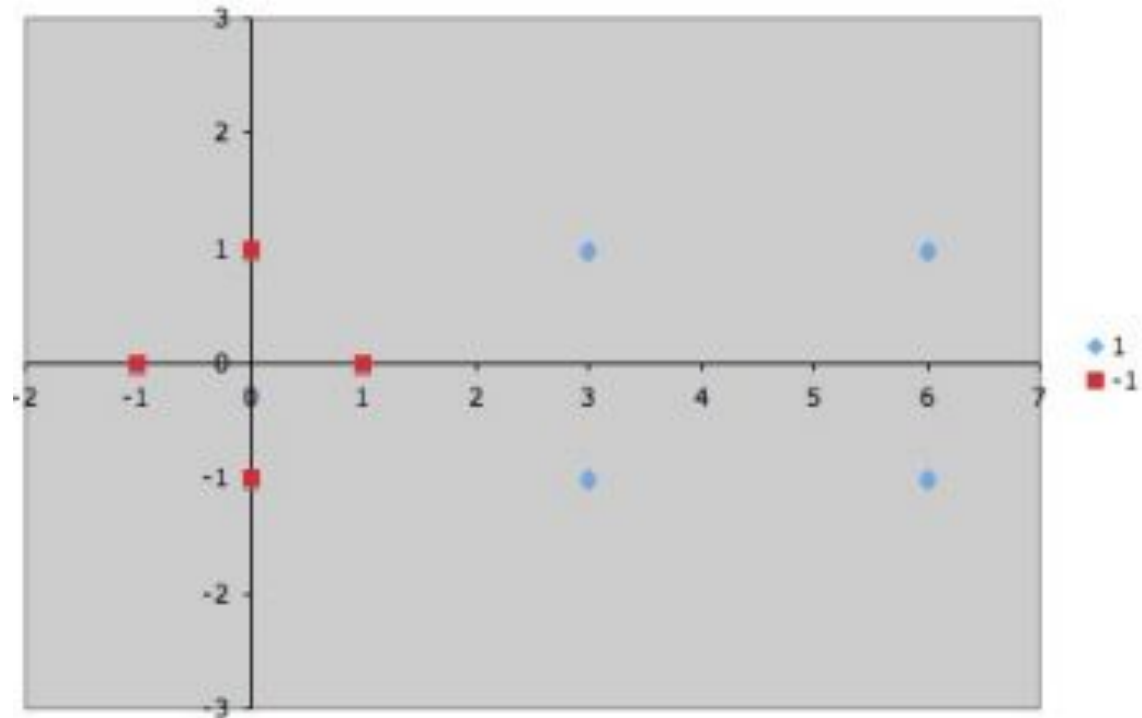
$$\begin{aligned} \min f(x) &= q^T x + \frac{1}{2} x^T Q x \\ \text{s.t. } Ax &= a \\ Bx &\leq b \\ x &\geq \underline{0} \end{aligned}$$

A Geometrical Interpretation

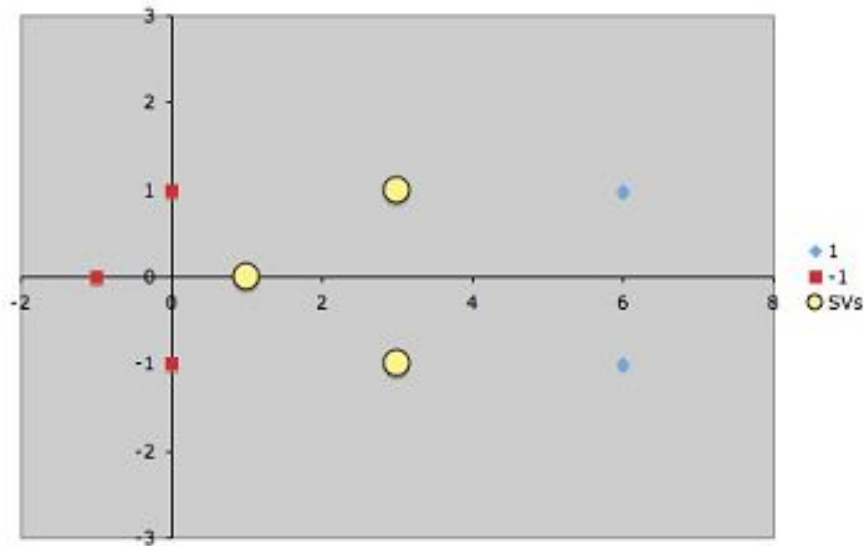


Contoh

x1	x2	Kelas
3	1	+1
3	-1	+1
6	1	+1
6	-1	+1
1	0	-1
0	1	-1
0	-1	-1
-1	0	-1



- Taken from:
<http://axon.cs.byu.edu/Dan/678/miscellaneous/SVM.example.pdf>



Support Vectors

$$f(\vec{x}) = \sum_{i=1}^{nsv} (\alpha_i y_i \vec{x}_i \vec{x}) + b$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow -1 = \alpha_1 \cdot -1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_2 \cdot 1 \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_3 \cdot 1 \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b = -\alpha_1 + 3\alpha_2 + 3\alpha_3 + b \dots (1)$$

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} \Rightarrow 1 = \alpha_1 \cdot -1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \alpha_2 \cdot 1 \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \alpha_3 \cdot 1 \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} + b = -3\alpha_1 + 10\alpha_2 + 8\alpha_3 + b \dots (2)$$

$$\begin{pmatrix} 3 \\ -1 \end{pmatrix} \Rightarrow 1 = \alpha_1 \cdot -1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} + \alpha_2 \cdot 1 \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} + \alpha_3 \cdot 1 \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} + b = -3\alpha_1 + 8\alpha_2 + 10\alpha_3 + b \dots (3)$$

$$-\alpha_1 + 3\alpha_2 + 3\alpha_3 + b = -1 \dots (1)$$

$$-3\alpha_1 + 10\alpha_2 + 8\alpha_3 + b = 1 \dots (2)$$

$$-3\alpha_1 + 8\alpha_2 + 10\alpha_3 + b = 1 \dots (3)$$

$$-\alpha_1 + \alpha_2 + \alpha_3 = 0 \dots (4)$$

$$(2) - (3): 2\alpha_2 - 2\alpha_3 = 0 \rightarrow \alpha_2 = \alpha_3 \dots (5)$$

$$(5) \text{subs}(1): -\alpha_1 + 6\alpha_2 + b = -1 \dots (6)$$

$$(5) \text{subs}(2): -3\alpha_1 + 18\alpha_2 + b = 1 \dots (7)$$

$$(6) - (7): 2\alpha_1 - 12\alpha_2 = -2$$

$$\alpha_1 - 6\alpha_2 = -1$$

$$\alpha_1 = 6\alpha_2 - 1 \dots (8)$$

$$(8,5) \text{subs}(4): -6\alpha_2 + 1 + \alpha_2 + \alpha_2 = 0$$

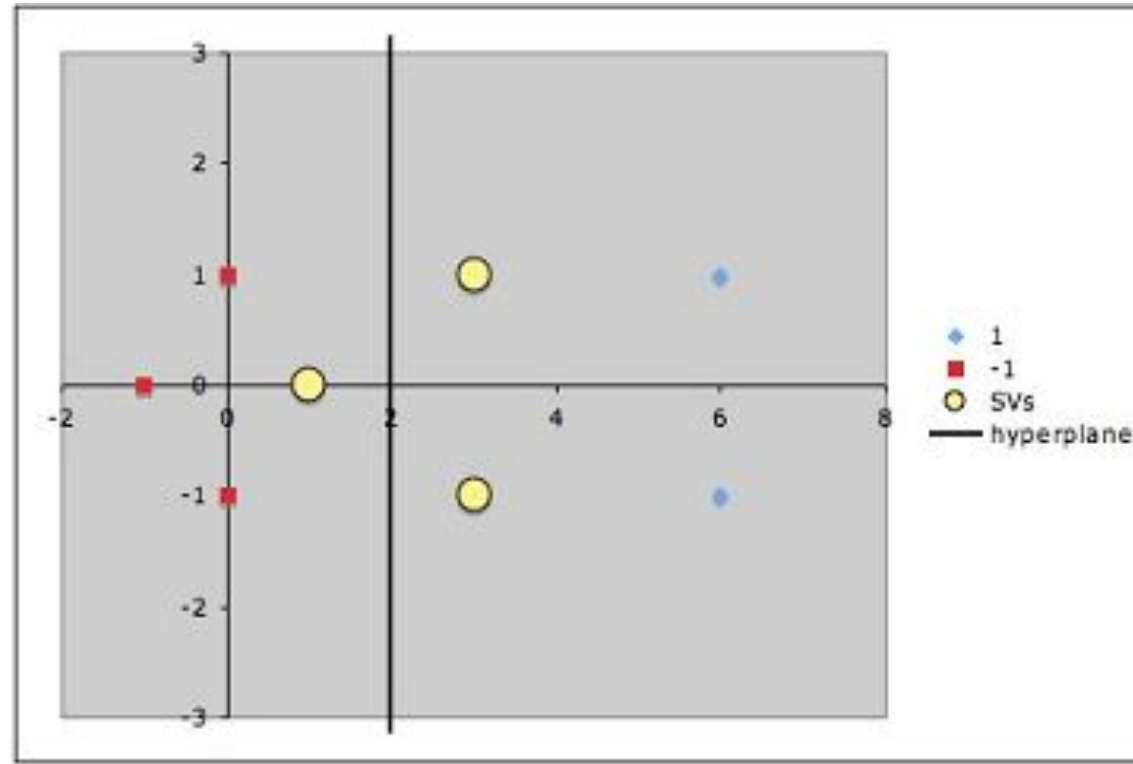
$$-4\alpha_2 = -1 \rightarrow \alpha_2 = \frac{1}{4} = 0.25$$

$$\alpha_1 = 0.5; \alpha_2 = 0.25; \alpha_3 = 0.25$$

$$\text{substitusi}(6): -0.5 + 6 * 0.25 + b = -1$$

$$b = -2$$

Hipotesis



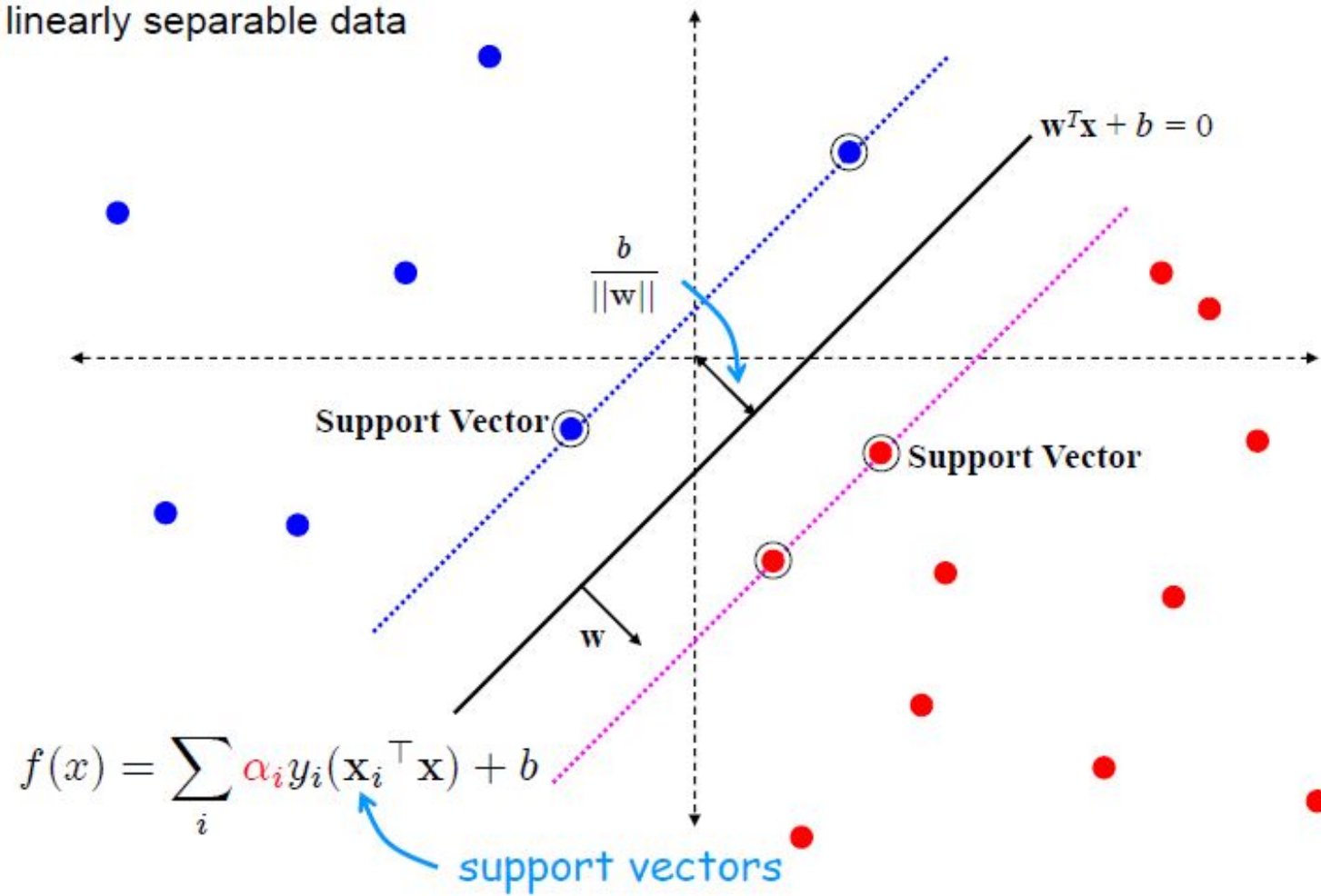
$$f(\vec{x}) = \sum_{i=1}^{nsv} (\alpha_i y_i \vec{x}_i \cdot \vec{x}) - 2; \alpha_1 = 0.5; \alpha_2 = \alpha_3 = 0.25$$

$$f\left(\begin{pmatrix} 6 \\ 1 \end{pmatrix}\right) = \text{sign}\left((0.5)(-1)\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 1 \end{pmatrix} + (0.25)(1)\begin{pmatrix} 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 1 \end{pmatrix} + (0.25)(1)\begin{pmatrix} 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 1 \end{pmatrix} - 2\right) = \text{sign}(-3 + 4.75 + 4.25 - 2) = \text{sign}(4) = 1$$



Summary

linearly separable data



Inner Product and Similarity

Why should inner product kernels be involved in pattern recognition using SVMs, or at all?

- Intuition is that inner products provide some measure of ‘similarity’
- Inner product in 2D between 2 vectors of unit length returns the cosine of the angle between them = how ‘far apart’ they are

e.g. $\mathbf{x} = [1, 0]^T$, $\mathbf{y} = [0, 1]^T$

i.e. if they are parallel their inner product is 1 (completely similar)

$$\mathbf{x}^T \mathbf{y} = \mathbf{x} \bullet \mathbf{y} = 1$$

If they are perpendicular (completely unlike) their inner product is 0 (so should not contribute to the correct classifier)

$$\mathbf{x}^T \mathbf{y} = \mathbf{x} \bullet \mathbf{y} = 0$$



03 SVM for Non-linearly Separable Data

IF3170 Artificial Intelligence

Modul: Supervised Learning

03 SVM for Non-linearly Separable Data

IF3170 - Inteligensi Artifisial

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Outline

Slack Variable

Non-linearly
Separable

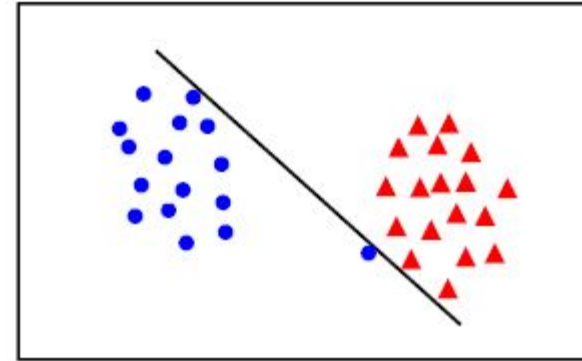
Optimization
Problem

SVM for Non-linearly
Separable Data

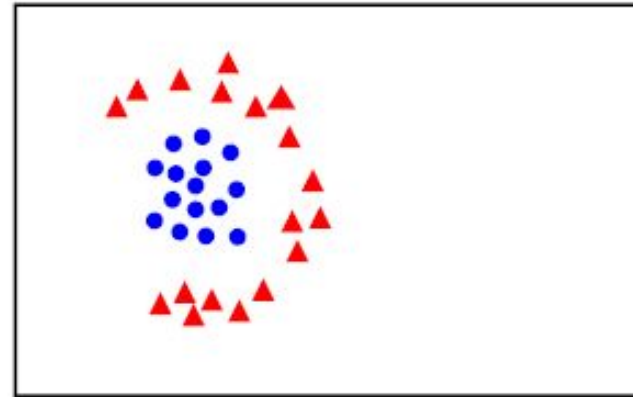
Non-linear Boundary
Transformation

Non-Linearly Separable

- Existence of the noise



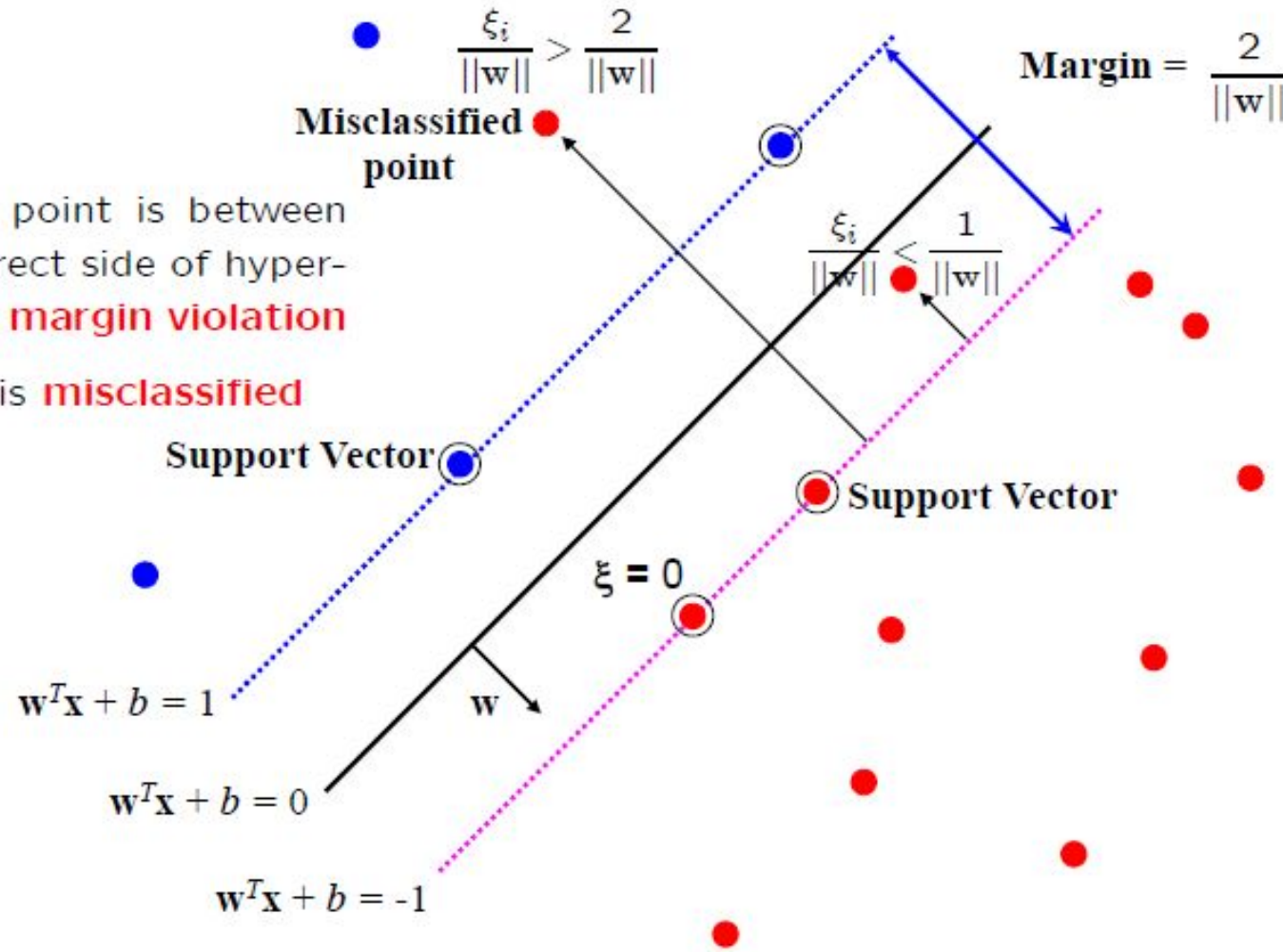
- Nature of data : Non linear boundary



Slack Variable

$$\xi_i \geq 0$$

- for $0 < \xi \leq 1$ point is between margin and correct side of hyper-plane. This is a **margin violation**
- for $\xi > 1$ point is **misclassified**



Noise in Data

Minimize : $\Phi(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w}$

Subject to : $d_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 \quad \forall i$

Introduce slack variables $\xi_i \geq 0$

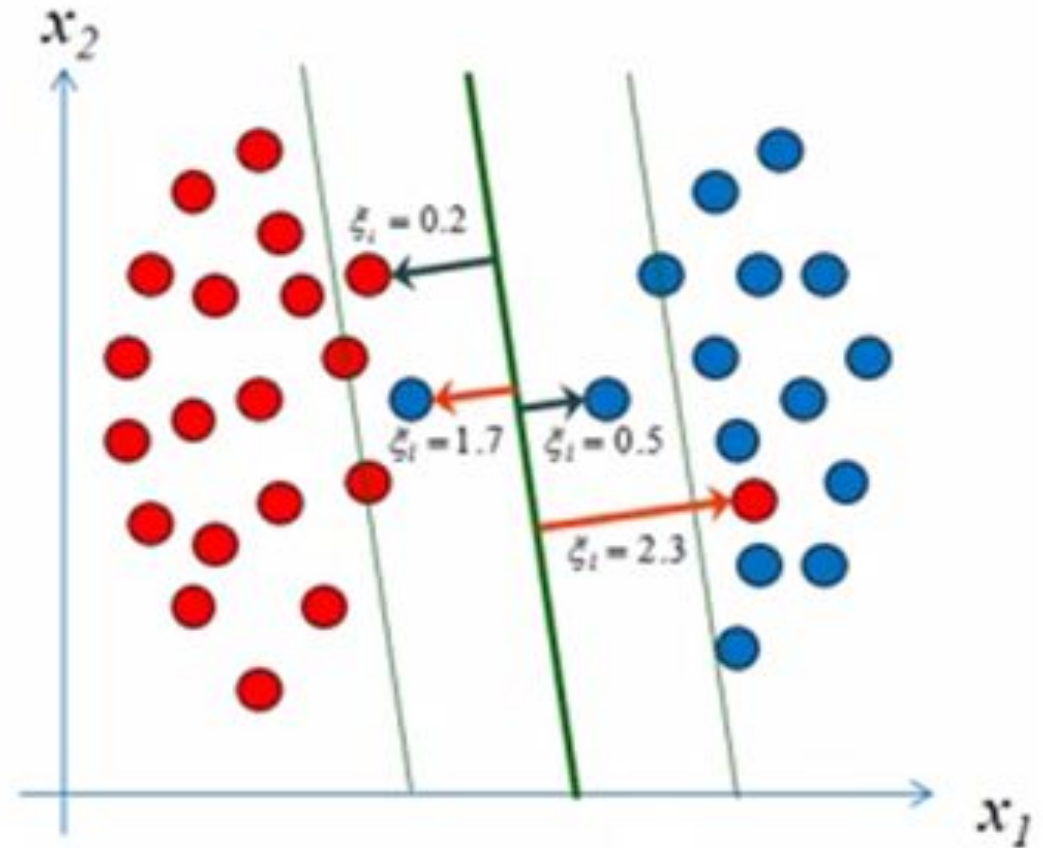
Minimize : $\Phi(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w}$

Subject to : $d_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i \quad \forall i$

Also minimize training error $\sum_{i=1}^N I(\xi_i \geq 1)$ or $\sum_{i=1}^N \xi_i$

Minimize : $\Phi(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^N \xi_i$

Subject to : $d_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i ; \quad \xi_i \geq 0, \quad \forall i$



Dual Form dengan Slack

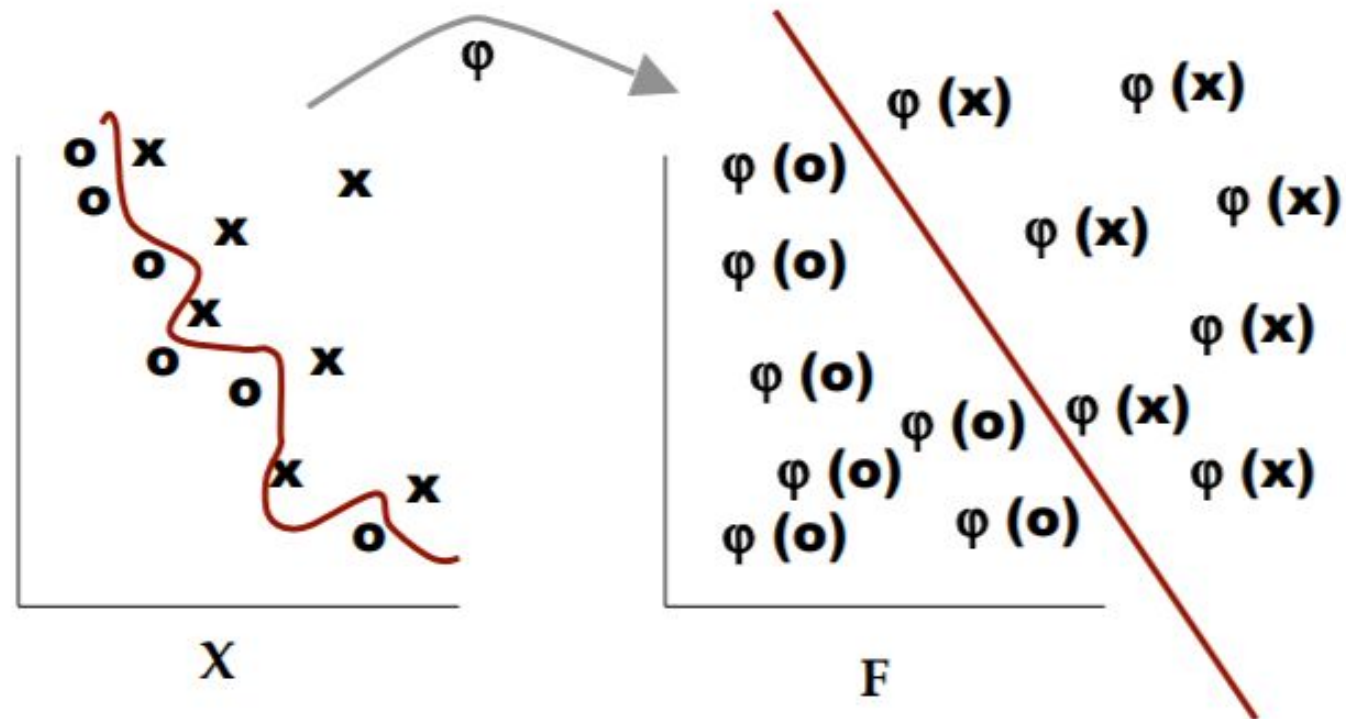
- Forming the Lagrangian and converting to dual, we get:

$$Q(\boldsymbol{\alpha}) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j \mathbf{x}_i^T \mathbf{x}_j$$

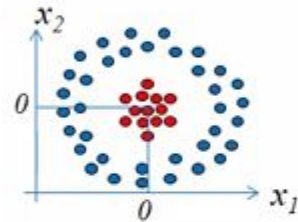
$$\text{Subject to } 0 \leq \alpha_i \leq C \quad \forall_i \quad \text{and} \quad \sum_{i=1}^N \alpha_i d_i = 0$$

- Note that neither the slack variables, nor their Lagrange multipliers appear in the dual.
- The only change is the additional constraint on α_i
- The parameter C controls the relative weight between training error and the VC dimension.

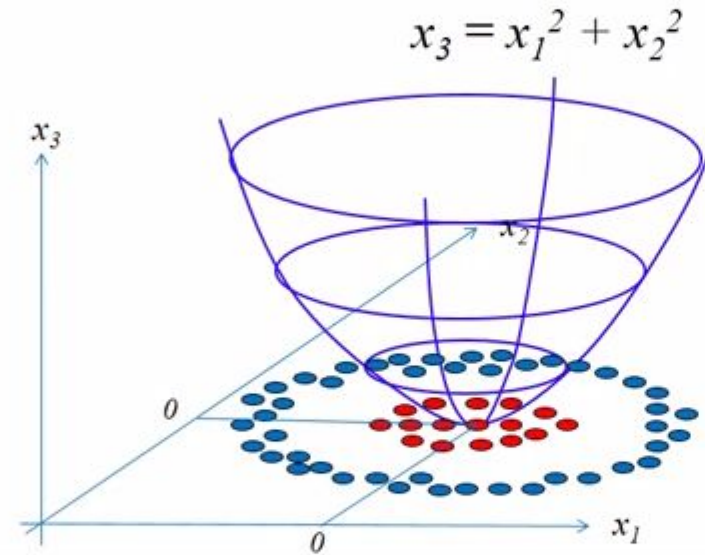
Non-Linear Boundary Transformation



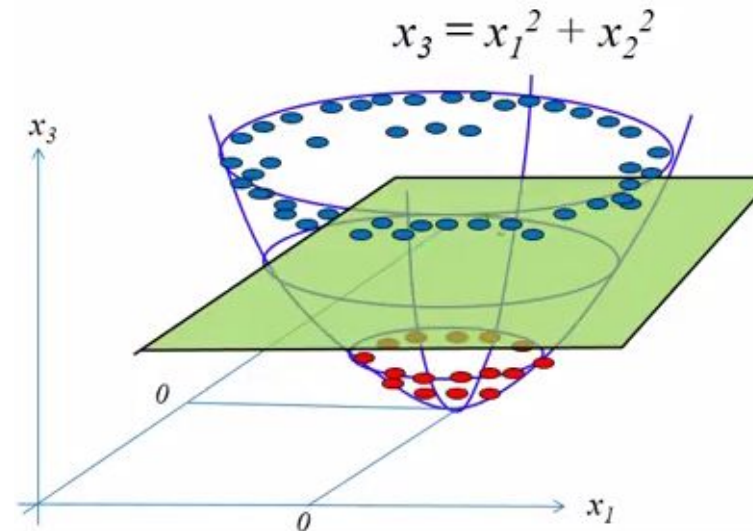
Non-Linear Boundary Transformation



Lower Dimension: Non-linearly Separable Data



Higher Dimension: Linearly Separable Data

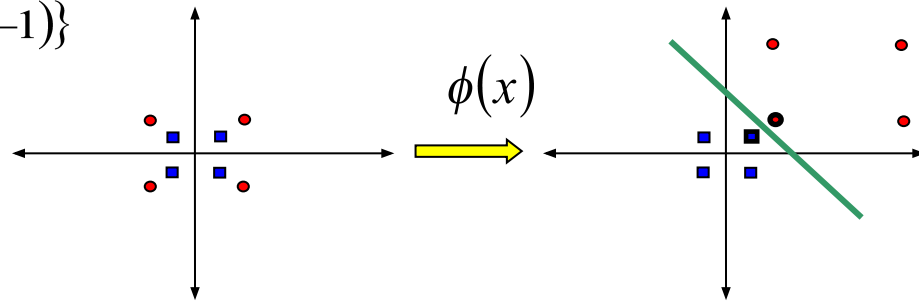


SVM pada *non-linearly separable data* (2)

- Contoh:

Misalkan dataset

- Data kelas positif $\{(2, 2), (2, -2), (-2, 2), (-2, -2)\}$
- Data kelas negatif $\{(1, 1), (1, -1), (-1, 1), (-1, -1)\}$



$$\phi(x_1, x_2) = \begin{cases} \sqrt{x_1^2 + x_2^2} > 2 \rightarrow (4 - x_2 + |x_1| - |x_2|, 4 - x_1 + |x_1| - |x_2|) \\ \sqrt{x_1^2 + x_2^2} \leq 2 \rightarrow (x_1, x_2) \end{cases}$$

- Dengan transformasi diperoleh

- Data kelas positif $\{(2, 2), (6, 2), (6, 6), (2, 6)\}$
- Data kelas negatif $\{(1, 1), (1, -1), (-1, 1), (-1, -1)\}$

SVM pada *non-linearly separable data* (3)

Klasifikasi: $f(x) = \sum_{i=1}^{ns} \alpha_i y_i x_i \cdot x + b \quad \Rightarrow \quad f(x) = \sum_{i=1}^{ns} \alpha_i y_i \phi(x_i) \phi(x) + b$

- Sulit untuk mengetahui $\phi(x)$ dan feature space biasanya memiliki dimensi yang lebih besar
- Solusinya “*kernel trick*”, yang perlu diketahui adalah $K(x_i, x) = \phi(x_i) \phi(x)$
- Dengan fungsi K (fungsi Kernel), maka fungsi $\phi(x)$ tidak perlu diketahui

Klasifikasi: $f(x) = \sum_{i=1}^{ns} \alpha_i y_i K(x_i, x) + b$

Fungsi Kernel yang umum digunakan:

Linear Kernel ☐ $K(x_i, x_j) = x_i^T x_j$

Polynomial kernel ☐ $K(x_i, x_j) = (\gamma \cdot x_i^T x_j + r)^p, \gamma > 0$

RBF kernel ☐ $K(x_i, x_j) = \exp(-\gamma |x_i - x_j|^2), \gamma > 0$

Sigmoid kernel ☐ $K(x_i, x_j) = \tanh(\gamma x_i^T x_j + r)$

Example

- Suppose we have 5 (1D) data points
 - $x_1=1, x_2=2, x_3=4, x_4=5, x_5=6,$
 - with 1, 2, 5 as class 1 and 3, 4 as class 2 $\Rightarrow y_1=1, y_2=1, y_3=-1, y_4=-1, y_5=1$
- We use the polynomial kernel of degree 2
 - $K(x,y) = (xy+1)^2$
 - C is set to 100
- We first find a_i ($i=1, \dots, 5$) by

$$\max. \sum_{i=1}^5 \alpha_i - \frac{1}{2} \sum_{i=1}^5 \sum_{j=1}^5 \alpha_i \alpha_j y_i y_j (x_i x_j + 1)^2$$

$$\text{subject to } 100 \geq \alpha_i \geq 0, \sum_{i=1}^5 \alpha_i y_i = 0$$

Example

- By using a QP solver, we get
 - $\alpha_1=0, \alpha_2=2.5, \alpha_3=0, \alpha_4=7.333, \alpha_5=4.833$
 - Note that the constraints are indeed satisfied
 - The support vectors are $\{x_2=2, x_4=5, x_5=6\}$

- $x_1=1, x_2=2, x_3=4, x_4=5, x_5=6,$
- $y_1=1, y_2=1, y_3=-1, y_4=-1, y_5=1$

- The discriminant function is

$$\begin{aligned}
 f(z) &= 2.5(1)(2z + 1)^2 + 7.333(-1)(5z + 1)^2 + 4.833(1)(6z + 1)^2 + b \\
 &= 0.6667z^2 - 5.333z + b
 \end{aligned}$$

α_5 y_5 $K(z, x_5)$

- b is recovered by solving $f(2)=1$ or by $f(5)=-1$ or by $f(6)=1$, as x_2 and x_5 lie on the line $\phi(w)^T \phi(x) + b = 1$ and x_4 lies on the line $\phi(w)^T \phi(x) + b = -1$
- All three give $b=9$
 $\rightarrow f(z) = 0.6667z^2 - 5.333z + 9$



04 SVM for Multi-class Data

IF3170 Artificial Intelligence

Modul: Supervised Learning

04 SVM for Multi-class Data

IF3170 - Inteligensi Artifisial

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Multi Class SVM

- SVM dirancang untuk mengklasifikasikan data ke dalam dua kelas → SVM biner
- Untuk klasifikasi multiclass gabungkan beberapa SVM biner.
- Ada 3 metode umum:
 - *One-against-all*
 - *One-against-one*
 - DAGSVM

Multi Class SVM → one-against-all

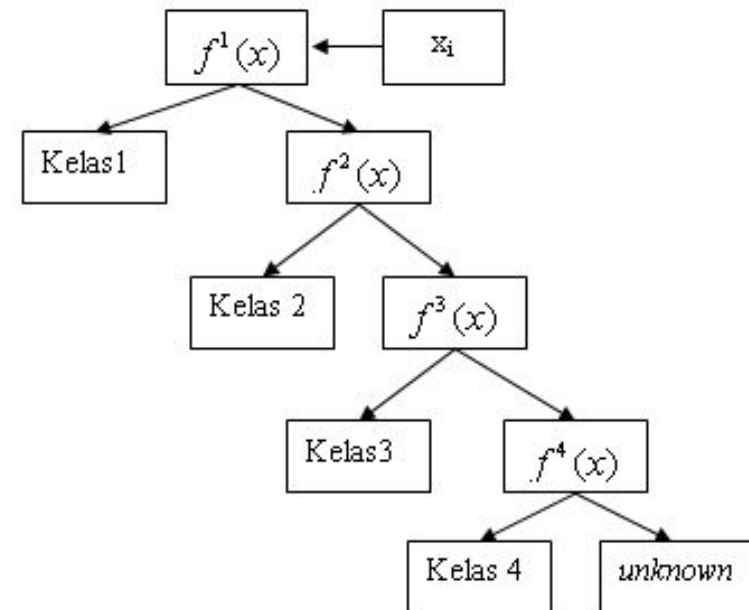
- Dibangun k model klasifikasi (k adalah jumlah kelas)
- Setiap pelatihan model menggunakan data dari semua kelas
- Prediksi kelas data umumnya seperti gambar sebelah kanan atau berdasarkan nilai maksimum $f(X)$

Pembelajaran:

$y_i = 1$	$y_i = -1$	Hipotesis
Kelas 1	Bukan kelas 1	$f^1(x) = (w^1)x + b^1$
Kelas 2	Bukan kelas 2	$f^2(x) = (w^2)x + b^2$
Kelas 3	Bukan kelas 3	$f^3(x) = (w^3)x + b^3$
Kelas 4	Bukan kelas 4	$f^4(x) = (w^4)x + b^4$



Klasifikasi:



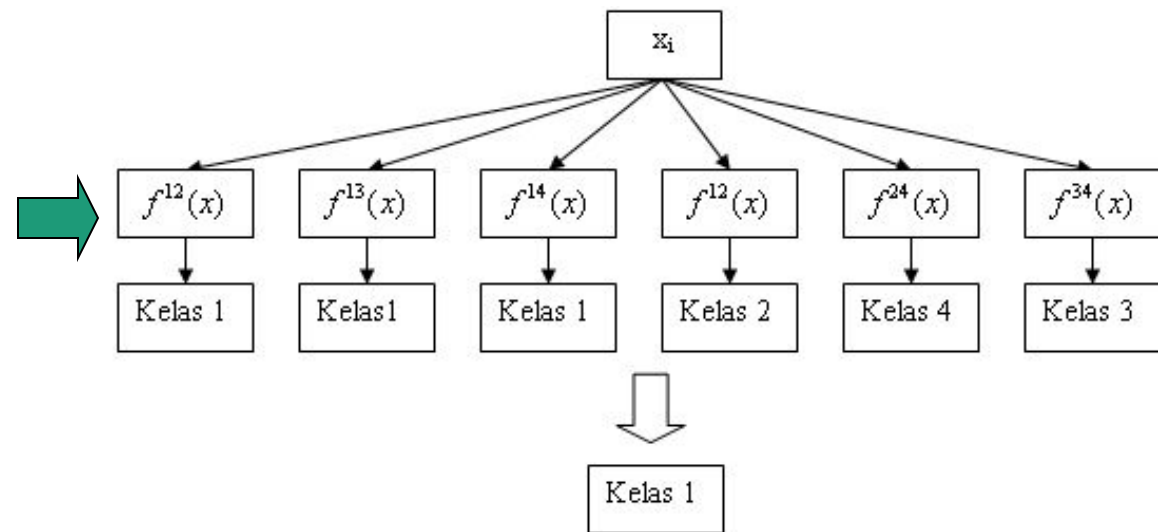
Multi Class SVM → one-against-one

- Dibangun $\frac{k(k-1)}{2}$ model klasifikasi (k adalah jumlah kelas)
- Setiap pelatihan model menggunakan data dari dua kelas
- Prediksi Kelas data dengan metode voting

Pembelajaran:

$y_i = 1$	$y_i = -1$	Hipotesis
Kelas 1	Kelas 2	$f^{12}(x) = (w^{12})x + b^{12}$
Kelas 1	Kelas 3	$f^{13}(x) = (w^{13})x + b^{13}$
Kelas 1	Kelas 4	$f^{14}(x) = (w^{14})x + b^{14}$
Kelas 2	Kelas 3	$f^{23}(x) = (w^{23})x + b^{23}$
Kelas 2	Kelas 4	$f^{24}(x) = (w^{24})x + b^{24}$
Kelas 3	Kelas 4	$f^{34}(x) = (w^{34})x + b^{34}$

Klasifikasi:



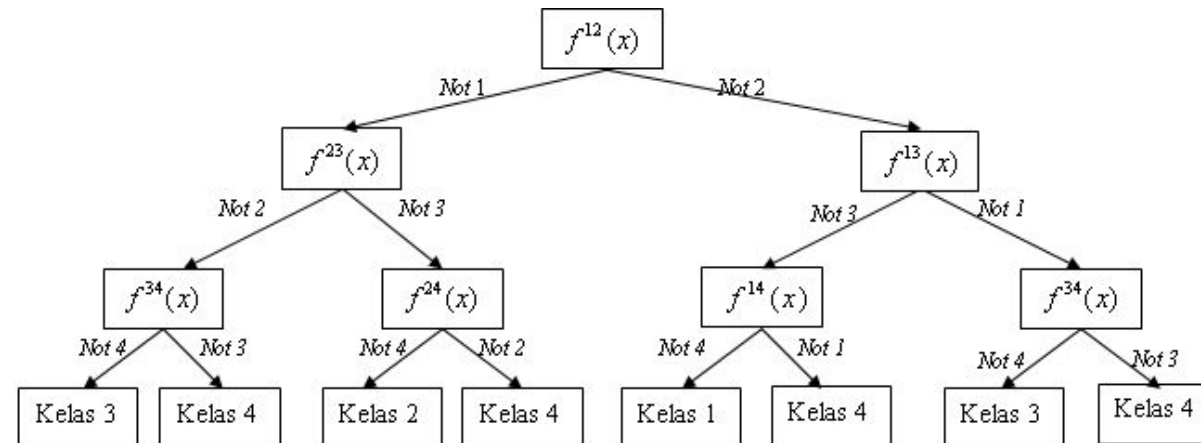
Multi Class SVM → DAG (Directed Acyclic Graph) SVM

- Proses pembelajaran sama dengan One-Against-One

Pembelajaran:

$y_i = 1$	$y_i = -1$	Hipotesis
Bukan Kelas 2	Bukan Kelas 1	$f^{12}(x) = (w^{12})x + b^{12}$
Bukan Kelas 3	Bukan Kelas 1	$f^{13}(x) = (w^{13})x + b^{13}$
Bukan Kelas 4	Bukan Kelas 1	$f^{14}(x) = (w^{14})x + b^{14}$
Bukan Kelas 3	Bukan Kelas 2	$f^{23}(x) = (w^{23})x + b^{23}$
Bukan Kelas 4	Bukan Kelas 2	$f^{24}(x) = (w^{24})x + b^{24}$
Bukan Kelas 4	Bukan Kelas 3	$f^{34}(x) = (w^{34})x + b^{34}$

Klasifikasi:



SVM Software

- LibSVM
 - Umum, dapat digunakan untuk berbagai aplikasi, tidak dioptimasi untuk SVM Linier (fungsi kernel linier)
 - Mendukung Multi Class SVM One-Against-One, One Class SVM dan Support Vector Regression
 - C++, Java, Python, C#, Matlab
- LibLinear
 - versi LibSVM yang dioptimasi untuk kernel linier
 - Mendukung Multi Class SVM One-Against-All
 - C++
- SVMLight
 - C++, populer dalam aplikasi klasifikasi teks
- Info tambahan tentang SVM:
 - <http://www.kernel-machines.org>
 - www.svms.org
 - <http://agbs.kyb.tuebingen.mpg.de>
 - <http://support-vector-machines.org>

Terima Kasih

Nur ULFA Maulidevi

KK IF - Teknik Informatika- STEI ITB

Modul : Supervised Learning

Prediction Measurement

**Inteligensi Buatan
(Artificial Intelligence)**



Prediction Measurement

Supervised
Learning



“Correct Class”

Prediction based
on Model/
Hypothesis from
Learning

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative



Example

Instance	Correct Class	Prediction	
1	+	+	TP
2	-	-	TN
3	-	+	FP
4	+	-	FN
5	-	+	FP
6	+	-	FN
7	+	+	TP
8	-	-	TN
9	-	-	TN
10	+	+	TP



Accuracy

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative

Fraction of all correct predictions over all predicted instances

$$Accuracy = \frac{Tp + Tn}{Tp + Fp + Tn + Fn}$$



Precision

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative

Fraction of positive predictions that are correct

$$Precision = \frac{Tp}{Tp + Fp}$$

Recall

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative

Fraction of positive instances that are correctly predicted
(retrieved/caught)

$$Recall = \frac{Tp}{Tp + Fn}$$



F1-Score

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Harmonic mean of precision and recall

Exercise: Find accuracy, precision and recall

Instance	Correct Class	Prediction
1	+	+
2	-	-
3	-	+
4	+	-
5	-	+
6	+	-
7	+	+
8	-	-
9	-	-
10	+	+
11	+	-
12	-	-
13	-	+
14	+	-



MultiClass Problem

No	Actual	Predicted	Match
1	Airplane	Airplane	✓
2	Car	Boat	✗
3	Car	Car	✓
4	Car	Car	✓
5	Car	Boat	✗
6	Airplane	Boat	✗
7	Boat	Boat	✓
8	Car	Airplane	✗
9	Airplane	Airplane	✓
10	Car	Car	✓

		Predicted		
		 Airplane	 Boat	 Car
Actual	 Airplane	2	1	0
	 Boat	0	1	0
	 Car	1	2	3

Source: <https://iamirmasoud.com/2022/06/19/understanding-micro-macro-and-weighted-averages-for-scikit-learn-metrics-in-multi-class-classification-with-example/>

Label	True Positive (TP)	False Positive (FP)	False Negative (FN)	Precision	Recall	F1 Score
 Airplane	2	1	1	0.67	0.67	$2 * (0.67 * 0.67) / (0.67 + 0.67)$ = 0.67
 Boat	1	3	0	0.25	1.00	$2 * (0.25 * 1.00) / (0.25 + 1.00)$ = 0.40
 Car	3	0	3	1.00	0.50	$2 * (1.00 * 0.50) / (1.00 + 0.50)$ = 0.67



MultiClass Problem

No	Actual	Predicted	Match
1	Airplane	Airplane	✓
2	Car	Boat	✗
3	Car	Car	✓
4	Car	Car	✓
5	Car	Boat	✗
6	Airplane	Boat	✗
7	Boat	Boat	✓
8	Car	Airplane	✗
9	Airplane	Airplane	✓
10	Car	Car	✓

micro-F1 = micro-precision = micro-recall = accuracy

Label	Per-Class F1 Score	Macro-Averaged F1 Score
 Airplane	0.67	$\frac{0.67 + 0.40 + 0.67}{3}$ = 0.58
 Boat	0.40	
 Car	0.67	

Label	Per-Class F1 Score	Support	Support Proportion	Weighted Average F1 Score
 Airplane	0.67	3	0.3	$(0.67 * 0.3) + (0.40 * 0.1) + (0.67 * 0.6)$ = 0.64
 Boat	0.40	1	0.1	
 Car	0.67	6	0.6	
Total	-	10	1.0	

Label	True Positive (TP)	False Positive (FP)	False Negative (FN)	Micro-Averaged Values
 Airplane	2	1	1	Precision = $\frac{6}{6+4} = \mathbf{0.60}$ Recall = $\frac{6}{6+4} = \mathbf{0.60}$ F1 Score = $\frac{6}{6 + \frac{1}{2}(4+4)} = \mathbf{0.60}$
 Boat	1	3	0	
 Car	3	0	3	
TOTAL	6	4	4	



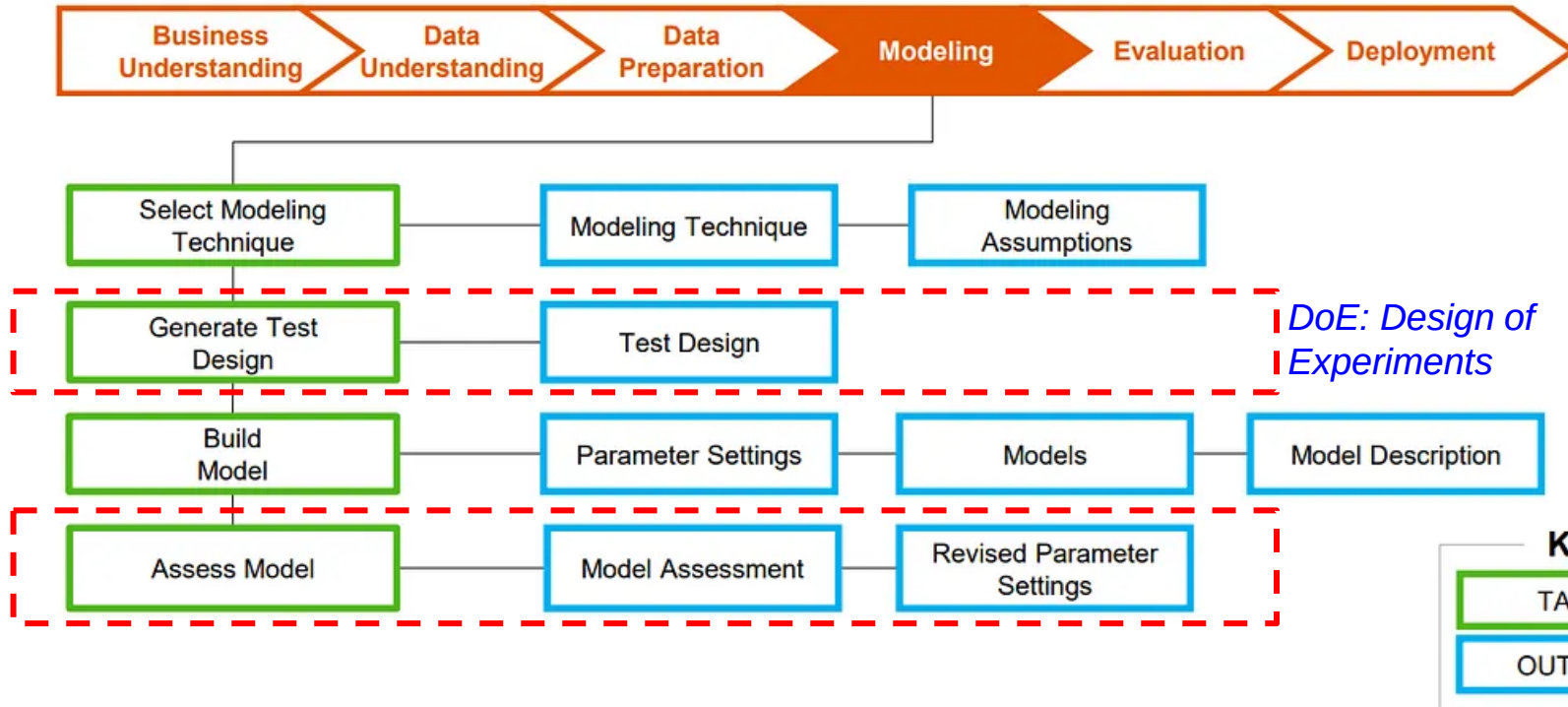
THANK YOU



Pengukuran Kinerja Model

Tim Dosen IF3170 Teknik Informatika ITB

CRISP-DM – Phase 4: Modeling



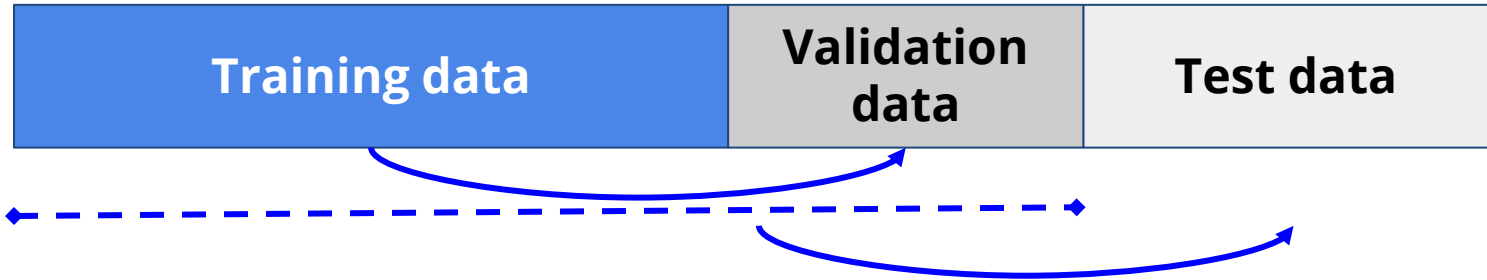
Pemilihan Skema Validasi

- Hold-out validation
- K-fold cross validation
- Bootstrapping Validation



Pemilihan Skema Validasi:

Hold-out Validation



1. Random split dataset menjadi training data, validation data, dan test data.
2. Untuk setiap konfigurasi parameter, lakukan pembelajaran, lalu hitung kinerja terhadap validation data.
3. Menggunakan model dengan konfigurasi terbaik hasil (2) (training dengan training data saja atau training+validation data), hitung kinerjanya terhadap test data.



Contoh Implementasi Python: Hold-out Validation

```
from sklearn.model_selection import train_test_split
from sklearn import datasets
from sklearn import svm
```

```
X, y = datasets.load_iris(return_X_y=True)
print(X.shape, y.shape)
```

```
#split 60:20:20
```

```
X_hold, X_test, y_hold, y_test = train_test_split(X, y, test_size=0.2,
random_state=0)
```

```
X_train, X_val, y_train, y_val = train_test_split(X_hold, y_hold,
test_size=0.25, random_state=0)
```

```
print('Training data: ',X_train.shape, y_train.shape)
print('Validation data: ',X_val.shape, y_val.shape)
print('Testing data: ',X_test.shape, y_test.shape)
```

```
clf = svm.SVC(kernel='linear', C=1).fit(X_train, y_train)
print('Validation score:',clf.score(X_val, y_val))
```

5

(150, 4) (150,)

Training data: (90, 4) (90,)

Validation data: (30, 4) (30,)

Testing data: (30, 4) (30,)

Validation score: 0.933333

Test score: 1.0

```
# tahap pengujian, setelah eksperimen semua konfigurasi selesai
```

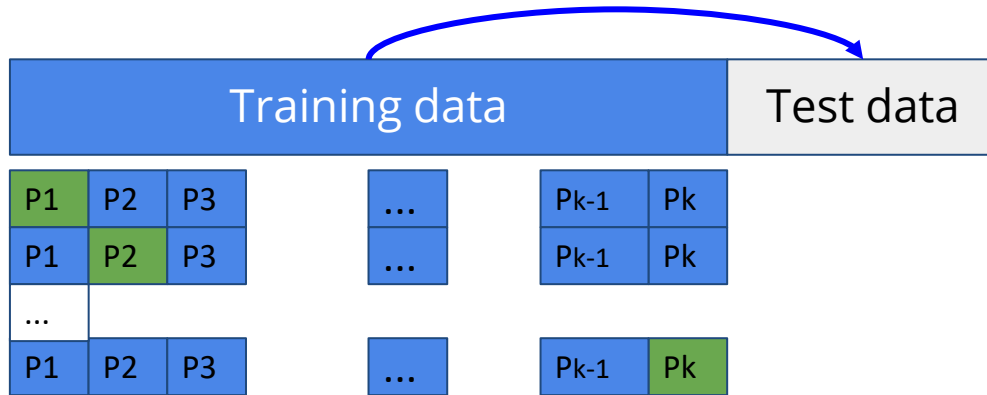
```
clf = svm.SVC(kernel='linear', C=1).fit(X_hold, y_hold)
print('Test score:',clf.score(X_test, y_test))
```



Pemilihan Skema Validasi:

K-fold Cross Validation

1. Random split dataset menjadi training dan test data.
2. Untuk setiap konfigurasi lakukan k-fold cross validation (1 partisi validation data, k-1 partisi training data).
3. Menggunakan model dengan konfigurasi terbaik hasil (2) (all training data), hitung kinerjanya terhadap test data.



Keterangan:

Validation data



Contoh Implementasi Python: K-Fold Cross Validation

```
from sklearn.model_selection import train_test_split
from sklearn.model_selection import KFold
from sklearn.model_selection import cross_val_score

X_hold, X_test, y_hold, y_test = train_test_split(X, y,
test_size=0.2, random_state=0)
clf = svm.SVC(kernel='linear', C=1)
scores=cross_val_score(clf, X, y, cv=5)
print('Validation scores:',scores)

clf = svm.SVC(kernel='linear', C=1).fit(X_hold, y_hold)
print('Test score:',clf.score(X_test, y_test))
```

Validation scores: [0.96666667 1. 0.96666667 0.96666667 1.]
Test score: 1.0

`sklearn.model_selection.cross_val_score(estimator, X, y=None, *, groups=None, scoring=None, cv=None, n_jobs=None, verbose=0, fit_params=None, params=None, pre_dispatch='2*n_jobs', error_score=nan)`

https://scikit-learn.org/1.5/modules/cross_validation.html



Contoh Implementasi Python: K-Fold Cross Validation

```
from sklearn.model_selection import train_test_split
from sklearn.model_selection import KFold
from sklearn.model_selection import cross_val_score
```

```
X_hold, X_test, y_hold, y_test = train_test_split(X, y, test_size=0.2,
random_state=0)
kf = KFold(n_splits=5)
for train, val in kf.split(X_hold):
    X_train, X_val, y_train, y_val = X[train], X[val], y[train], y[val]
    print('Train vs valid: ', y_train.shape[0], ' - ',
y_val.shape[0],'; Indeks data-1 val: ',val[0])
```

```
clf = svm.SVC(kernel='linear', C=1)
scores=cross_val_score(clf, X, y, cv=kf)
print('Validation scores:',scores)
```

```
Train vs valid: 96 - 24 ; Indeks data-1 val: 0
Train vs valid: 96 - 24 ; Indeks data-1 val: 24
Train vs valid: 96 - 24 ; Indeks data-1 val: 48
Train vs valid: 96 - 24 ; Indeks data-1 val: 72
Train vs valid: 96 - 24 ; Indeks data-1 val: 96
Validation scores: [1.  1.  0.86666667 1.  0.8667]
Test score: 1.0
```

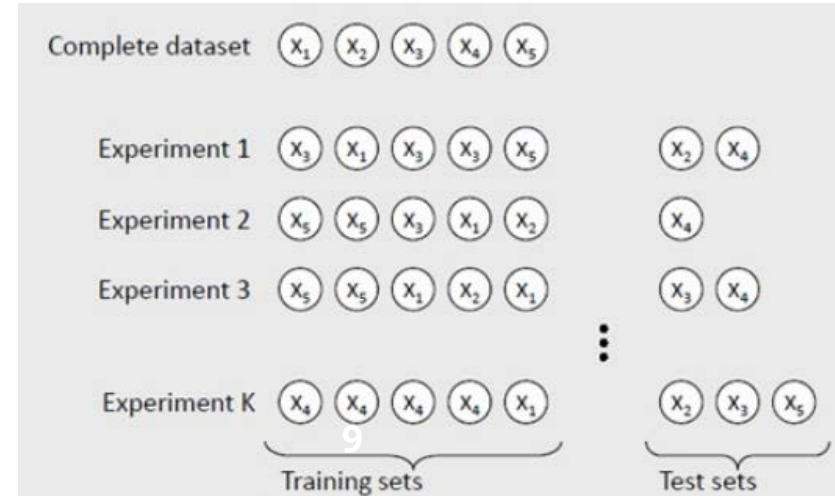
```
# tahap pengujian, setelah eksperimen semua konfigurasi selesai
clf = svm.SVC(kernel='linear', C=1).fit(X_hold, y_hold)
print('Test score:',clf.score(X_test, y_test))
```



Pemilihan Skema Validasi:

Bootstrapping Validation

- Sampel bootstrap mungkin tumpang tindih lebih banyak dibandingkan dengan sampel cross-validation
- Cara terbaik untuk melakukan resampling untuk dataset yang sangat kecil



Contoh Implementasi Python: Bootstrapping Validation

```
num_splits = 5

X = np.arange(10)
n_samples = len(X)
test_size = int(n_samples * 0.2)

for i in range(num_splits):
    # Bootstrap sampling
    bootstrap_idx = resample(np.arange(n_samples), replace=True, n_samples=n_samples)
    test_idx = np.setdiff1d(np.arange(n_samples), bootstrap_idx)[:test_size]
    print("%s %s" % (bootstrap_idx, test_idx))
```

[1 0 8 3 4 8 0 8 0 3] [2 5]

[7 2 3 3 7 8 4 2 8 1] [0 5]

[2 4 1 3 6 4 8 8 5 9] [0 7]

[9 2 2 9 8 6 7 8 0 3] [1 4]

[8 0 6 0 7 8 5 2 3 8] [1 4]



Metrik Evaluasi

Metrik untuk Klasifikasi

- Akurasi (Accuracy)
- Precision
- Recall (Sensitivity atau True Positive Rate)
- F1-Score
- ROC-AUC (Receiver Operating Characteristic - Area Under Curve)

Metrik untuk Regresi

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)

Metrik untuk Clustering

- Silhouette Score
- Davies-Bouldin Index
- Inertia (Sum of Squared Distances)

Metrik untuk Pemodelan Waktu (Time Series Forecasting)

- Mean Absolute Percentage Error (MAPE)
- Mean Squared Logarithmic Error (MSLE)



Recall: Prediction Measurements - Klasifikasi

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative

$$Accuracy = \frac{Tp + Tn}{Tp + Fp + Tn + Fn}$$

$$Precision = \frac{Tp}{Tp + Fp} \quad Recall = \frac{Tp}{Tp + Fn}$$

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- **Akurasi:**

Persentase prediksi yang benar dari total prediksi yang dilakukan oleh model.

- **Precision:**

Persentase prediksi positif yang benar dari semua prediksi positif yang dibuat oleh model.

- **Recall:**

Persentase kasus positif yang benar-benar diidentifikasi sebagai positif oleh model.

- **F1-Score:**

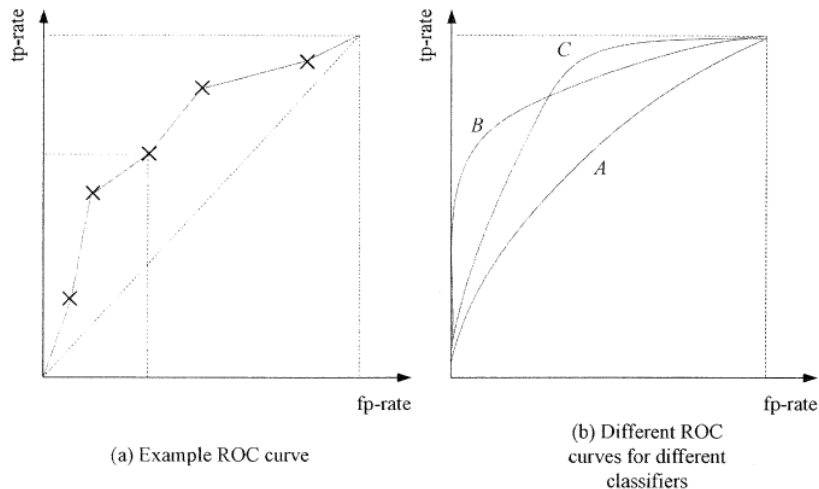
Harmonic mean dari precision dan recall. Memberikan keseimbangan antara precision dan recall.



Metrik Evaluasi:

ROC (Receiver operating characteristic) - AUC

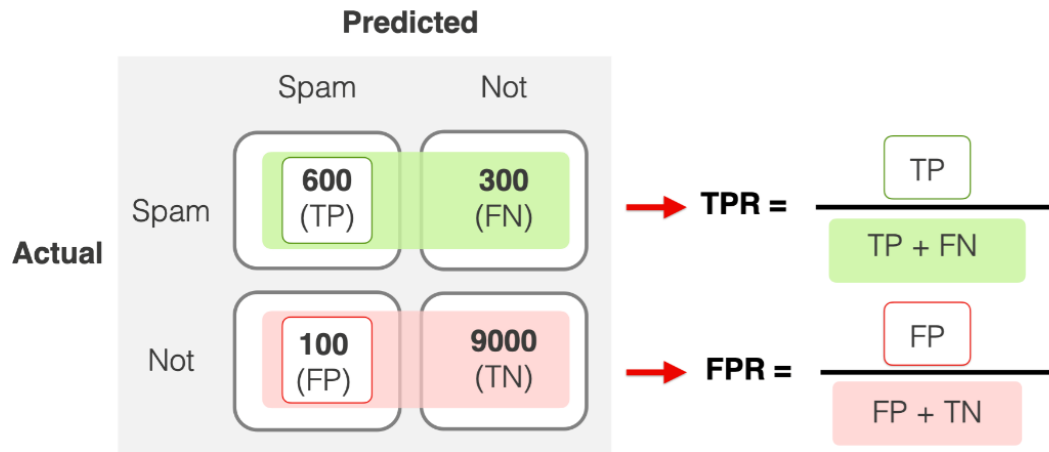
- AUC mengukur area di bawah kurva ROC, yang merupakan grafik dari true positive rate (Recall) versus false positive rate.
- **Nilai AUC:**
 - 1.0: Kinerja sempurna.
 - 0.5: Kinerja acak (setara dengan tebakan acak).
- **Kelebihan** : Berguna untuk mengevaluasi model pada berbagai threshold klasifikasi.
- **Kekurangan** : Lebih sulit diinterpretasikan daripada metrik sederhana seperti akurasi.



Metrik Evaluasi:

ROC (Receiver operating characteristic) -AUC

- The ROC curve plots the True Positive rate (TPR) against the False Positive rate (FPR) at various classification thresholds.



Metrik Evaluasi:

ROC (Receiver operating characteristic) -AUC

- The ROC curve plots the True Positive rate (TPR) against the False Positive rate (FPR) at various classification thresholds.

For a probabilistic classification model, you can vary the decision threshold that defines how to convert the model predictions into labels.

data	label	predicted	confidence
	1	1	[0.20,0.80]
	0	1	[0.40,0.60]
	1	0	[0.50,0.50]
	0	0	[0.50,0.50]
	0	0	[0.60,0.40]
	0	0	[0.60,0.40]
	0	0	[0.70,0.30]
	0	0	[0.75,0.25]
	0	0	[0.80,0.10]
	0	0	[1.00,0.00]

put most confident positive predictions at top

put most confident negative predictions at bottom

calculate precision/recall at each break point/threshold

classify everything above threshold as positive and everything else negative

Akurasi = 9/10
Precision₀ = 8/9
Recall₀ = 8/8
Precision₁ = 1/1
Recall₁ = 1/2

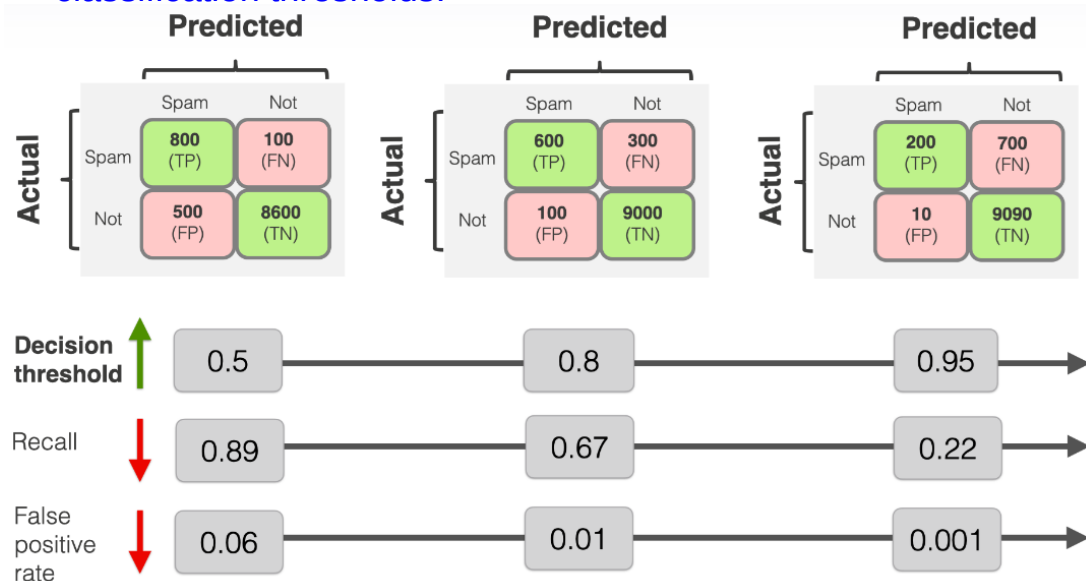
Label\Pred	0	1
0	8	0
1	1	1



Metrik Evaluasi:

ROC (Receiver operating characteristic) -AUC

- The ROC curve plots the True Positive rate (TPR) against the False Positive rate (FPR) at **various classification thresholds**.



0.5 threshold:

$$\text{TPR: } 800/(800+100)=0.89$$

$$\text{FPR: } 500/(500+8600)=0.06$$

0.8 threshold:

$$\text{TPR: } 600/(600+300)=0.67$$

$$\text{FPR: } 100/(100+9000)=0.01$$

0.95 threshold:

$$\text{TPR: } 200/(200+700)=0.22$$

$$\text{FPR: } 10/(10+9090)=0.001$$

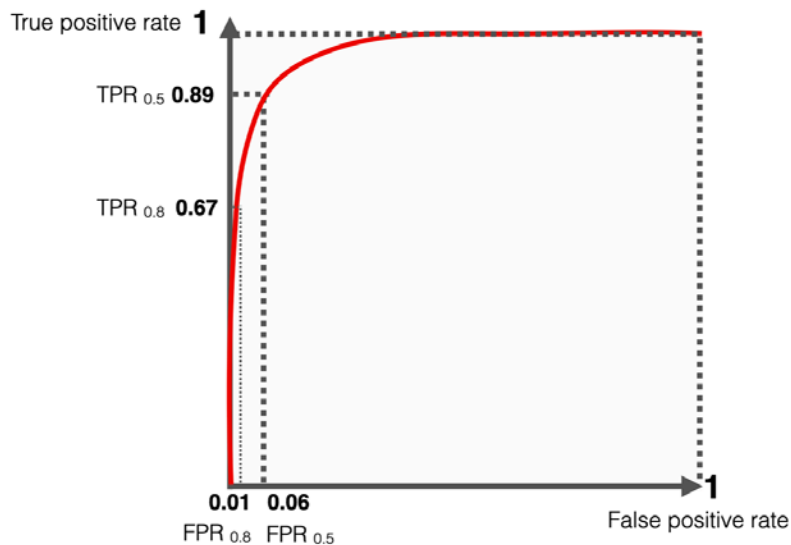
- The **left side** of the curve corresponds to the more "confident" thresholds: a higher threshold leads to lower recall and fewer false positive errors.
- The **right side** of the curve represents the "less strict" scenarios when the threshold is low.



Metrik Evaluasi:

ROC (Receiver operating characteristic) -AUC

- The ROC curve plots the True Positive rate (TPR) against the False Positive rate (FPR) at various classification thresholds.



0.5 threshold:

TPR: 0.89

FPR: 0.06

0.8 threshold:

TPR: 0.67

FPR: 0.01

0.95 threshold:

TPR: 0.22

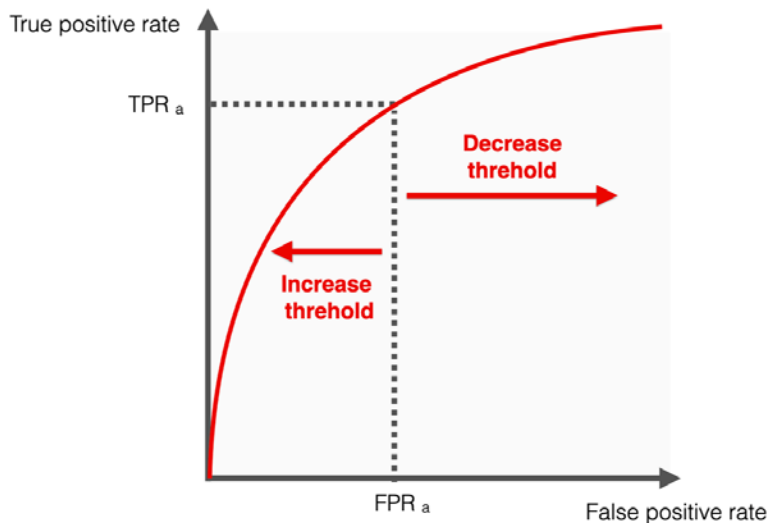
FPR: 0.001



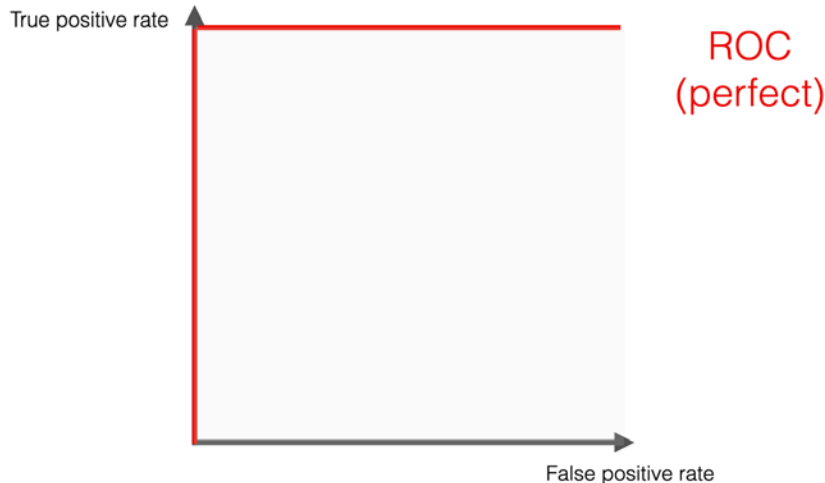
Metrik Evaluasi:

ROC (Receiver operating characteristic) - AUC

- The ROC curve plots the True Positive rate (TPR) against the False Positive rate (FPR) at various classification thresholds.



If our model is correct in all the predictions, all the time, it means that the TPR is always 1.0, and FPR is 0.



Recall: Prediction Measurements - Regresi

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

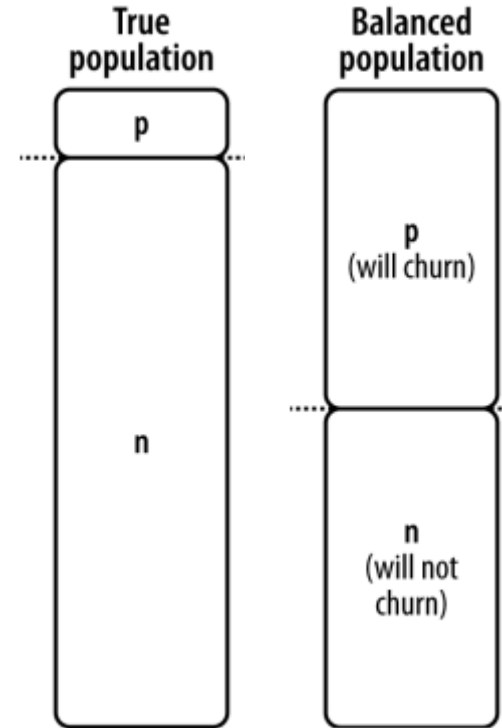
$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Data	Target	Prediksi	Abs Err	Abs % Err	Sq Err
R1	1	0.99	0.01	1.00	0.00
R2	1	0.48	0.52	52.00	0.27
R3	0.5	0.51	0.01	2.00	0.00
R4	1	0.9	0.1	10.00	0.01
R5	1	0.4	0.6	60.00	0.36
R6	0.5	0.6	0.1	20.00	0.01
			0.22	24.17	0.11
			MAE	MAPE	MSE



Example: Churn Prediction

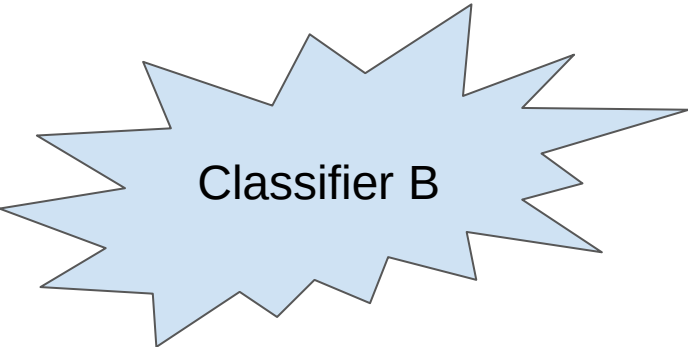
- Let's consider a customer who churns to be a positive example, so within our population of customers we expect a positive to negative class ratio of 1:9.
- We have two datasets:
 - Representative dataset. If the training population is 1000 customers, a representative sample from the population have 100 positive examples and 900 negative examples.
 - Artificially balanced dataset. If the training population is 1000 customers, a balanced sample from the population have 500 positive examples and 500 negative examples.



Comparing Models: Description



Classifier A



Classifier B

- Both models correctly classify 80% of the balanced population (80% accuracy).
- Classifier A often falsely predicts that customers will churn when they will not, while classifier B makes many opposite errors of predicting that customers will not churn when in fact they will.
- Training with balanced dataset, and testing with representative dataset.

Comparing Models: Balanced Dataset

- Classifier A achieves 80% accuracy on the balanced sample by correctly identifying **all positive** examples but only **30% of the negative** examples.
- Classifier B also achieves 80% accuracy on the balanced sample by correctly identifying **all the negative** examples but only **30% of the positive** examples.

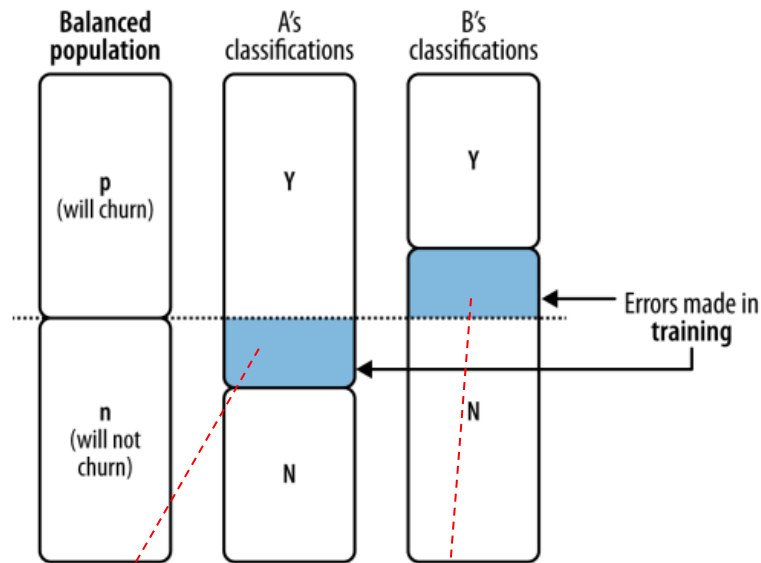


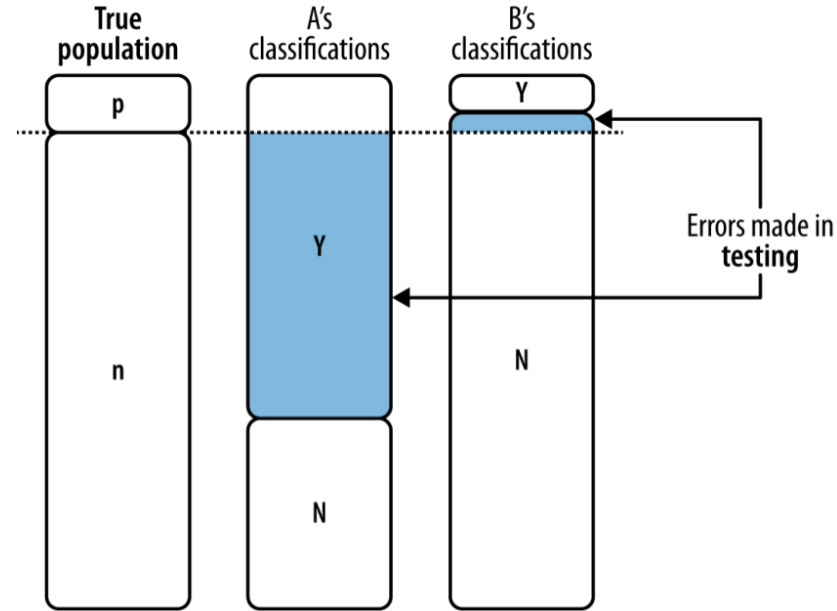
Table 7-2. Confusion matrix of A Table 7-3. Confusion matrix of B

	churn	not churn
Y	500	200
N	0	300

	churn	not churn
Y	300	0
N	200	500

Comparing Models: Representative Dataset

- Both models correctly classify 80% of the balanced population (80% accuracy).
- When applied to the original, unbalanced population of customers, model A's accuracy declines to 37% while model B's rises to 93%.
- Which model is better?
 - model (B) now appears to be better than A because B seems to have greater performance on the true population



Evaluasi & Analisis Hasil

Setelah eksperimen selesai dikerjakan, lakukan:

- **Tinjau hasil evaluasi model berdasarkan metrik kinerja yang dipilih.**
 - Apakah model mencapai target kinerja yang diinginkan?
 - Jika tidak, mengapa?
 - Apakah model cukup baik untuk digunakan dalam produksi?
- **Jika menggunakan beberapa model atau konfigurasi dalam eksperimen, bandingkan kinerja model-model tersebut.**
 - Model mana yang terbaik berdasarkan kinerja keseluruhan?
 - Tinjau pentingnya fitur yang digunakan oleh model untuk membuat prediksi. Fitur mana yang paling berpengaruh pada hasil model?
 - Apakah ada trade-off yang harus dipertimbangkan, seperti model dengan kinerja terbaik tetapi dengan waktu komputasi yang lebih lama?



Contoh Penggunaan Statistical Test

Statistical Test bisa digunakan untuk mengecek apakah hasil dua model klasifikasi kategorikal itu berbeda secara signifikan atau tidak

1. Tulis null hypothesis H_0 yang berisi perbedaan tidak signifikan,
2. Pilih level of significance $\alpha \in (0, 1)$,
3. Tetapkan uji statistik yang sesuai. Misalnya menggunakan McNemar Test
4. Hitung nilai statistik dan bandingkan dengan nilai dengan critical value pada tabel (misalnya untuk McNemar, menggunakan tabel distribusi chi-square dengan derajat kebebasan = 1). Jika nilai statistik lebih besar dari critical value untuk level of significance α yang dipilih, maka null hypothesis ditolak dan perbedaan antara dua model dianggap signifikan

Contoh Penggunaan Uji Statistik McNemar

1. Buat confusion matrix yang membandingkan hasil dari kedua model
2. Hitung nilai statistik
3. Bandingkan dengan tabel chi-square

	Test 2 positive	Test 2 negative	Row total
Test 1 positive	a	b	$a + b$
Test 1 negative	c	d	$c + d$
Column total	$a + c$	$b + d$	N

a. $\chi^2 > \chi^2_{\alpha}$: H0 ditolak

b. $\chi^2 \leq \chi^2_{\alpha}$: H0 diterima

$$\chi^2 = \frac{(b - c)^2}{b + c}$$

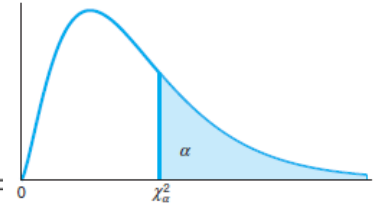


Table A.5 (continued) Critical Values of the Chi-Squared Distribution

v	α									
	0.30	0.25	0.20	0.10	0.05	0.025	0.02	0.01	0.005	0.001
1	1.074	1.323	1.642	2.706	3.841	5.024	5.412	6.635	7.879	10.827
2	1.386	1.578	1.876	2.915	3.841	5.024	5.412	6.635	7.879	10.827



Contoh Penggunaan Uji Statistik McNemar

Table A.5 (continued) Critical Values of the Chi-Squared Distribution

$$\chi^2 = \frac{(b - c)^2}{b + c}$$

v	α									
	0.30	0.25	0.20	0.10	0.05	0.025	0.02	0.01	0.005	0.001
1	1.074	1.323	1.642	2.706	3.841	5.024	5.412	6.635	7.879	10.827
2	0.446	0.578	0.810	1.833	2.779	3.828	4.015	5.024	5.991	7.378

	M1: correct	M1: incorrect	Row total
M2: correct	101	121	222
M2: incorrect	59	33	92
Column total	160	154	314

Statistics = $(121-59)^2 / 121+59 = 21.35$
 Untuk alpha = 0.001, batas p-value adl 10.827 maka H_0 ditolak (ada perbedaan yang signifikan)

	M1: correct	M1: incorrect	Row total
M2: correct	59	6	65
M2: incorrect	16	80	96
Column total	75	86	161

Statistics = $(6-16)^2 / 6+16 = 4.55$
 untuk alpha = 0.01, batas p-value adl 6.635 maka H_0 diterima (tidak ada perbedaan yang signifikan)



Evaluasi dan Analisis Hasil (2)

Setelah eksperimen selesai dikerjakan, lakukan:

- **Analisis Error**
 - Analisis kesalahan yang dibuat oleh model, misalnya dengan melihat confusion matrix untuk klasifikasi atau dengan meninjau residuals untuk regresi.
 - Apakah ada pola dalam kesalahan model? Apakah model salah dalam kasus-kasus tertentu yang bisa dihindari? Apakah model mengalami overfitting atau underfitting?



Evaluasi dan Analisis Hasil (3)

- Example: confusion matrix of cat classifier
- But it does not help understand what is being misclassified, so we isolate the 50 and 100 images respectively, so we look at the images that have been misclassified and create some hypotheses/ observations on grouping these errors.
- Identify common errors across all erroneous cases:
 - 1. White cats not being correctly identified
 - 2. Dogs like cats being identified as cats
 - 3. Dark background leads to misclassification
 - and so on
- Map the exact distribution of the errors based on common errors
- Improve model by working on top common errors data correction first to have the most impact.

Cat Classifier - Confusion Matrix			
Predicted	Actual		
	Cat	Not Cat	
	Not Cat		
	Cat	700	100
	Not Cat	50	150

Cat Classification Model Error Distribution		
Possible Hypothesis for error	Count	% of total Errors
1. White cats	27	18.0%
2. Dogs like cats being identified as cats	28	18.7%
3. Dark background	61	40.7%
4. Sphinx Cats !	22	14.7%

Evaluasi dan Analisis Hasil (6)

- **Konsistensi dengan Tujuan Bisnis**
 - Evaluasi hasil model dalam konteks tujuan bisnis atau penelitian awal. Apakah model membantu mencapai tujuan bisnis? Jika tidak, apa yang perlu diperbaiki?
- **Risiko dan Mitigasi**
 - Identifikasi potensi risiko atau masalah yang mungkin muncul jika model diimplementasikan dalam lingkungan nyata, seperti bias model atau sensitivitas terhadap outliers.
 - Apakah ada risiko yang perlu diperhatikan?
 - Bagaimana cara mitigasinya?



Menarik Kesimpulan

Berdasarkan hasil evaluasi dan analisis hasil, buat kesimpulan tentang:

- Hasil evaluasi kinerja model, termasuk evaluasi fitur dalam model dan identifikasi fitur yang signifikan.
- Analisis Error
- Evaluasi apakah hasil model mendukung tujuan bisnis atau eksperimen.
- Risiko dalam implementasi model



Hal-hal yang Perlu Diperhatikan

- Pembelajaran mesin bersifat iteratif
 - Pada eksperimen pertama, tidak lebih dari 25% sumber daya
 - *It is good idea not to start with high expectations, or promises.*
- Pengujian statistik tidak menunjukkan bahwa hipotesis yang dibangun benar, tetapi menunjukkan sebanyak apa samples yang mendukung hipotesis tersebut.
- Dokumentasikan seluruh proses eksperimen, hasil, dan kesimpulan



Hal-hal yang Perlu Diperhatikan

- Secara umum, kita membandingkan algoritma pembelajaran dengan menggunakan nilai error atau akurasi. Terdapat kriteria lain selain nilai error dalam memutuskan pemilihan algoritma:
 - Risiko
 - Waktu pelatihan dan kompleksitas ruang
 - Waktu pengujian dan kompleksitas ruang
 - Interpretabilitas (kemudahan untuk dipahami)
 - Kemudahan untuk diprogram



Hal-hal yang Perlu Diperhatikan

- Perhatikan aspek etis dalam pembangunan model *machine learning*, perhatikan bias yang terjadi yang berkaitan dengan etis, misal diskriminasi atau stereotyping kelompok tertentu



Pertanyaan ?

Knowledge Representation and Reasoning

Introduction to Probabilistic Reasoning System

Nur Ulfa Maulidevi
(ulfa@informatika.org)

KK IF – Teknik Informatika – STEI

Inteligensi Artifisial
(Artificial Intelligence)

Probabilistic Reasoning System (PRS)

Agent in real world need to handle uncertainty

Logical Agents handle uncertainty by disjunction → cannot tell us how likely the different conditions are

Probability theory provides a quantitative way of encoding likelihood

Given state(s) e , what is the probability that x happens → $P(x|e)$

Joint Probability Distribution

Bayesian/ Belief Network

Joint Probability Distribution

- Random variables

- Function: discrete domain $\rightarrow [0, 1]$
- Sums to 1 over the domain
 - Raining is a propositional random variable
 - $\text{Raining}(\text{true}) = 0.2$
 - $P(\text{Raining} = \text{true}) = 0.2$
 - $\text{Raining}(\text{false}) = 0.8$
 - $P(\text{Raining} = \text{false}) = 0.8$

- Joint distribution

- Probability assignment to all combinations of values of random variables

Inference using Joint Probability Distribution

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- For any proposition ϕ , sum the atomic events where it is true: $P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$
- $P(\textit{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$
- $P(\textit{cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2$
- $P(\textit{cavity} \cup \textit{toothache}) = ?$
- $P(\neg \textit{cavity} \mid \textit{toothache}) = P(\neg \textit{cavity} \cap \textit{toothache}) / P(\textit{toothache})$
 $= (0.016 + 0.064) / (0.108 + 0.012 + 0.016 + 0.064)$
 $= 0.4$

Inference using Joint Probability Distribution

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- If you have n binary propositional variables $\rightarrow$ requires 2^n numbers to build Joint Probability Distribution
- $\rightarrow$ Bayesian Network (We want to exploit independences in the domain)

Bayes' Rule:

$$\begin{aligned}P(A \mid B) &= P(A \cap B) / P(B) \\&= P(B \mid A) P(A) / P(B)\end{aligned}$$

Conditioning:

$$\begin{aligned}P(A) &= P(A \mid B) P(B) + P(A \mid \neg B) P(\neg B) \\&= P(A \cap B) + P(A \cap \neg B)\end{aligned}$$

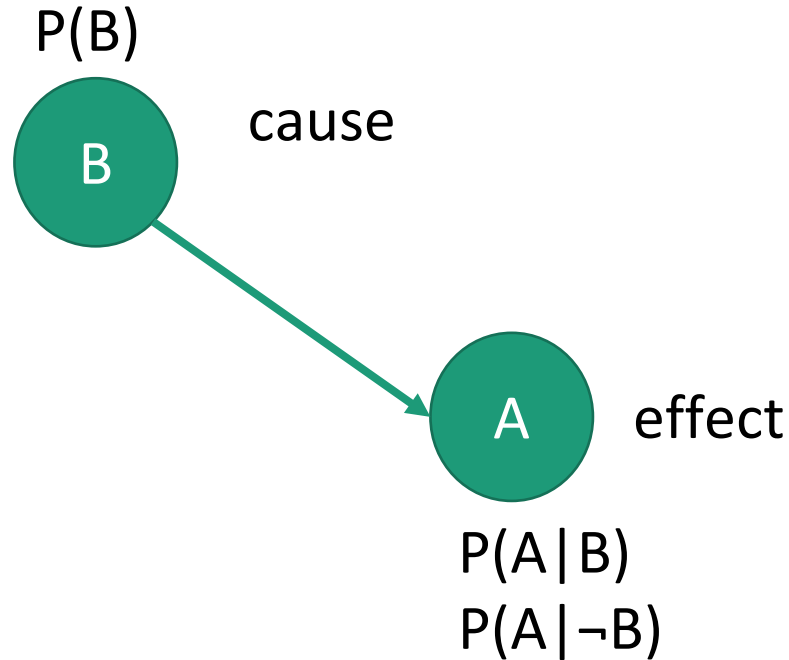
Structure of Bayesian Network

Nodes (variable)

Directed arc

Numerical
Parameters

- Prior probability
- Probability conditional tables



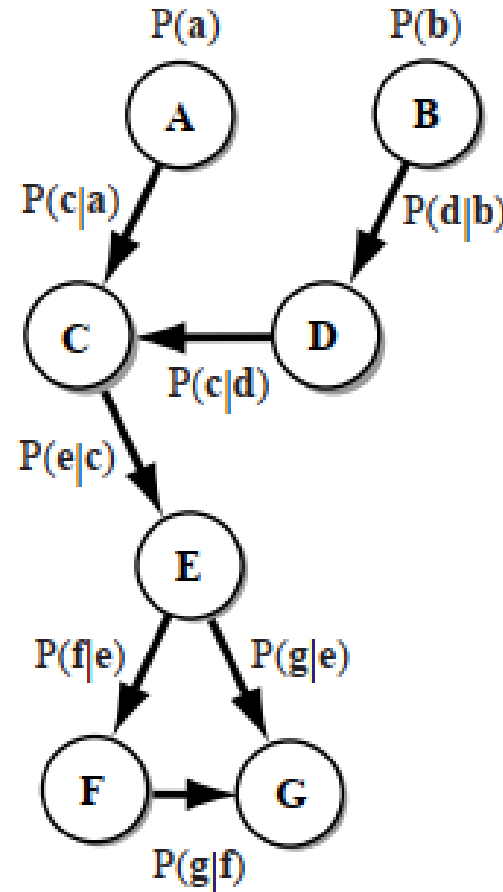
Should be
Directed Acyclic Graph (DAG)

Structure of Bayesian Network

Representation of causal dependencies graphically (Hart et al., 2001)

BN have capability probabilistic reasoning like full joint probability distribution. It can answer any question about the domain.

-- How we exploit Independence?



Bayesian belief nets

Causal networks

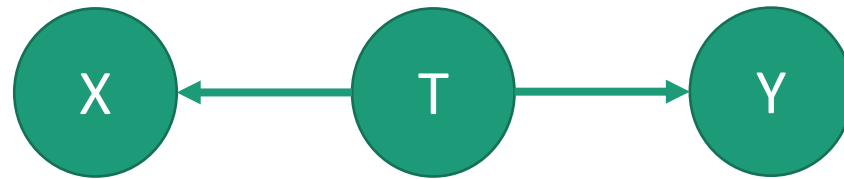
Belief nets

Independence

- A and B are independent iff
 - $P(A \cap B) = P(A) \cdot P(B)$
 - $P(A \mid B) = P(A)$
 - $P(B \mid A) = P(B)$
- Independence is essential for efficient probabilistic reasoning
- A and B are conditionally independent given C iff
 - $P(A \mid B, C) = P(A \mid C)$
 - $P(B \mid A, C) = P(B \mid C)$
 - $P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$

Example of Independence

- X is late (X)
- Traffic Jam (T)
- Y is late (Y)
- None of these propositions are independent of one other
- X and Y are conditionally independent given T

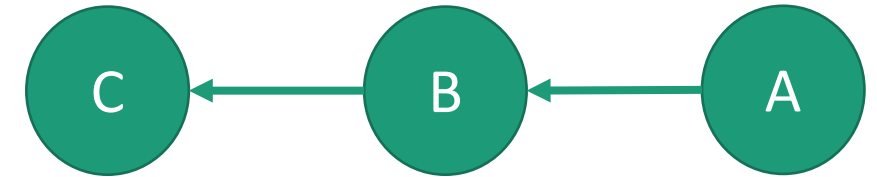
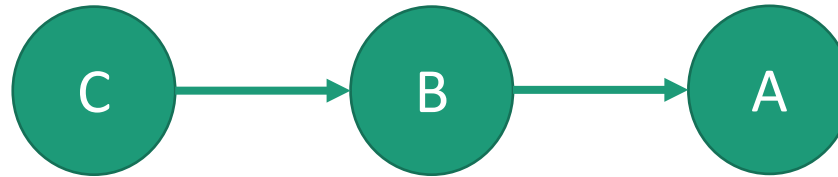


Types of Connections in Bayesian Network

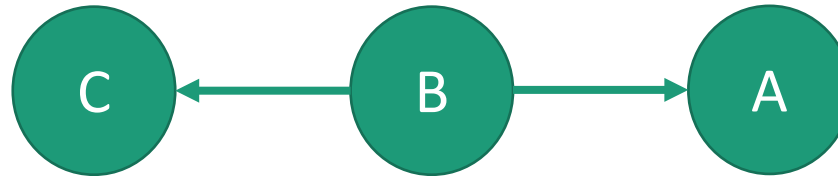
Forward

Backward

Serial

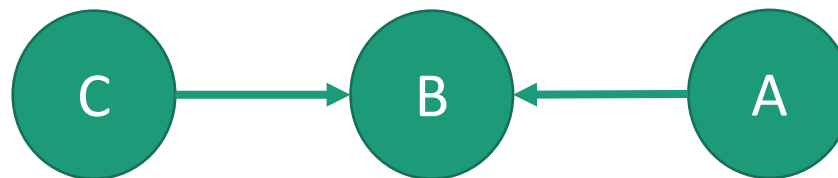


Diverging



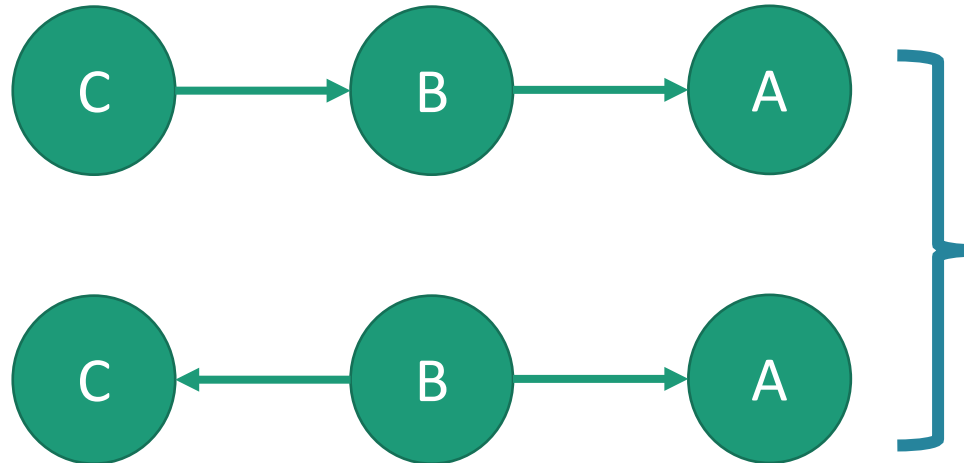
Why? To exploit Independence

Converging



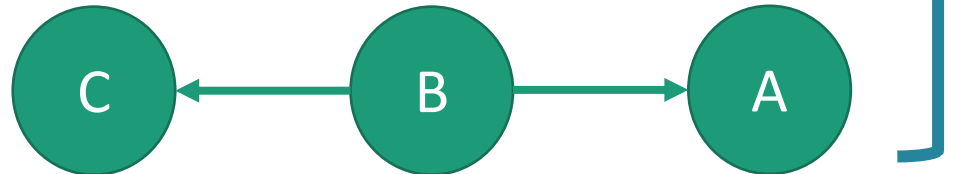
Independence in Connection

Serial



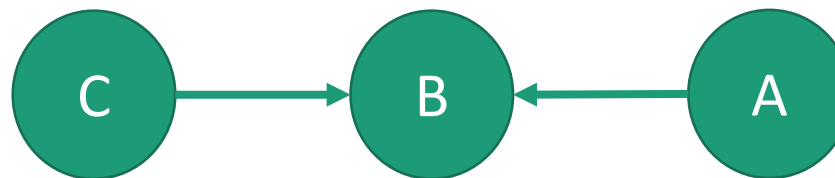
Knowing C will tell us about A, but if we know B, knowing C will tell us nothing about A (C and A conditionally independent or d-separated)

Diverging



Knowing C will tell us nothing about A without knowing B, but if we see evidence about B, C and A becomes dependent

Converging

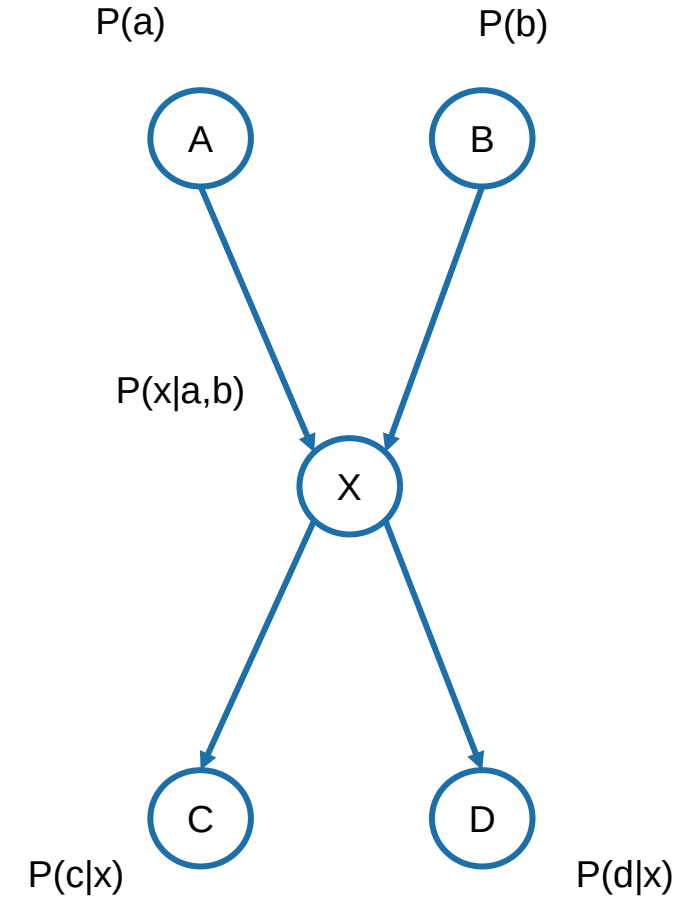


Reduction Number of Probabilities

In a domain with N binary propositional variables (2 possibilities value), one needs 2^N numbers to specify the joint probability distribution. $N=5$: need 32 probabilities

Independence and conditional independence relationships among variables can greatly reduce the number of probabilities that need to be specified in order to define the full joint distribution (Russel & Norvig, 2013)

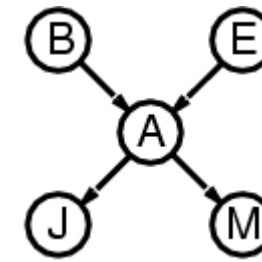
For 5 binary variables with causal networks: need $2+2+8+4+4=20$ probabilities (or 10 with complements).



Bayesian Network as Joint Probability Distribution (chain rule)

The full joint distribution is defined as the product of the local conditional distributions:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i))$$

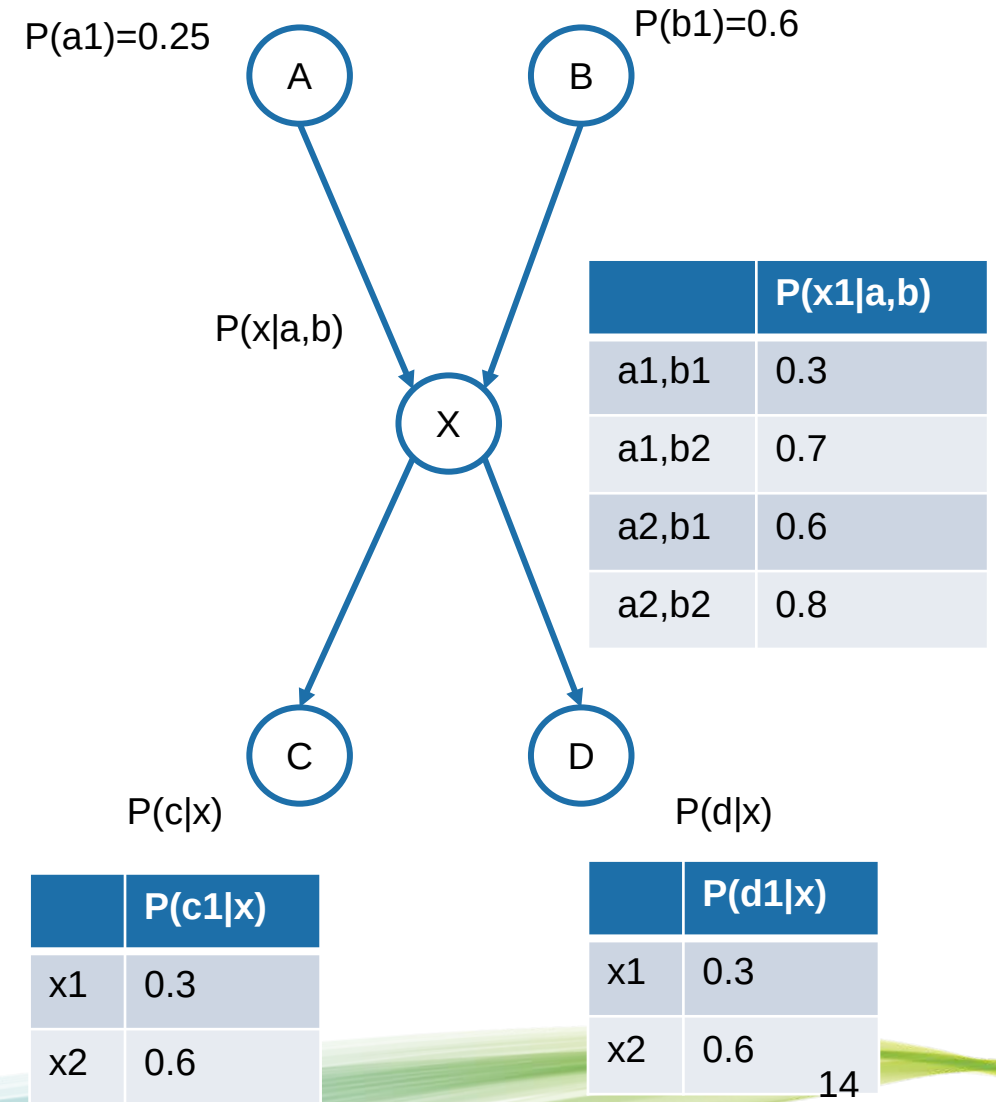


e.g., $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$

$$= P(j | a) P(m | a) P(a | \neg b, \neg e) P(\neg b) P(\neg e)$$

Example

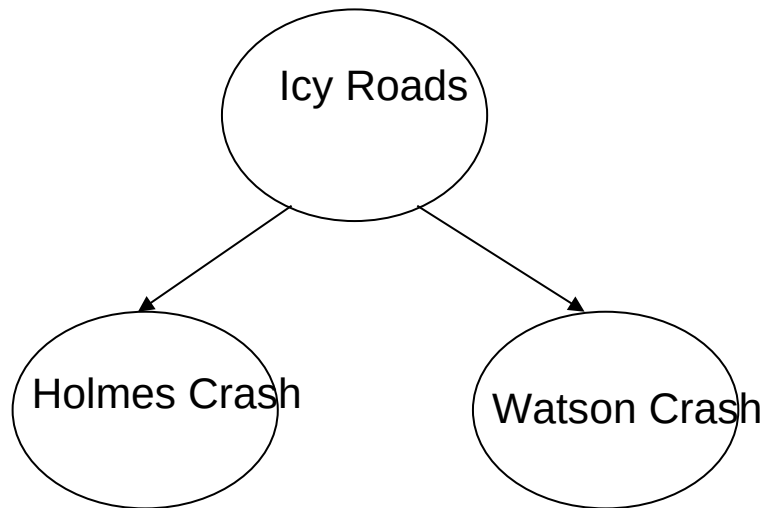
- We can determine the value of any entry in the joint probability.
- $P(a_2, b_1, x_2, c_2, d_1)$
 $= P(a_2)P(b_1)P(x_2|a_2, b_1)P(c_2|x_2)P(d_1|x_2)$
 $= 0.75 * 0.6 * 0.4 * 0.4 * 0.6$
 $= 0.0432$



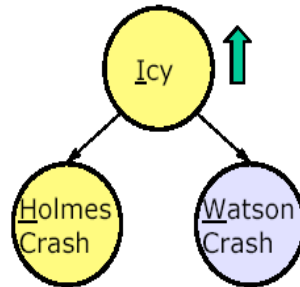
Example of Bayesian Network

Icy Roads

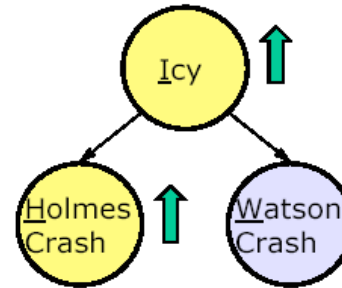
Inspector Smith is waiting for Holmes and Watson, who are driving (separately) to meet him. It is winter. His secretary tells him that Watson has had an accident. He says, "It must be that the roads are icy. I bet that Holmes will have an accident too. I should go to lunch." But, his secretary says, "No, the roads are not icy, look at the window." So, he says, "I guess I better wait for Holmes."



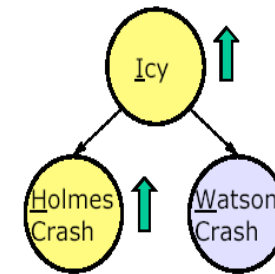
"Causal" Component



"Causal" Component

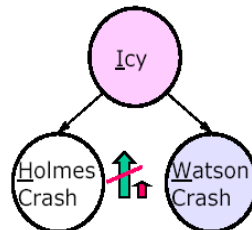


"Causal" Component



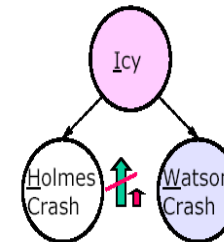
H and W are dependent,

"Causal" Component



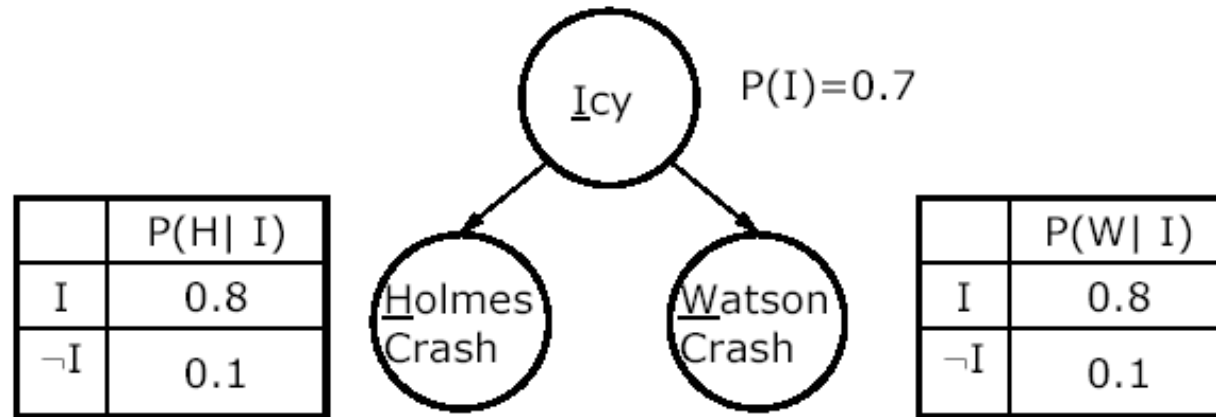
H and W are dependent,

"Causal" Component



H and W are dependent, but conditionally independent given I

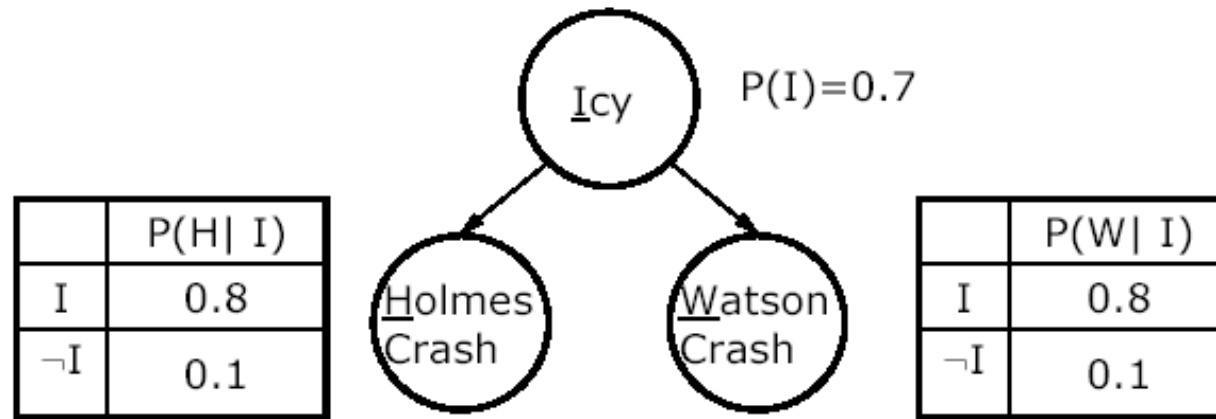
Icy Roads with Numbers



Probability that Watson Crashes:

$$\begin{aligned} P(W) &= P(W|I) P(I) + P(W|\neg I) P(\neg I) \\ &= 0.8 \cdot 0.7 + 0.1 \cdot 0.3 \\ &= 0.56 + 0.03 \\ &= 0.59 \end{aligned}$$

Icy Roads with Numbers

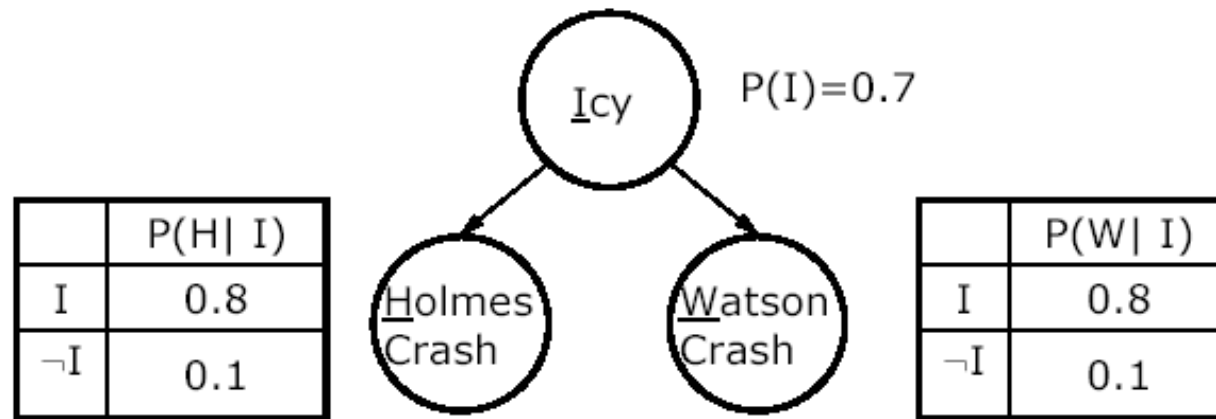


Probability of Icy given Watson (Bayes' Rule):

$$\begin{aligned} P(I | W) &= P(W | I) P(I) / P(W) \\ &= 0.8 \cdot 0.7 / 0.59 \\ &= 0.95 \end{aligned}$$

We started with $P(I) = 0.7$; knowing that Watson crashed raised the probability to 0.95

Icy Roads with Numbers



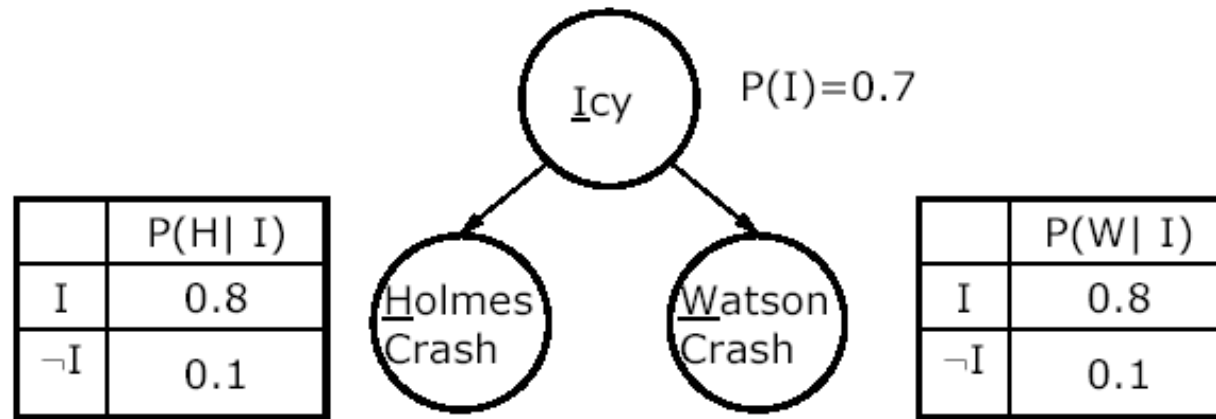
Probability of Holmes given Watson :

$$\begin{aligned} P(H|W) &= P(H|W, I)P(I|W) + P(H|W, \neg I) P(\neg I| W) \\ &= P(H|I)P(I|W) + P(H|\neg I) P(\neg I| W) \\ &= 0.8 \cdot 0.95 + 0.1 \cdot 0.05 \\ &= 0.765 \end{aligned}$$

We started with $P(H) = 0.59$; knowing that

Watson crashed raised the probability to 0.765

Icy Roads with Numbers



Probability of Holmes given Icy and Watson :
 $P(H|W, \neg I) = P(H|\neg I) = 0.1$

H and W are d-separated given I, so H and W are conditionally independent given I

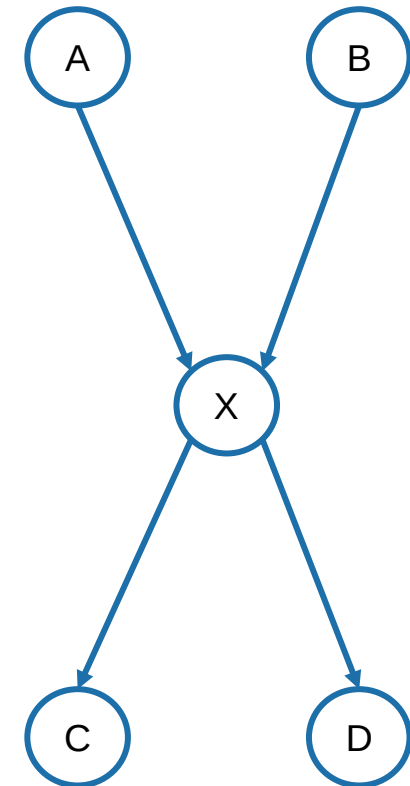
Classification using Bayesian Network

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KK IF – Teknik Informatika – STEI ITB

Inteligensi Artifisial
(*Artificial Intelligence*)

- X represents the fish : x1=salmon and x2=sea bass.
- X is influenced by A and B.
- A represents time of year: a1 = winter, a2 = spring, a3 = summer and a4 = autumn. Probability distribution on A is uniform.
- B represents geographical area where the fish was caught: b1 = north Atlantic and b2 = south Atlantic. The probabilities that any fish came from those areas are 0.6 and 0.4.
- C represents lightness with c1 = light, c2 = medium and c3 = dark
- D represents thickness with d1 = wide and d2 = thin.



The probability that the fish was caught in the summer in the north Atlantic and is a sea bass that is dark and thin.



The probability that the fish was caught in the summer (**a3**) in the north Atlantic (**b1**) and is a sea bass (**x2**) that is dark (**c3**) and thin (**d2**).



$P(a3, b1, x2, c3, d2)$

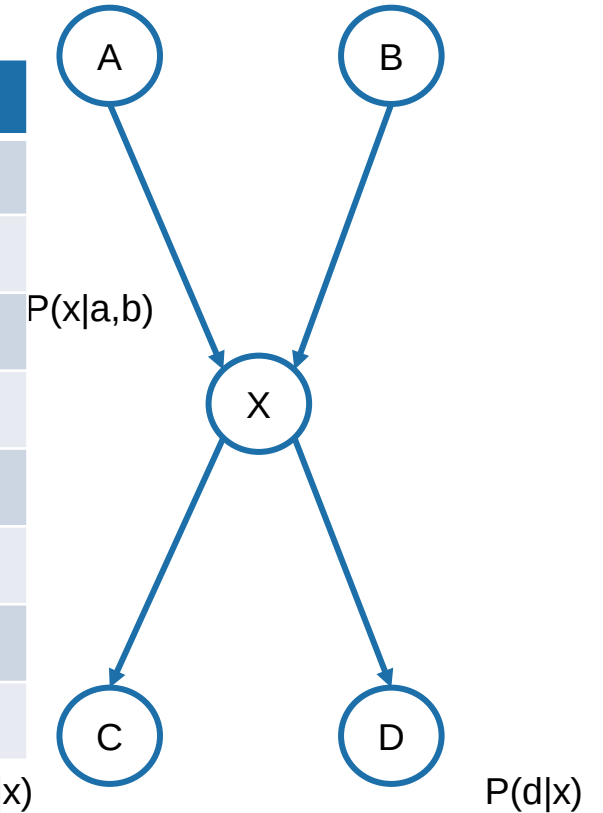
Inference: Example

$$P(a_3, b_1, x_2, c_3, d_2) = P(a_3)P(b_1)P(x_2|a_3, b_1)P(c_3|x_2)P(d_2|x_2) = 0.012$$

P(a)			
P(a1)	P(a2)	P(a3)	P(a4)
0.25	0.25	0.25	0.25

P(b)
P(b1)
0.6

	P(x1 a,b)
a1,b1	0.3
a1,b2	0.7
a2,b1	0.6
a2,b2	0.8
a3,b1	0.4
a3,b2	0.1
a4,b1	0.2
a4,b2	0.3



	P(c1 x)	P(c2 x)	P(c3 x)
x1	0.6	0.2	0.2
x2	0.2	0.3	0.5

	P(d1 x)
x1	0.3
x2	0.6

Classification

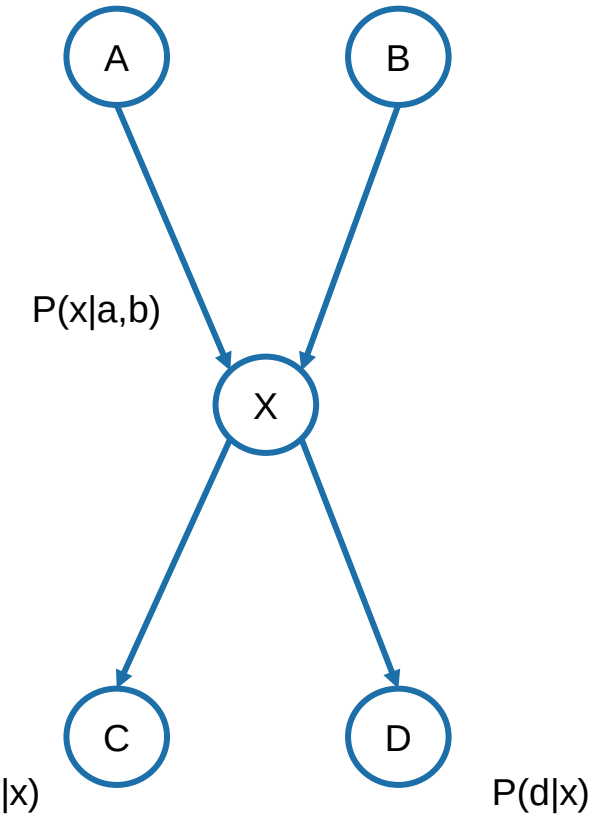
Classify the fish that is light (c1) and caught in the south Atlantic (b2), but we do not know what time of year the fish was caught nor its thickness.

Maximum a posterior probability:
 $P(x1|c1,b2)$ vs $P(x2|c1,b2)$

P(a)			
P(a1)	P(a2)	P(a3)	P(a4)
0.25	0.25	0.25	0.25

P(b)
P(b1)
0.6

	P(x1 a,b)
a1,b1	0.3
a1,b2	0.7
a2,b1	0.6
a2,b2	0.8
a3,b1	0.4
a3,b2	0.1
a4,b1	0.2
a4,b2	0.3



	P(c1 x)	P(c2 x)	P(c3 x)
x1	0.6	0.2	0.2
x2	0.2	0.3	0.5

	P(d1 x)
x1	0.3
x2	0.6

Classification (2)

Q: query

e: evidence of all variables

$$P(Q|e) = P(q,e)/P(e) = \alpha P(Q,e)$$

$$P(x_1|c_1,b_2) = P(x_1,c_1,b_2)/P(c_1,b_2)$$

$$= \alpha \sum P(x_1, \mathbf{a}, b_2, c_1, \mathbf{d})$$

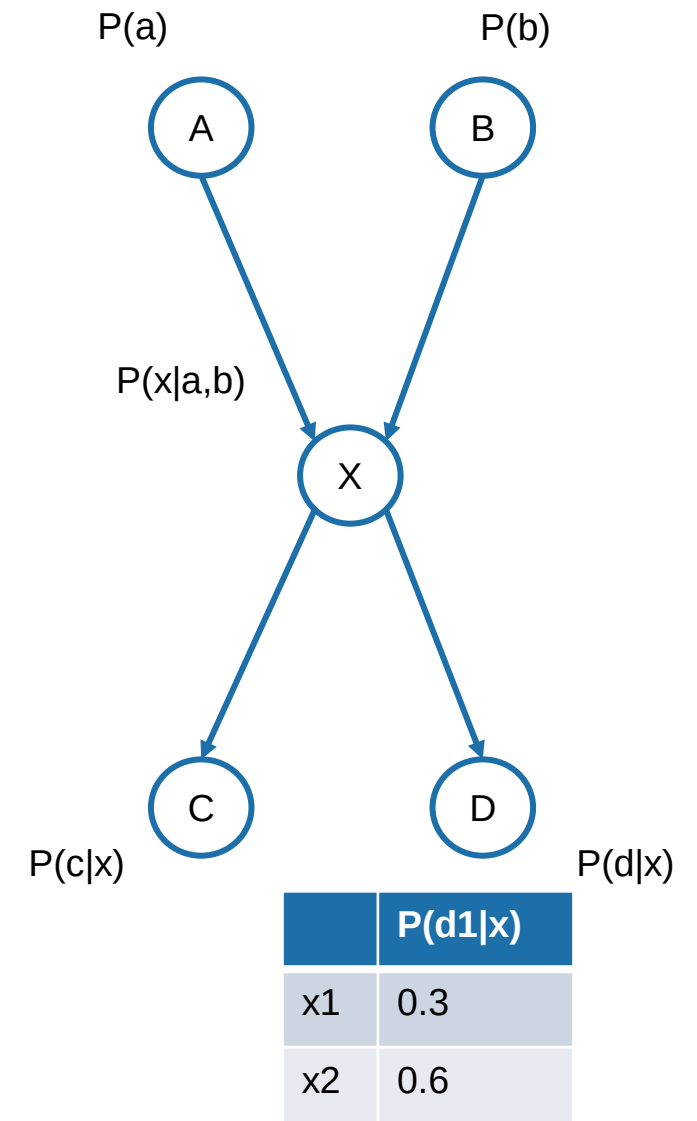
$$= \alpha \sum P(\mathbf{a}).P(b_2).P(x_1|\mathbf{a},b_2).P(c_1|x_1).P(\mathbf{d}|x_1)$$

$$= \alpha P(b_2).P(c_1|x_1) \sum P(\mathbf{a}).P(x_1|\mathbf{a},b_2).P(\mathbf{d}|x_1)$$

$$= \alpha P(b_2).P(c_1|x_1) [\sum P(\mathbf{a}).P(x_1|\mathbf{a},b_2)] [\sum P(\mathbf{d}|x_1)] = \alpha$$

$$\begin{aligned} &P(\mathbf{a}_1).P(x_1|\mathbf{a}_1,b_2) + \\ &P(\mathbf{a}_2).P(x_1|\mathbf{a}_2,b_2) + \\ &P(\mathbf{a}_3).P(x_1|\mathbf{a}_3,b_2) + \\ &P(\mathbf{a}_4).P(x_1|\mathbf{a}_4,b_2) \end{aligned}$$

$$P(d_1|x_1) + P(d_2|x_1) = 1.0$$



Classification (3)

$$P(x_1|c_1,b_2)=P(x_1,c_1,b_2)/P(c_1,b_2) \\ =\alpha P(b_2).P(c_1|x_1).[\sum P(\mathbf{a}).P(x_1|\mathbf{a},b_2) \\]. [\sum P(\mathbf{d}|x_1)]= \alpha 0.114$$

$$P(x_2|c_1,b_2)=P(x_2,c_1,b_2)/P(c_1,b_2) \\ =\alpha P(b_2).P(c_1|x_2) [\sum P(\mathbf{a}). \\ P(x_2|\mathbf{a},b_2)][\sum P(\mathbf{d}|x_2)]= \alpha 0.042$$

Normalize:

$$P(x_1|c_1,b_2)=0.73$$

$$P(x_2|c_1,b_2)=0.27$$

Decision: x_1 =salmon

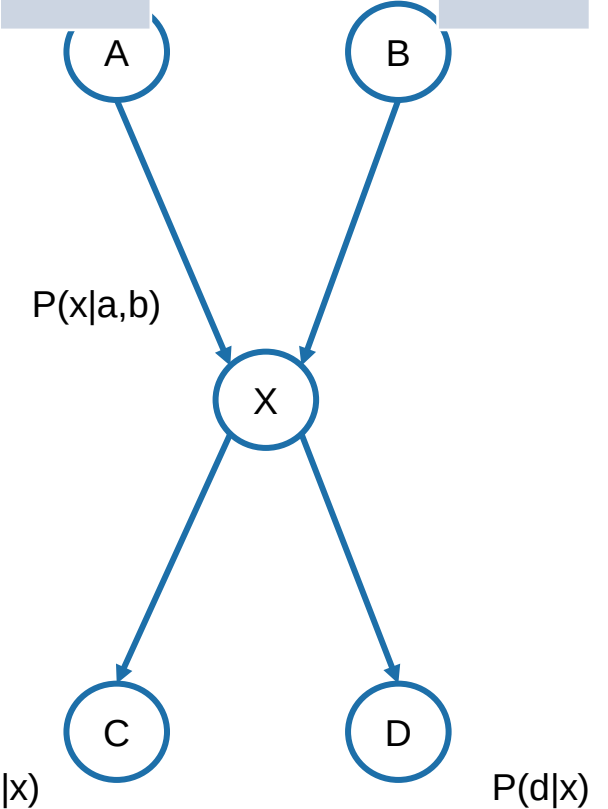
P(a1)	P(a2)	P(a3)	P(a4)
0.25	0.25	0.25	0.25

	P(x1 a,b)
a1,b1	0.3
a1,b2	0.7
a2,b1	0.6
a2,b2	0.8
a3,b1	0.4
a3,b2	0.1
a4,b1	0.2
a4,b2	0.3

P(a)

P(b)

P(b1)
0.6



	P(c1 x)	P(c2 x)	P(c3 x)
x1	0.6	0.2	0.2
x2	0.2	0.3	0.5

	P(d1 x)
x1	0.3
x2	0.6

Knowledge Representation and Reasoning

Learning from Data

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KK IF – Teknik Informatika – STEI ITB

Inteligensi Buatan
(*Artificial Intelligence*)

Problems in BN constructed by human expert

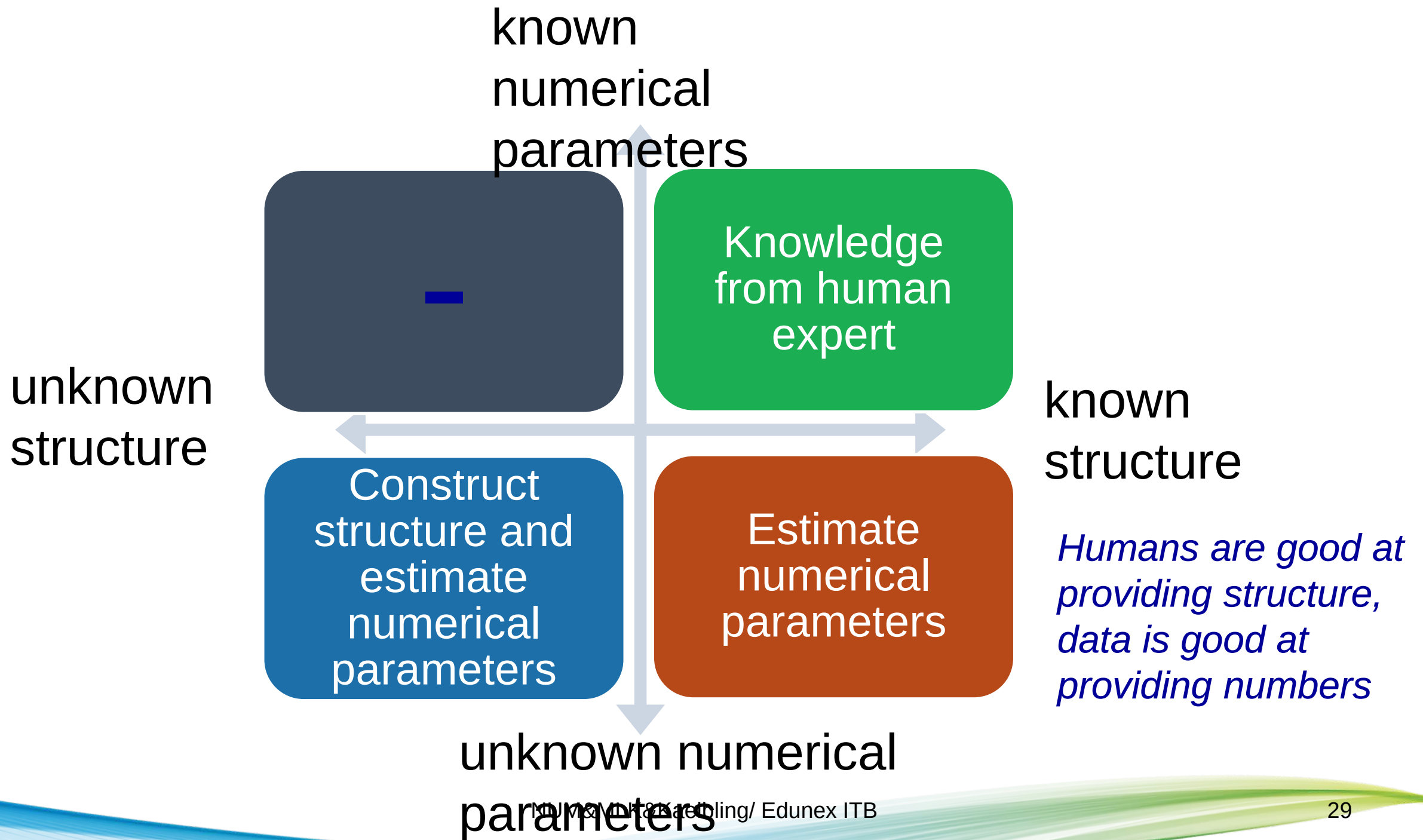
Knowledge acquisition bottleneck problem

Knowledge elicitation problem: slow speed, and inability of expert to express the knowledge they possess.

Very difficult in getting reliable probability estimates.

Ideally, determine probability by data distribution, experience, and assumption (subjective).

What is difference between $P(a)=0.5$ and $P(a)=0.6$?



Given structure
with m nodes

Given a data set

$D =$
 $\{ \langle v_1^1, \dots, v_m^1 \rangle, \dots, \langle v_1^k, \dots, v_m^k \rangle \}$

Count $\#(V_i=T), \#(V_i=F),$
 $\#(V_i=T, V_j=T), \#(V_i=T, V_j=F)$

Variable V_i with no parent
 $P(V_i) \approx \frac{\#(V_i = T)}{k}$

Variable V_i with parent V_j
 $P(V_i|V_j) \approx \frac{\#(V_i = T, V_j = T)}{\#(V_j = T)}$
 $P(V_i|\neg V_j) \approx \frac{\#(V_i = T, V_j = F)}{\#(V_j = F)}$

Smoothing: Avoid Probability = 0

Variable V_i with no parent

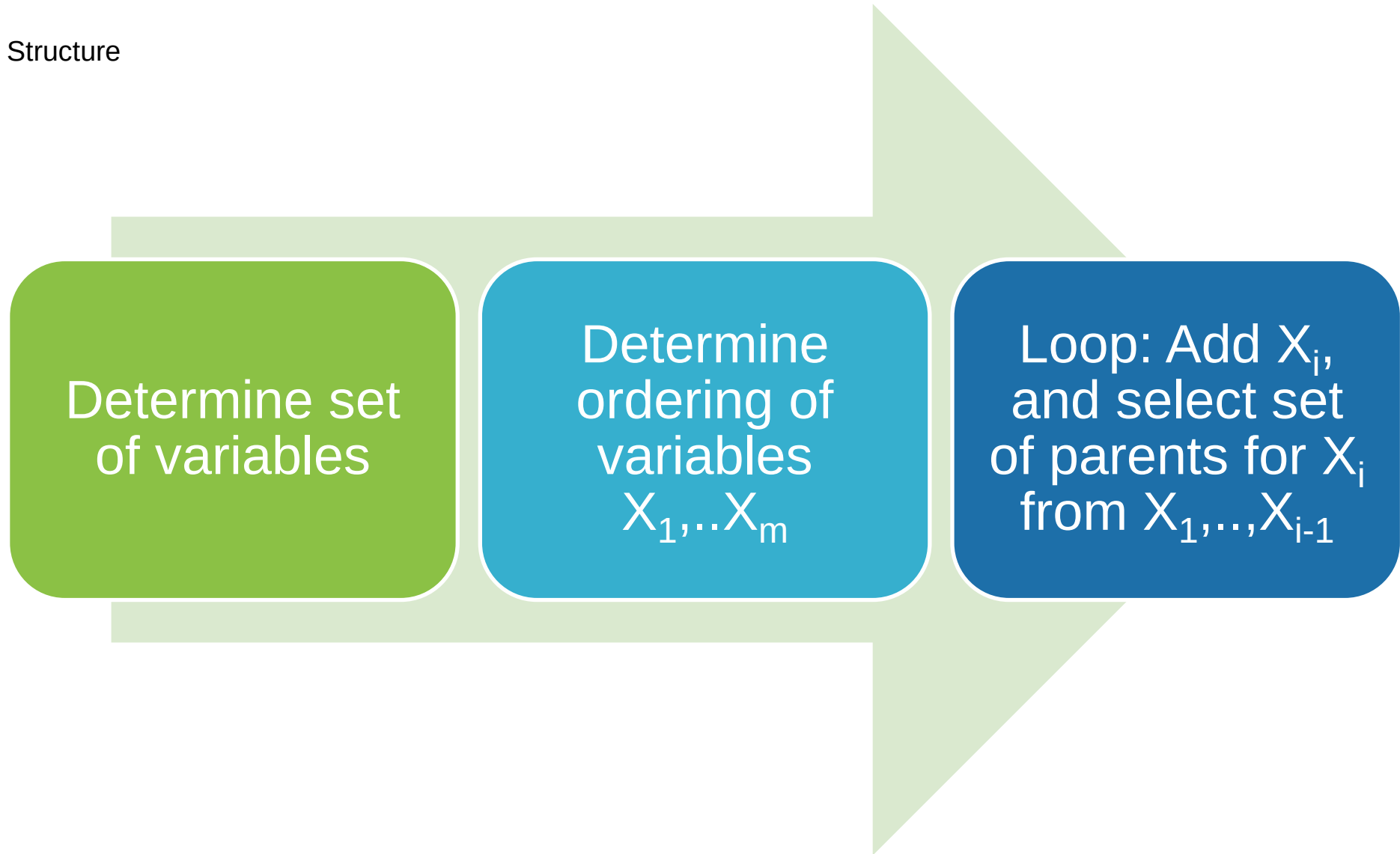
$$P(V_i) \approx \frac{\#(V_i = T) + 1}{k + 2}$$

Variable V_i with parent V_j

$$P(V_i|V_j) \approx \frac{\#(V_i = T, V_j = T) + 1}{\#(V_j = T) + 2}$$

$$P(V_i|\neg V_j) \approx \frac{\#(V_i = T, V_j = F) + 1}{\#(V_j = F) + 2}$$

Construct BN Structure



Construct BN Structure: Example

Based on causal knowledge:

- Set of variables: A,B,C,D,X
- Ordering of variables: A,B,C,D,X
- Add A: no parent.
- Add B: is A parent of B ?
- Add C: parent of C ?
- Add D: parent of D ?
- Add X: parent of X ?

Based on data:

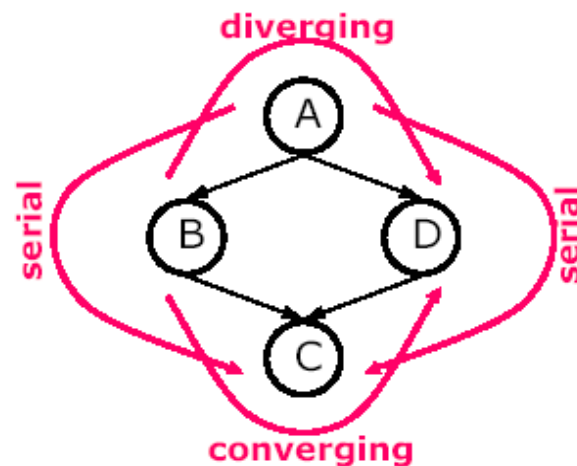
- Set of variables: A,B,C,D,X
- Ordering of variables: A,B,C,D,X
- Add A: no parent.
- Add B: $P(B|A)=P(B)$?
- Add C:
 - $P(C|A)=P(C)$?
 - $P(C|B)=P(C)$?
 - $P(C|A,B)=P(C|A)$?
 - $P(C|A,B)=P(C|B)$?
 - $P(C|A,B)=P(C)$?
- Etc.

D Separation

Two variables A and B are d-separated iff for every path between them, there is an intermediate variable V such that either

- The connection is serial or diverging and V is known
- The connection is converging and neither V nor any descendant is instantiated

Two variables are d-connected iff they are not d-separated



- A-B-C: serial, blocked when B is known, connected otherwise
- A-D-C: serial, blocked when D is known, connected otherwise
- B-A-D: diverging, blocked when A is known, connected otherwise
- B-C-D: converging, blocked when C has no evidence, connected otherwise

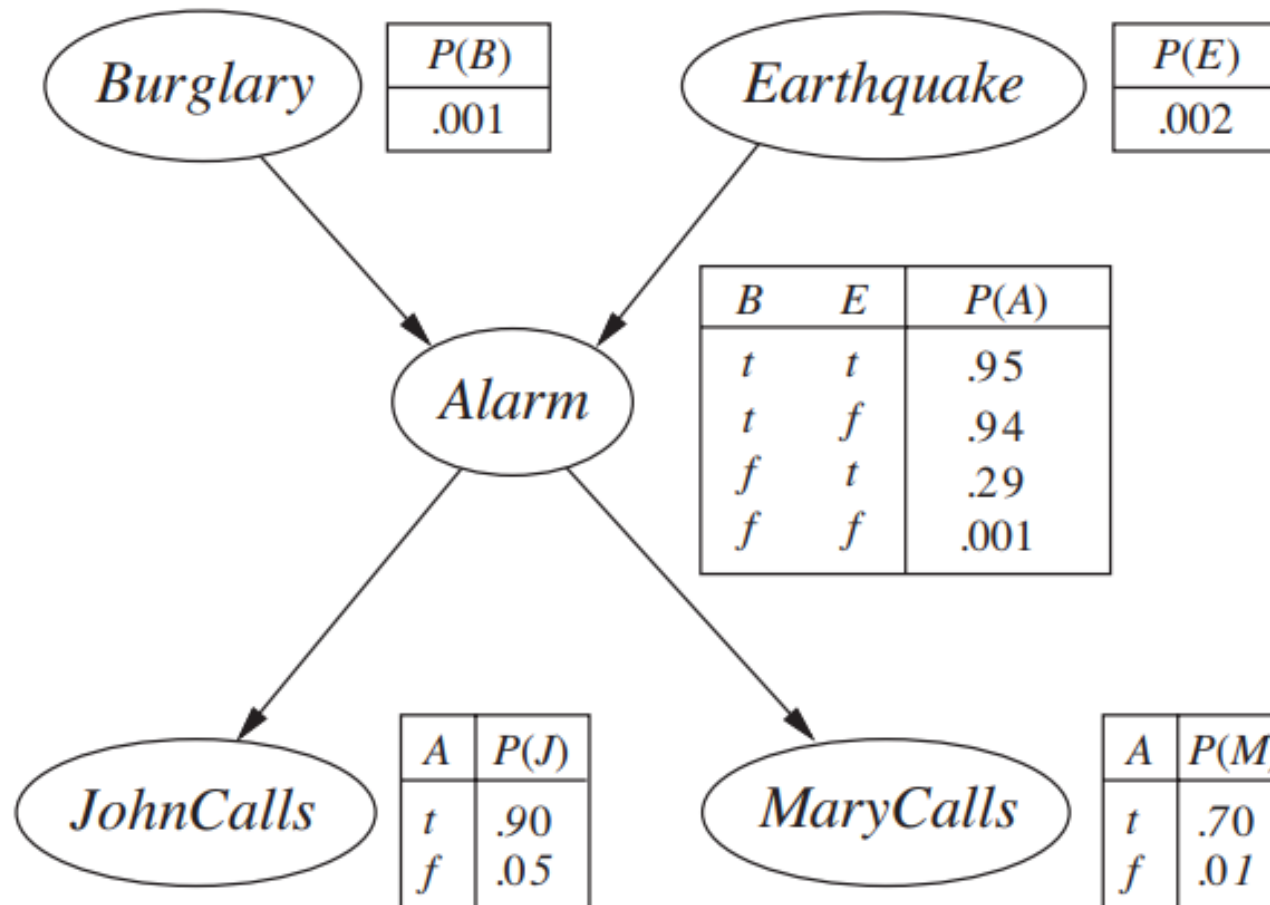
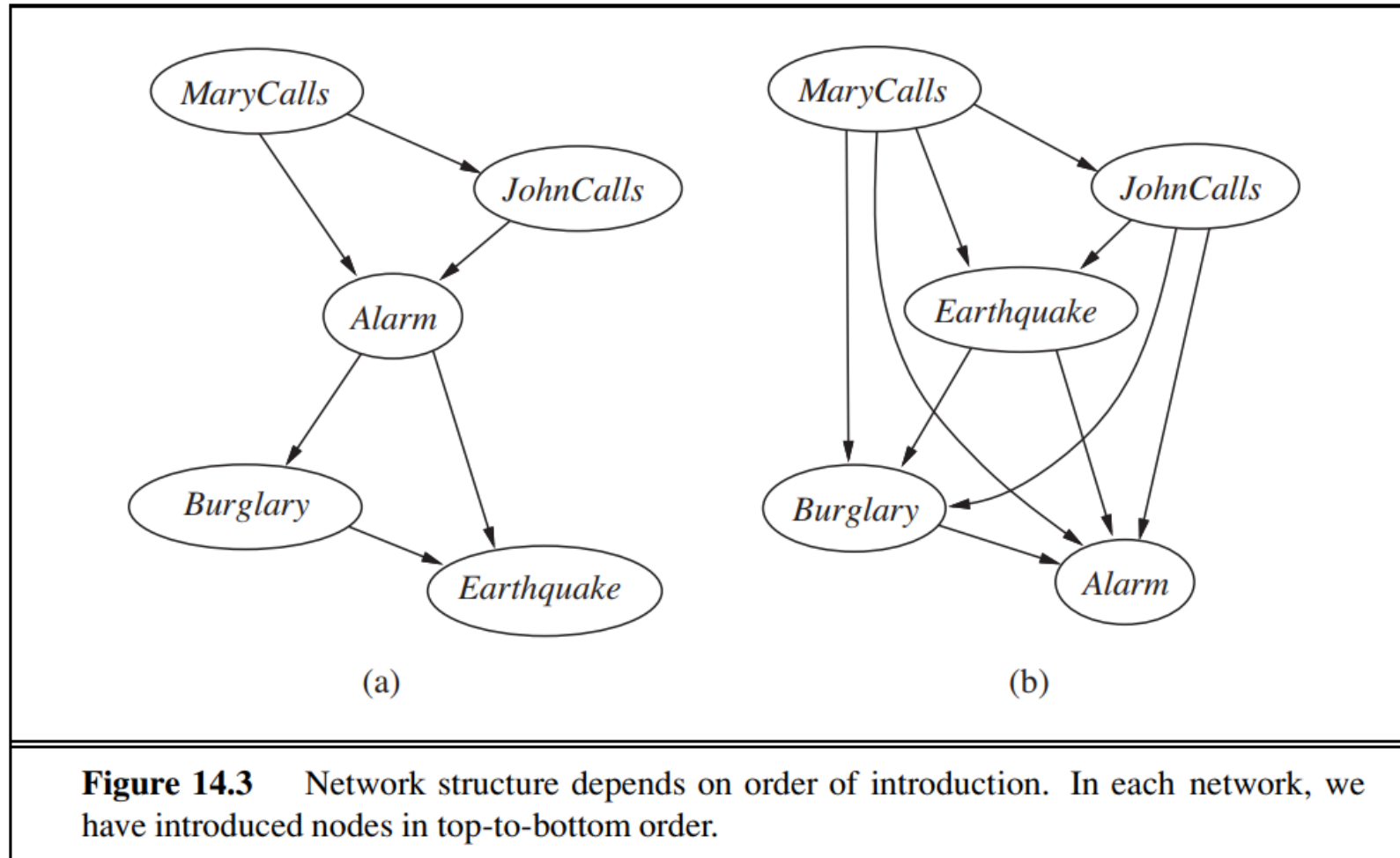


Figure 14.2 A typical Bayesian network, showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B , E , A , J , and M stand for *Burglary*, *Earthquake*, *Alarm*, *JohnCalls*, and *MaryCalls*, respectively.

Variable Ordering Influences Structure

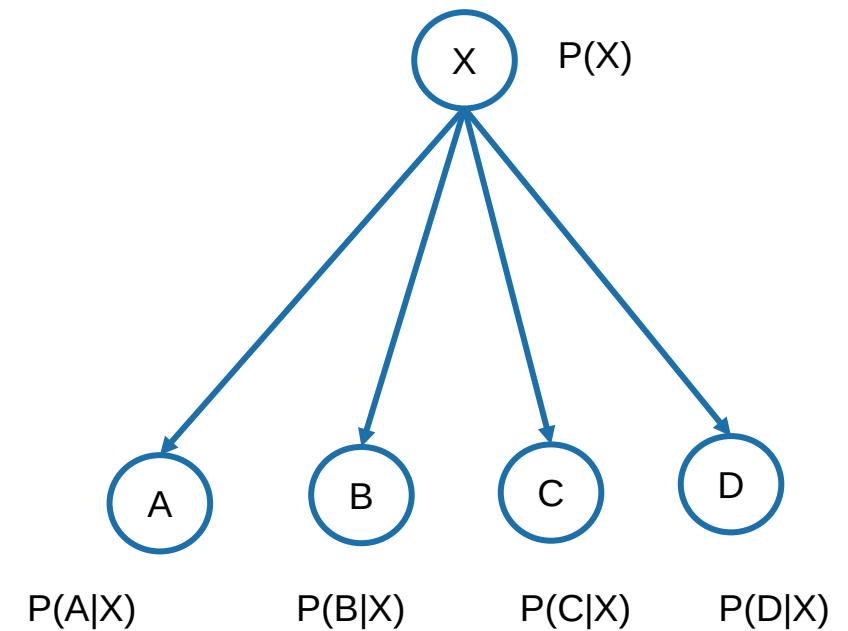


Russel & Norvig (2010). AI.

Construct BN Structure: Example 2

Based on causal knowledge:

- Set of variables: X, A, B, C, D
- Ordering of variables: X, A, B, C, D
- Add X : no parent.
- Add A : $\text{parent}(A)=X$.
- Add B : $\text{parent}(B)=X$. $P(B|A,X)=P(B|X)$
- Add C : $\text{parent}(C)=X$. $P(C|A,B,X)=P(C|X)$
- Add D : $\text{parent}(D)=X$. $P(D|A,B,C,X)=P(D|X)$

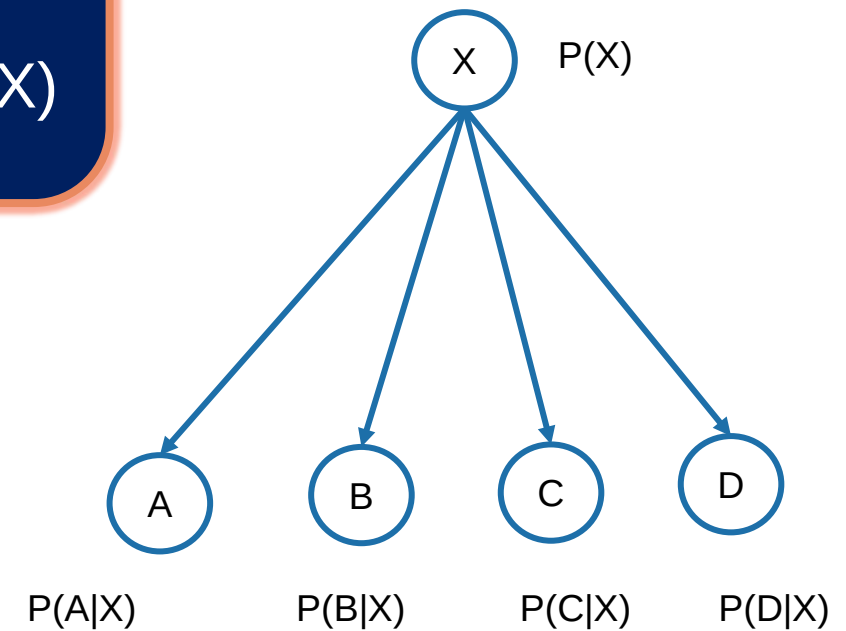


Naive Bayes as “Special” Bayesian Network

What is the Bayes Network for $X_1, \dots, X_n$ with NO assumed conditional independencies?

Probability Model of $P(X)$ and $P(a_i|X)$
 $P(X, A, B, C, D) = P(X) \cdot P(A|X) \cdot P(B|X) \cdot P(C|X) \cdot P(D|X)$

Classification:
Find the maximum $P(X | A, B, C, D)$



Example: Play Tennis Dataset

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes

outlook	temp.	humidity	windy	play
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Frequency of
(sunny|yes) $\rightarrow$ 2

Frequency of
(overcast|yes) $\rightarrow$ 4

Frequency of
(hot|yes) $\rightarrow$ 2

...

Frequency of
class 'yes' $\rightarrow$ 9

Frequency of
(sunny|no) $\rightarrow$ 3

Frequency of
(overcast|no) $\rightarrow$ 0

Frequency of
(hot|no) $\rightarrow$ 2

Frequency of
class 'no' $\rightarrow$ 5

Example: Play Tennis Dataset

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes

outlook	temp.	humidity	windy	play
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Frequency of
(sunny|yes) →

Frequency of
(sunny|no) →

	outlook		temperature				humidity				windy		play	
	yes	no	yes	no	yes	no	yes	no	yes	no	yes	no	yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5	
overcast	4	0	mild	4	2	normal	6	1	true	3	3			
rainy	3	2	cool	3	2									

Frequency of
class 'yes' → 9

Frequency of
class 'no' → 5

Example: Play Tennis Dataset

Frequency

outlook		temperature		humidity		windy		play	
yes	no	yes	no	yes	no	yes	no	yes	no
sunny	2 3	hot	2 2	high	3 4	false	6 2	9	5
overcast	4 0	mild	4 2	normal	6 1	true	3 3		
rainy	3 2	cool	3 1						

$P(a_i|v_j)$

outlook		temperature		humidity		windy	
yes	no	yes	no	yes	no	yes	no
sunny	2/9 3/5	hot	2/9 2/5	high	3/9 4/5	false	6/9 2/5
overcast	4/9 0/5	mild	4/9 2/5	normal	6/9 1/5	true	3/9 3/5
rainy	3/9 2/5	cool	3/9 1/5				

$P(v_j)$

play	
yes	no
9/14	5/14

Probability Model

Classify New Instance: <Sunny, Cool, High, True>

$P(a_i|v_j)$

outlook		temperature		humidity		windy	
	yes no		yes no		yes no		yes no
sunny	2/9 3/5	hot	2/9 2/5	high	3/9 4/5	false	6/9 2/5
overcast	4/9 0/5	mild	4/9 2/5	normal	6/9 1/5	true	3/9 3/5
rainy	3/9 2/5	cool	3/9 1/5				

$P(v_j)$

play	
	yes no
	9/14 5/14

$$P(v_j | a_1, a_2, \dots, a_n) = P(v_j) \cdot \prod_i P(a_i | v_j)$$

$P(\text{yes}|\text{sunny, cool, high, true})$

$= P(\text{yes}) \cdot P(\text{sunny}|\text{yes}) \cdot P(\text{cool}|\text{yes}) \cdot P(\text{high}|\text{yes}) \cdot P(\text{true}|\text{yes})$

$= 9/14 \cdot 2/9 \cdot 3/9 \cdot 3/9 \cdot 3/9 = 0.0053$

$P(\text{no}|\text{sunny, cool, high, true})$

$= P(\text{no}) \cdot P(\text{sunny}|\text{no}) \cdot P(\text{cool}|\text{no}) \cdot P(\text{high}|\text{no}) \cdot P(\text{true}|\text{no})$

$= 5/14 \cdot 3/5 \cdot 1/5 \cdot 4/5 \cdot 3/5 = 0.0206$

THANK YOU

Clustering: Unsupervised Learning

IF-3170 Inteligensi Artifisial
Teknik Informatika ITB

Fariska Z. Ruskanda, S.T., M.T.
(fariska@informatika.org)

KK IF -Teknik Informatika - STEI ITB

01 Clustering: What & Why?

Outline

Clustering: What?

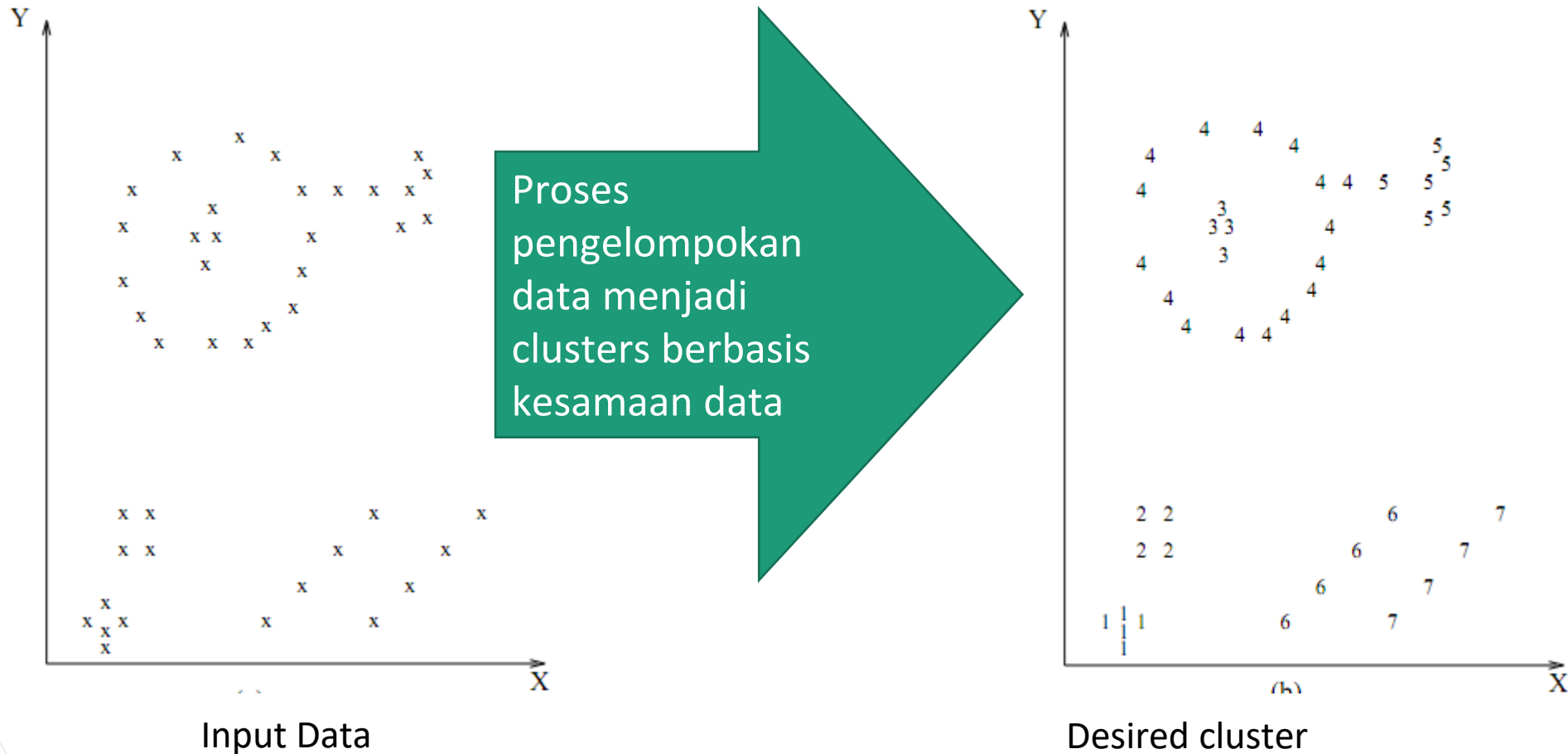
Tahapan Clustering

Representasi
Cluster

Clustering: Why?

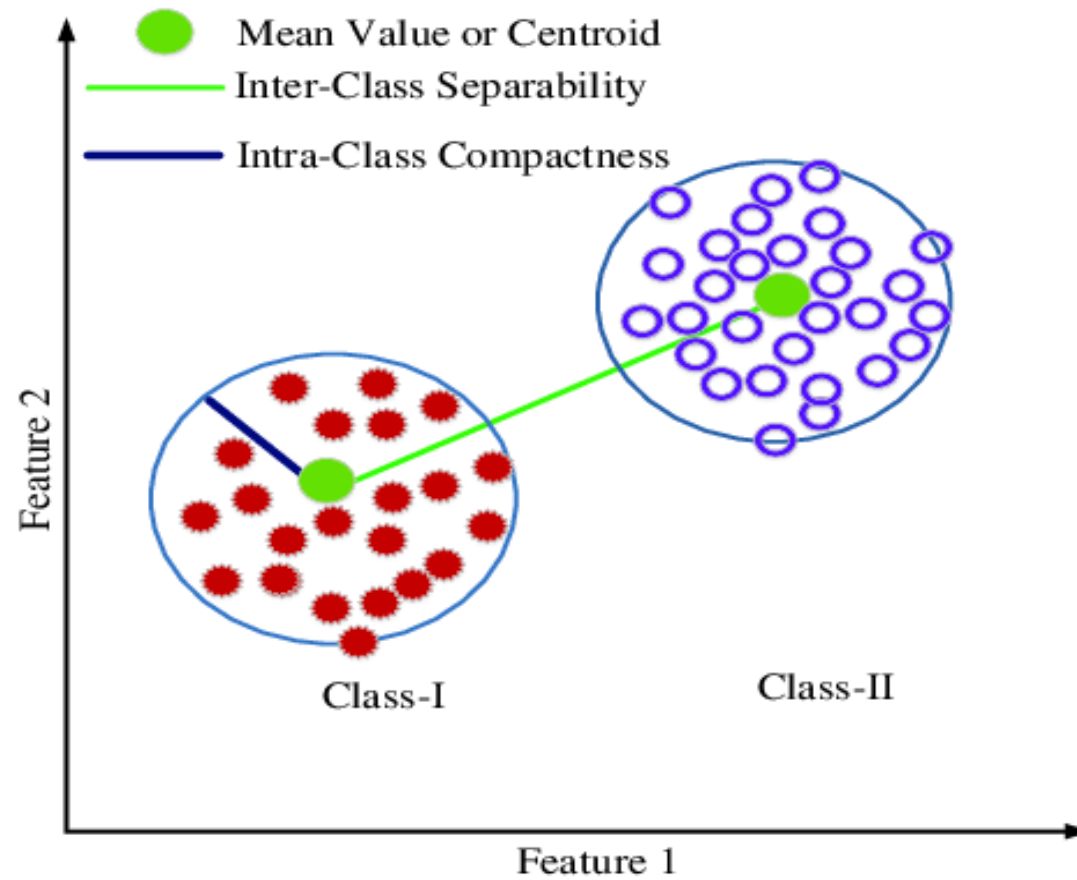
Kategori Metode

Clustering: What ?



Unsupervised learning = learning from raw data

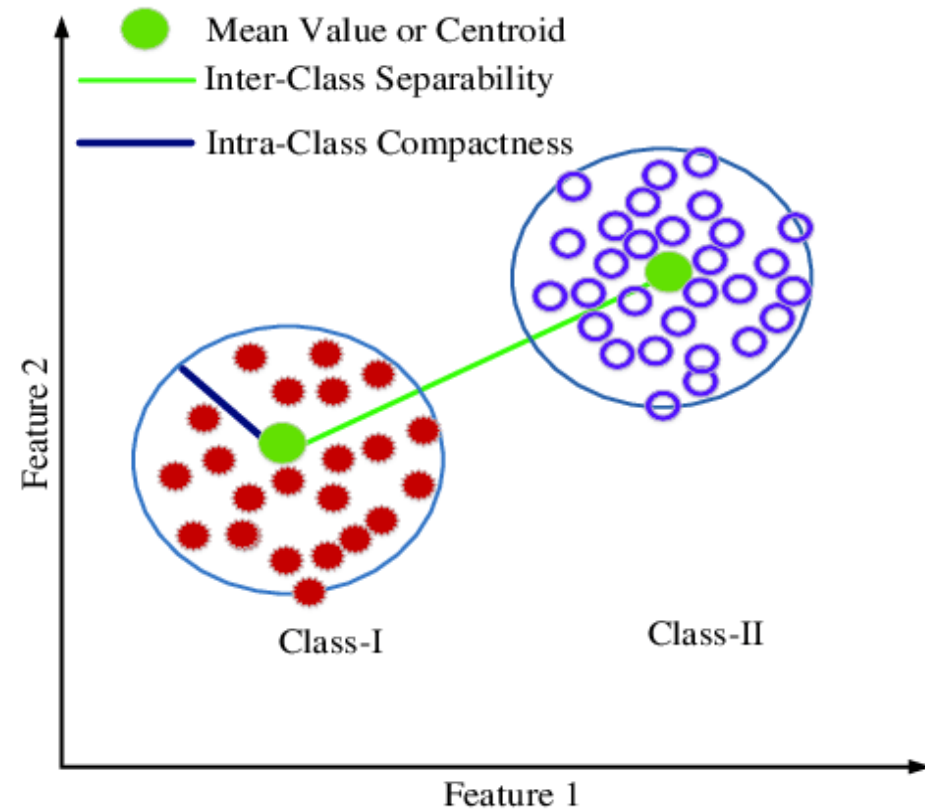
Clustering:



https://www.researchgate.net/figure/The-concept-of-intra-class-compactness-and-inter-class-separability-in-a-two-dimension_fig3_325095062

Clustering: Finding natural groups

- High intra-cluster similarity
 - Data pd cluster yang sama harus semirip mungkin
- Low inter-cluster similarity
 - Data pd cluster yang berbeda harus sejauh mungkin
- Pengukuran kemiripan dan jarak harus jelas dan punya semantik praktis (sesuai domain)

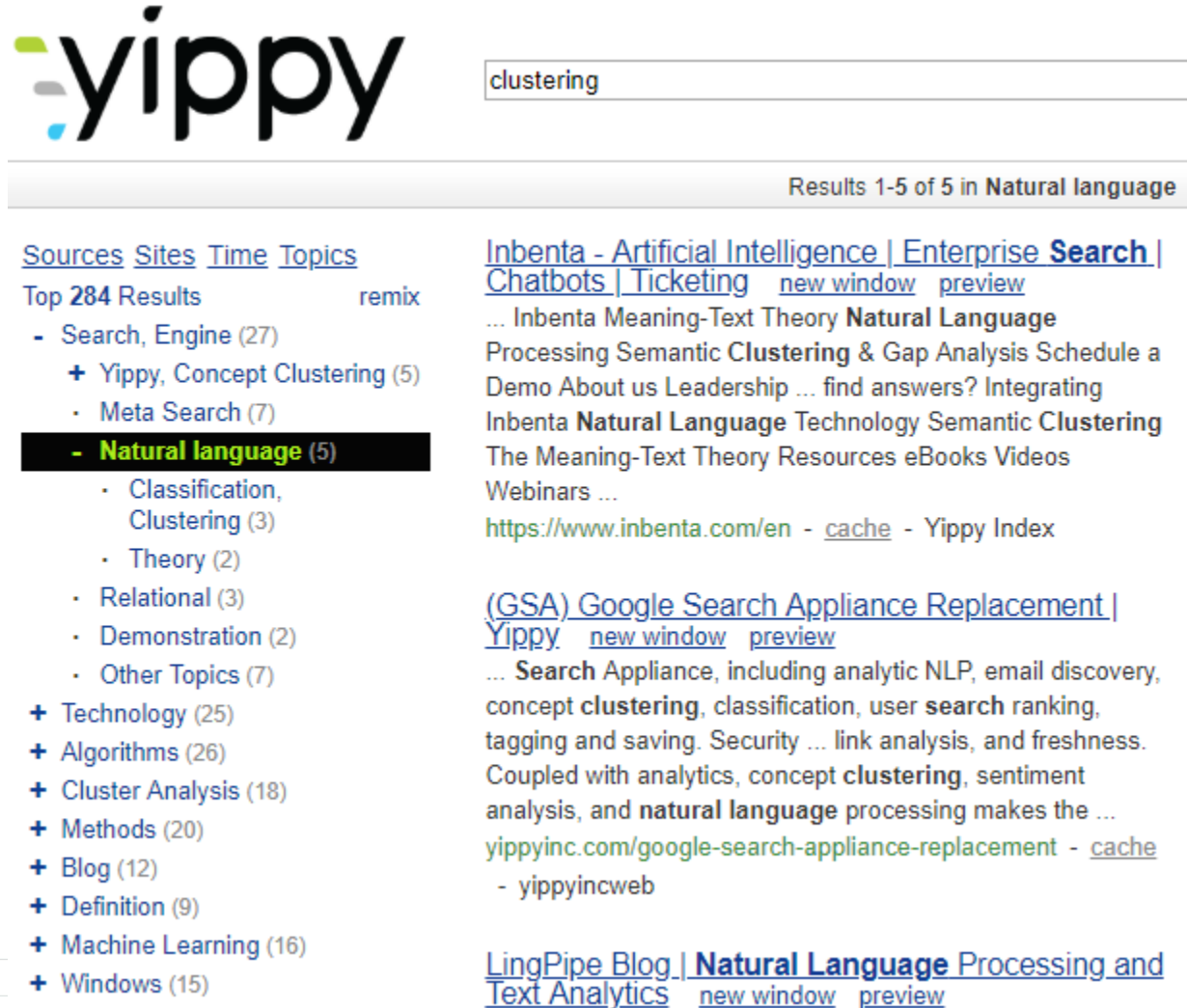


https://www.researchgate.net/figure/The-concept-of-intra-class-compactness-and-inter-class-separability-in-a-two-dimension_fig3_325095062

Clustering: Why ?

- Data discovery (cluster = struktur internal data)
 - Contoh: search engine, news aggregator, gen
- Tujuan awalnya partisi / pengelompokan
 - Contoh: segmentasi pasar, segmentasi gambar
- Bagian dari teknik lainnya
 - Contoh: peringkasan berbasis clustering

Why: Clustering pada Search Engine



yippy

Results 1-5 of 5 in Natural language

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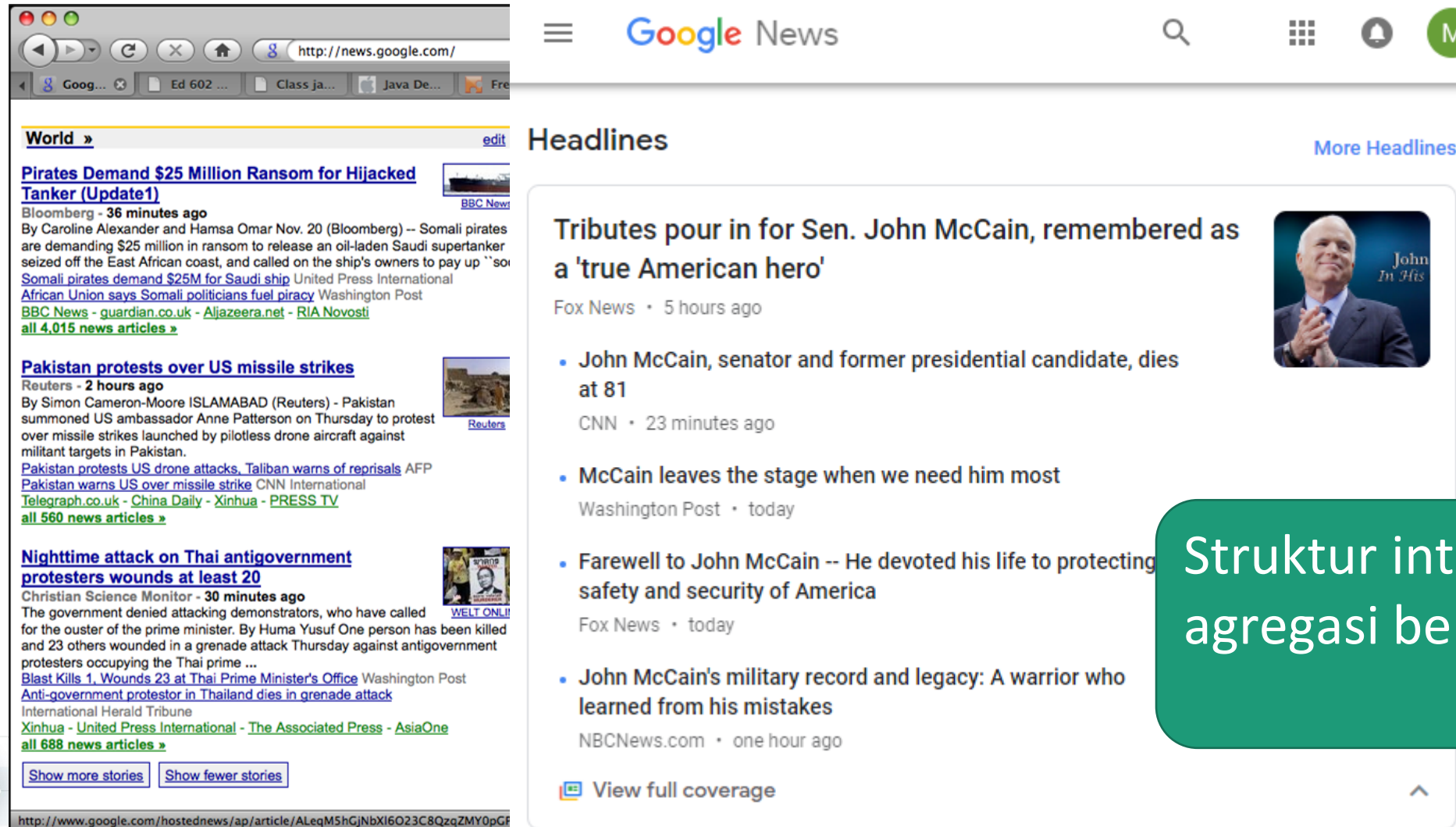
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Struktur internal
hasil pencarian



Why: Clustering pada News Aggregator



The screenshot shows the Google News homepage. The main section displays headlines under the 'Headlines' tab. A prominent headline reads 'Tributes pour in for Sen. John McCain, remembered as a 'true American hero'' from Fox News, 5 hours ago. Below it, other headlines include 'John McCain, senator and former presidential candidate, dies at 81' from CNN, 23 minutes ago; 'McCain leaves the stage when we need him most' from Washington Post, today; 'Farewell to John McCain -- He devoted his life to protecting safety and security of America' from Fox News, today; and 'John McCain's military record and legacy: A warrior who learned from his mistakes' from NBCNews.com, one hour ago. A 'View full coverage' link is at the bottom of the main section. On the left sidebar, under the 'World' tab, there are three news items: 'Pirates Demand \$25 Million Ransom for Hijacked Tanker (Update1)' from Bloomberg, 36 minutes ago; 'Pakistan protests over US missile strikes' from Reuters, 2 hours ago; and 'Nighttime attack on Thai antigovernment protesters wounds at least 20' from Christian Science Monitor, 30 minutes ago. Each item includes a brief summary and links to more articles. A 'Show more stories' button is at the bottom of the sidebar.

Headlines

Tributes pour in for Sen. John McCain, remembered as a 'true American hero'
Fox News • 5 hours ago

- John McCain, senator and former presidential candidate, dies at 81
CNN • 23 minutes ago
- McCain leaves the stage when we need him most
Washington Post • today
- Farewell to John McCain -- He devoted his life to protecting safety and security of America
Fox News • today
- John McCain's military record and legacy: A warrior who learned from his mistakes
NBCNews.com • one hour ago

[View full coverage](#)

World »

Pirates Demand \$25 Million Ransom for Hijacked Tanker (Update1)
Bloomberg - 36 minutes ago
By Caroline Alexander and Hamsa Omar Nov. 20 (Bloomberg) -- Somali pirates are demanding \$25 million in ransom to release an oil-laden Saudi supertanker seized off the East African coast, and called on the ship's owners to pay up -- so
[Somali pirates demand \\$25M for Saudi ship](#) United Press International
[African Union says Somali politicians fuel piracy](#) Washington Post
[BBC News](#) - [guardian.co.uk](#) - [Aljazeera.net](#) - [RIA Novosti](#)
[all 4,015 news articles »](#)

Pakistan protests over US missile strikes
Reuters - 2 hours ago
By Simon Cameron-Moore ISLAMABAD (Reuters) - Pakistan summoned US ambassador Anne Patterson on Thursday to protest over missile strikes launched by pilotless drone aircraft against militant targets in Pakistan.
[Pakistan protests US drone attacks, Taliban warns of reprisals](#) AFP
[Pakistan warns US over missile strike](#) CNN International
[Telegraph.co.uk](#) - [China Daily](#) - [Xinhua](#) - [PRESS TV](#)
[all 560 news articles »](#)

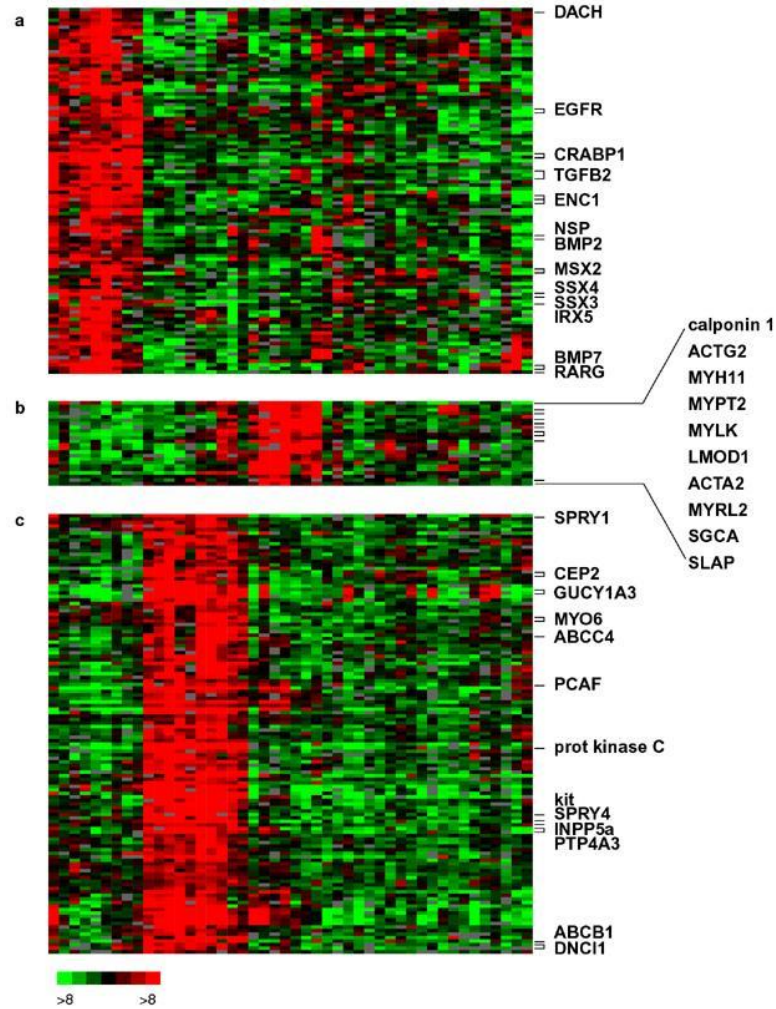
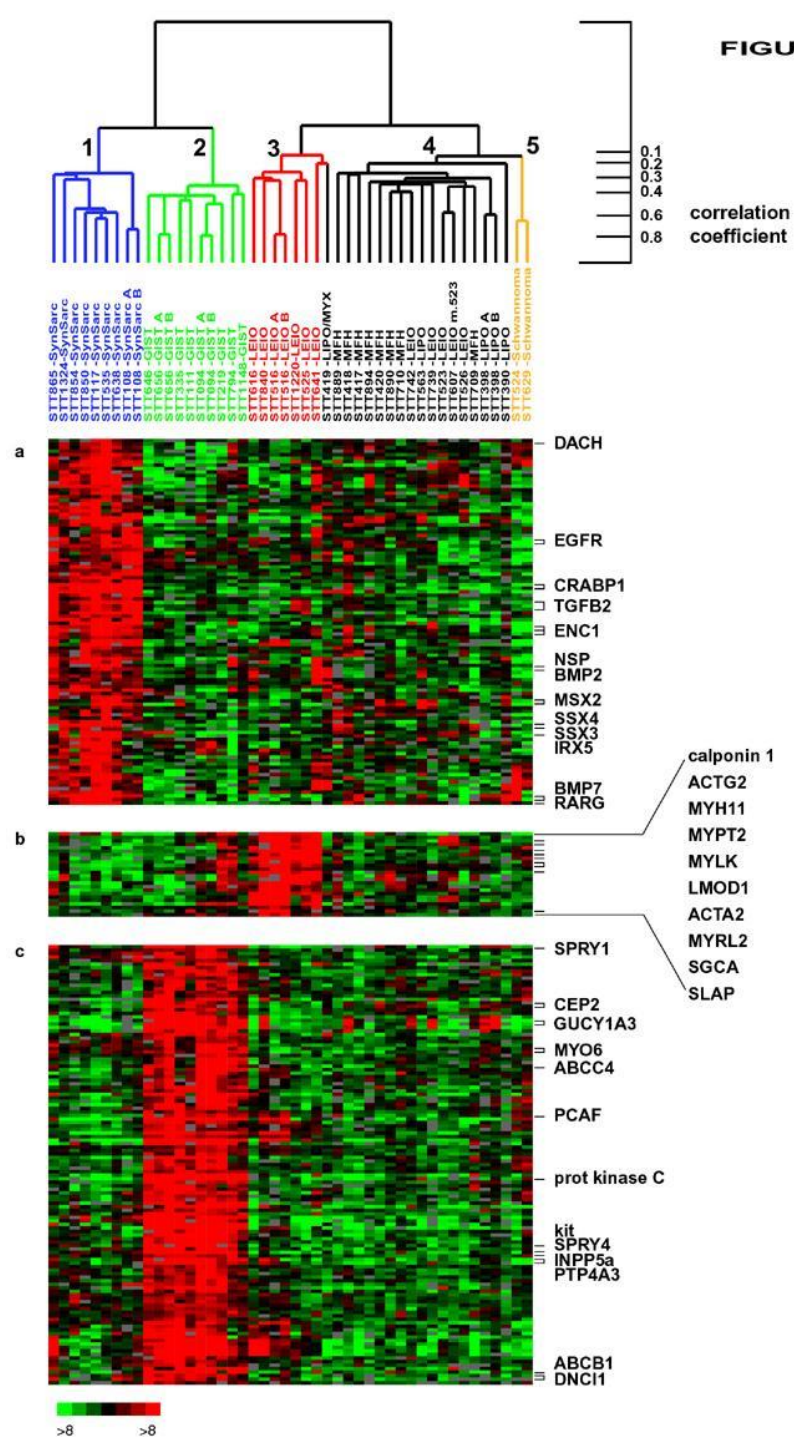
Nighttime attack on Thai antigovernment protesters wounds at least 20
Christian Science Monitor - 30 minutes ago
The government denied attacking demonstrators, who have called for the ouster of the prime minister. By Huma Yusuf One person has been killed and 23 others wounded in a grenade attack Thursday against antigovernment protesters occupying the Thai prime ...
[Blast Kills 1, Wounds 23 at Thai Prime Minister's Office](#) Washington Post
[Anti-government protestor in Thailand dies in grenade attack](#) International Herald Tribune
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Struktur internal agregasi berita

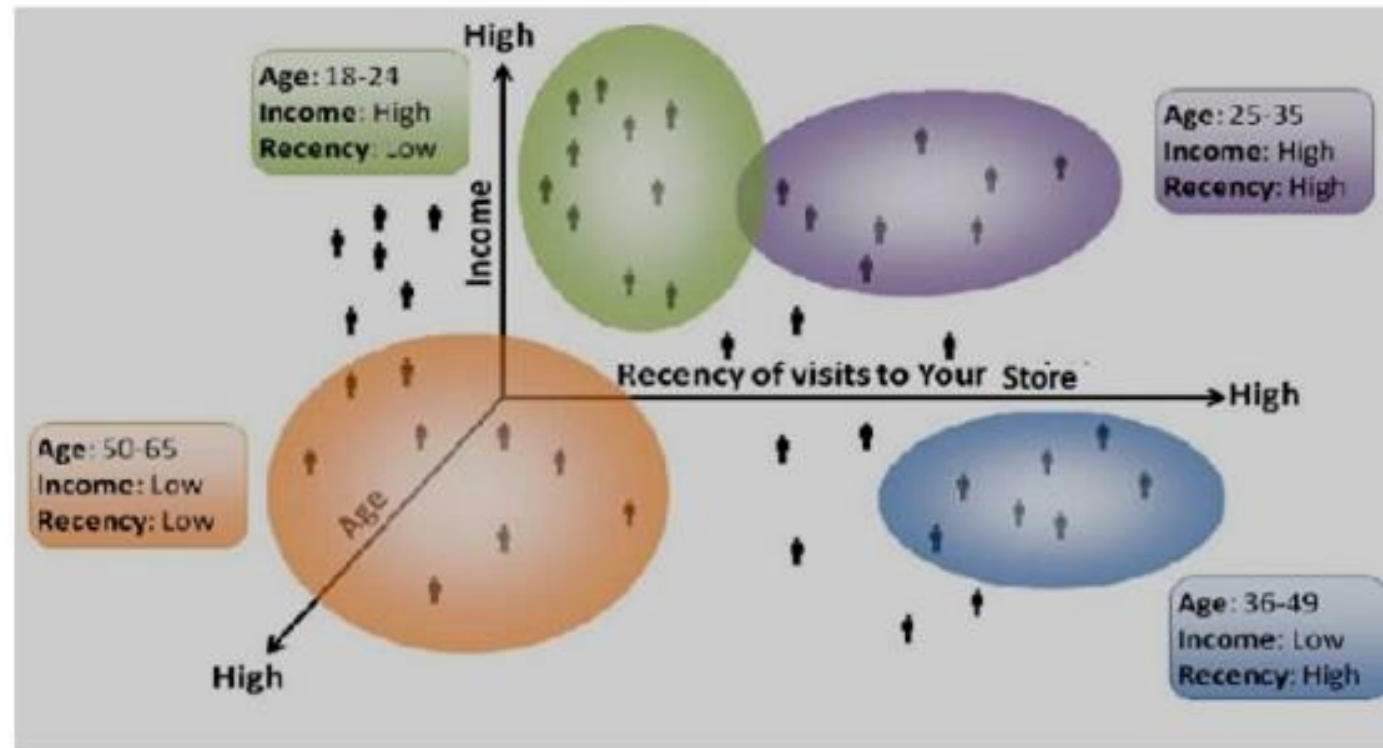


http://genome-www.stanford.edu/sarcoma/supplemental_data.html



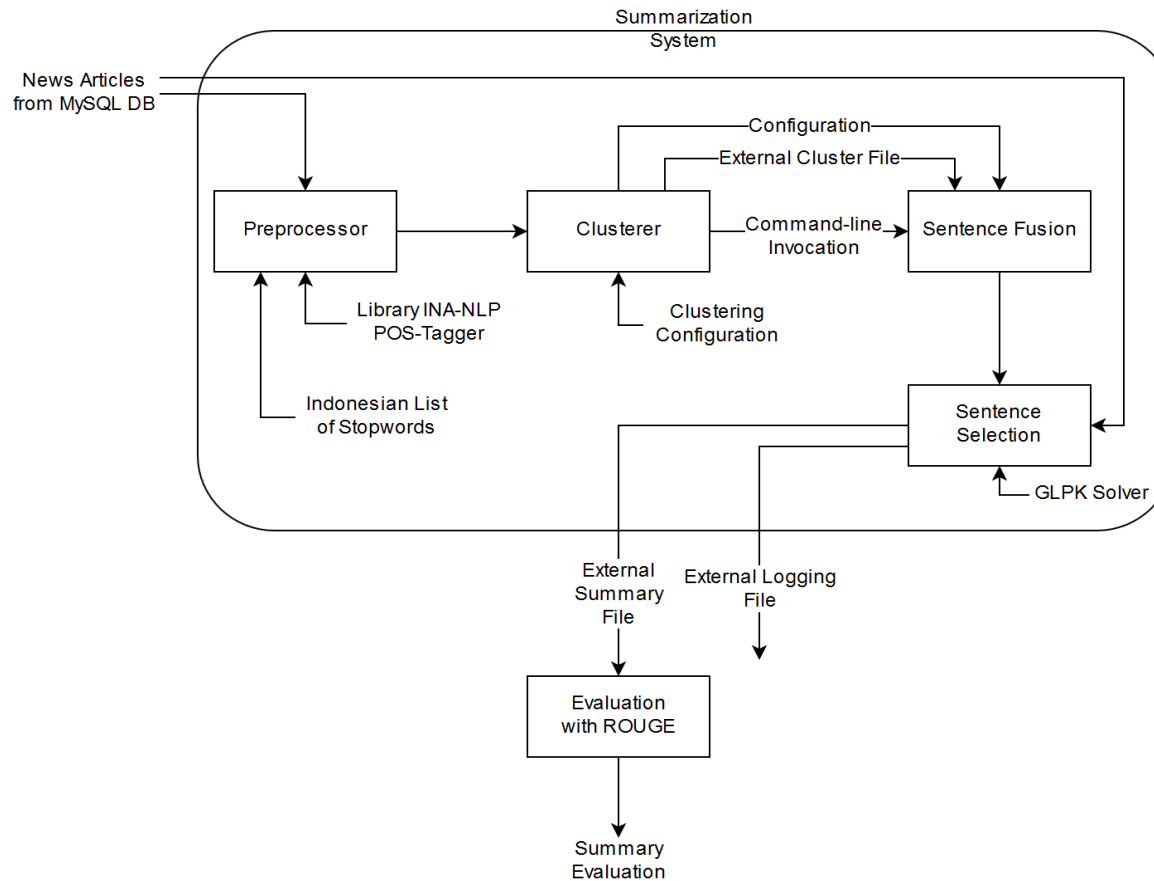
Why: Clustering untuk Segmentasi

Example - Clusters using Age, Income & Recency



Copyright: Canvass 2013-2016

Why: Clustering-based Approach

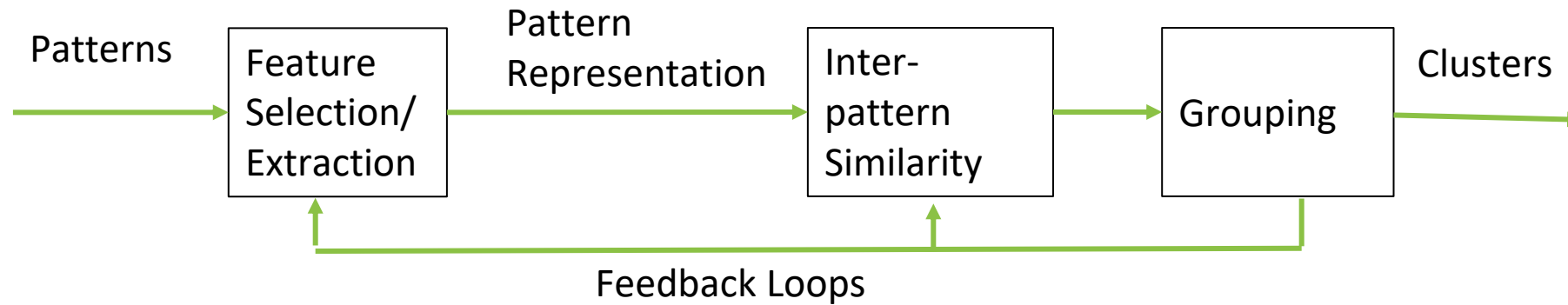


Clustering-based
summarization

Clustering-based
outlier detection

Clustering-based
analysis

Tahapan Clustering



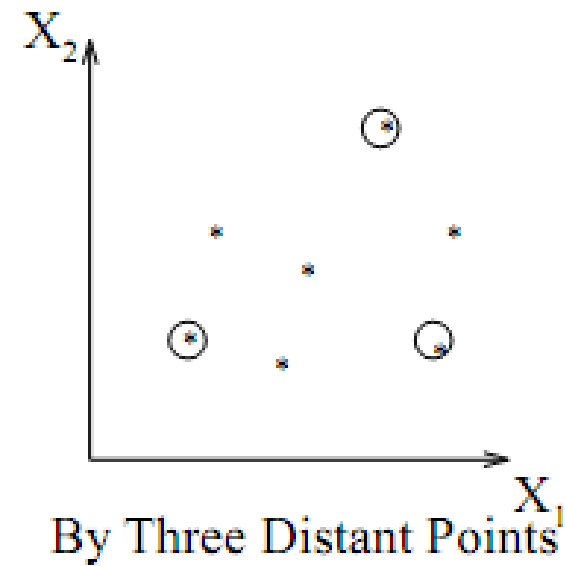
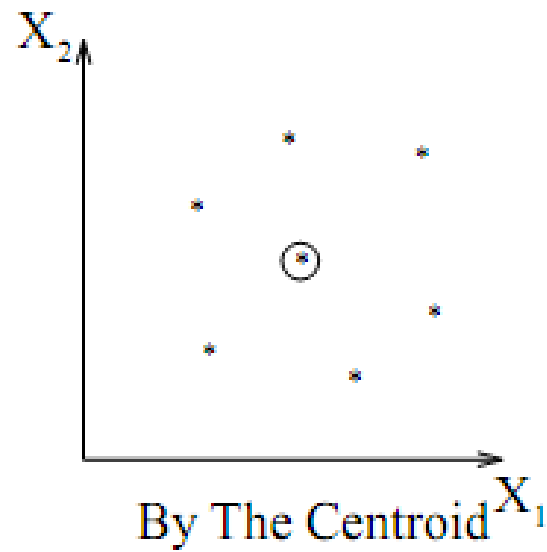
- Tahapan utama:
 - 1) Feature selection: original features $\square$ subset of features
Feature extraction: transformation into new features
 - 2) Pattern proximity/similarity measure
 - 3) Grouping
- Clustering output: hard atau soft (membership degree)

Tahapan Clustering (lanjutan)

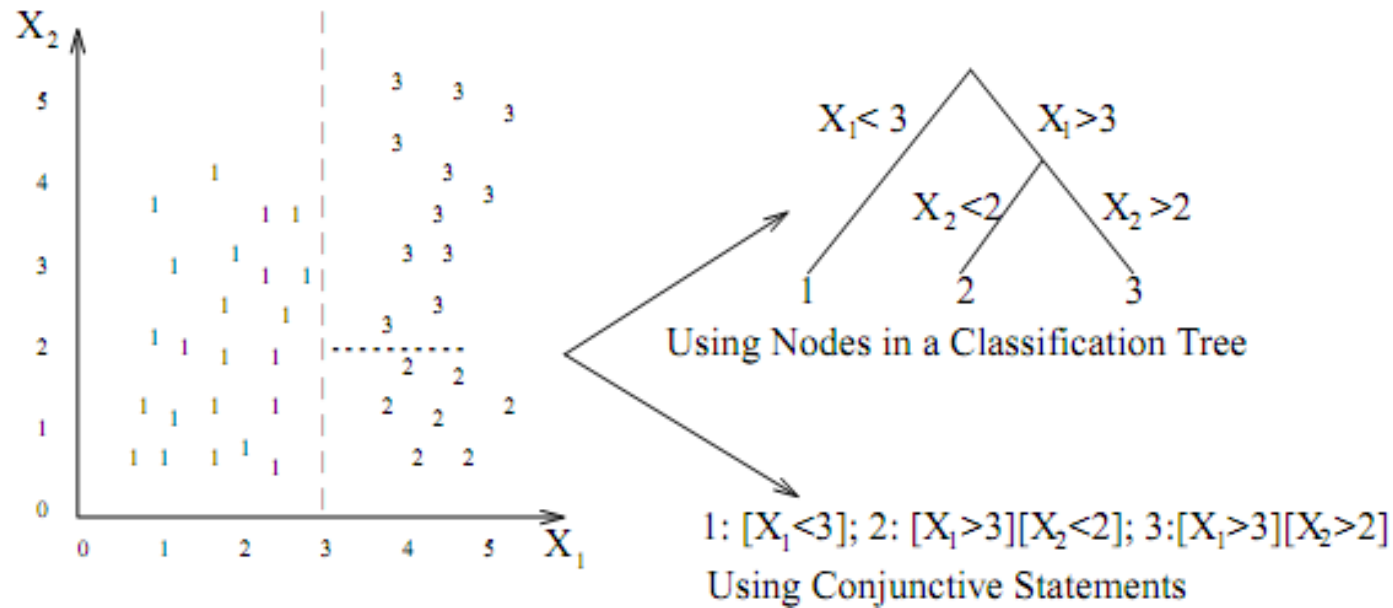
- Tahapan opsional:
 - 4) data abstraction
 - 5) assessment of output (good or poor)

Representasi Cluster (1)

- Centroid atau set of distant point

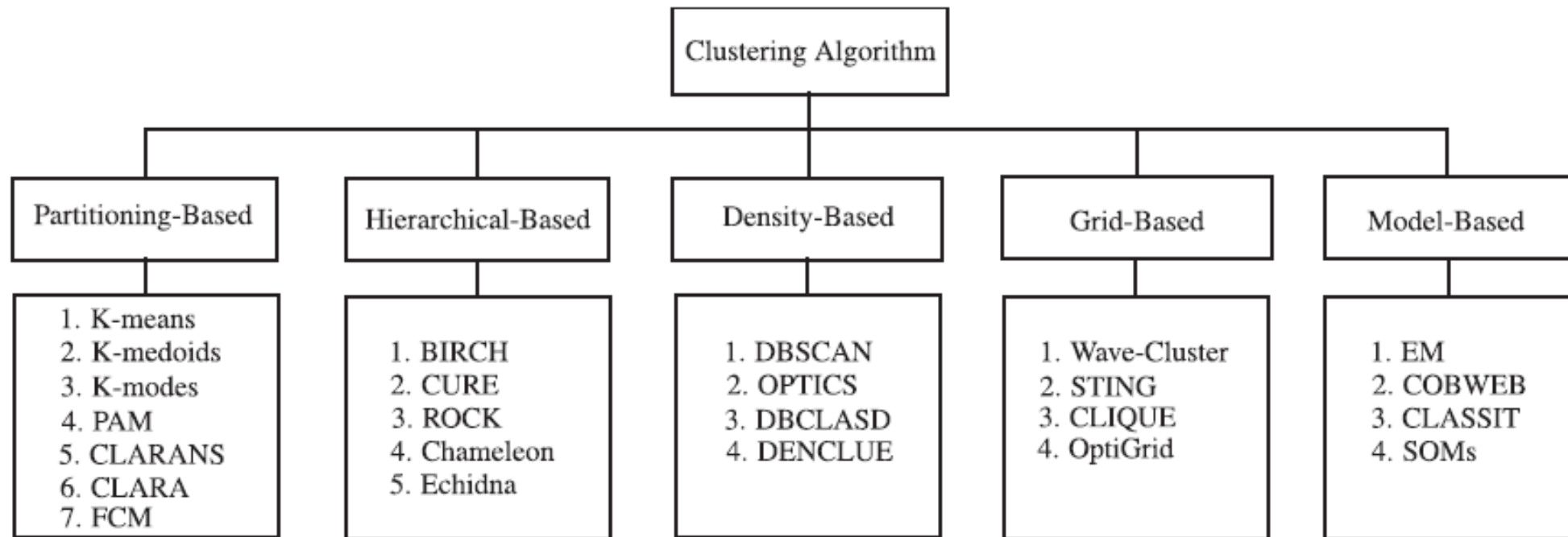


Representasi Cluster (2)



- Pohon klasifikasi
- Conjunctive statements

Kategori Metode Clustering (Fahad, 2014)



Kategori Metode Clustering

(Han & Kamber, 2006)

1. Metode *partitioning*

- mengidentifikasi partisi yang mengoptimalkan kriteria pengelompokan (squared error, absolute error)
- Konstruksi k-partisi data (partisi ~ cluster); $k \leq$ jumlah data
- Contoh: **K-means**, k-medoids

2. Metode *hierarchical*

menghasilkan rangkaian partisi bersarang

- **Agglomerative** (bottom-up, merge):
1 object ~ 1 cluster $\square$ 1 cluster n-object
- Divisive (top-down, split):
1 cluster n-object $\square$ 1 object ~ 1 cluster

Kategori Metode Clustering (lanj)

(Han & Kamber, 2006)

3. Metode berbasis density

- Densitas: jumlah objek
- Contoh: **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise)

4. Metode berbasis grid

- Struktur grid, cepat, bergantung jumlah sel, tidak dipengaruhi jumlah objek, perhitungan bisa dilakukan secara paralel
- Contoh: STING (STatistical INformation Grid)

5. Metode berbasis model

- Contoh: EM (Expectation-Maximization), SOM (self-organizing map)



02 Partitional Clustering

02 Partitional Clustering K-Means

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KK IF -Teknik Informatika - STEI ITB

Metode *Partitioning*

- Kriteria objektif: objek di dalam suatu cluster memiliki kemiripan yang lebih besar dibanding objek yang berada di cluster yang lain.
 - Minimize square-error function.
- Relokasi iteratif:
 - proses iteratif menempatkan objek ke kluster untuk memperbaiki partisi.
- Contoh: Squared-Error Clustering (**k-Means**), k-medoids, Graph theoretic

k-Means

Popular clustering algorithm

1. Initialization

Define k seeds (as centroid) for k cluster, randomly

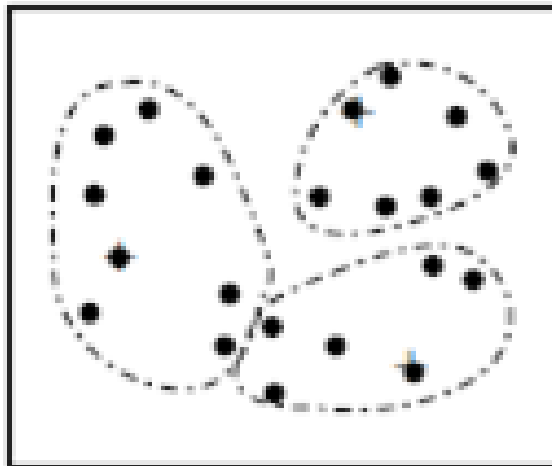
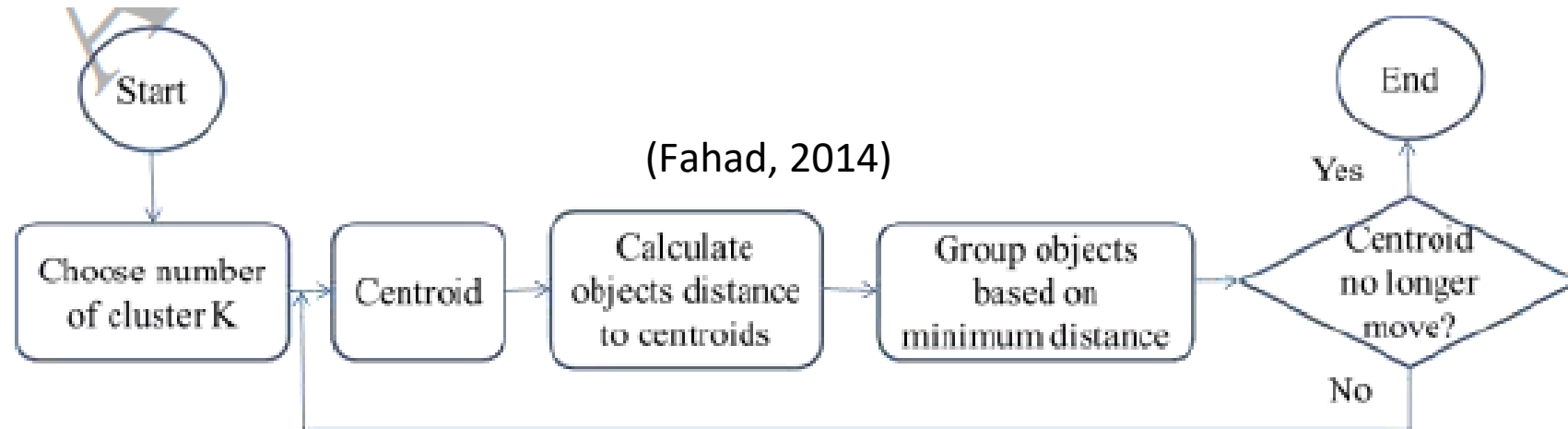
2. Iteration

Assign every instance from training data to the closest centroid, form k cluster

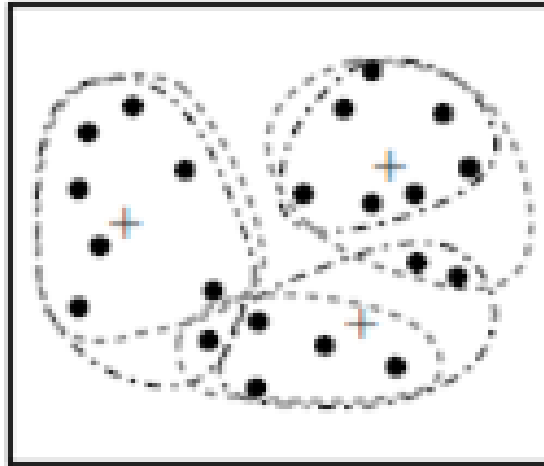
Specify the new centroid (M_i) of each cluster

Stop: Converge/ threshold/ number of iteration

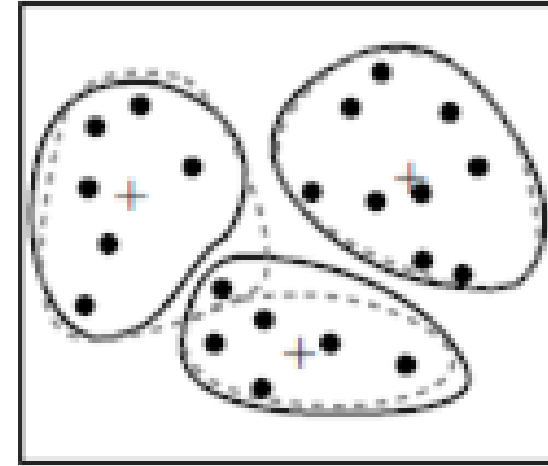
K-Means



(a)



(b)



(c)

Algoritma K-Means

(Han & Kamber, 2006)

- The k-means algorithm for partitioning, where each cluster's center is represented by mean value of the objects in the cluster.
- Input: **k** (number of clusters), D (data set containing **n** objects)
- Output: A set of k clusters
- Kompleksitas: **$O(nkt)$** ; t: jumlah iterasi; $k \ll n$
- Method:
 1. Arbitrarily choose k objects from D as the initial cluster centers
 2. Repeat
 - (re) assign each object to the cluster to which the object is the most similar based on the mean value of the objects in the clusters;
 - Update the cluster means, i.e. calculate the mean value of the objects for each cluster;
 3. Until no change (convergent)

Example

Data	Pengalaman	Programming	Bahasa Inggris	Buta Warna	Menikah
A	Tidak	Ya	Ya	Tidak	Tidak
B	Tidak	Tidak	Ya	Tidak	Ya
C	Ya	Ya	Tidak	Ya	Tidak
D	Tidak	Tidak	Tidak	Ya	Ya
E	Ya	Tidak	Ya	Ya	Ya
F	Tidak	Ya	Tidak	Ya	Tidak

$K = 3$

Seed: A,
B, C

Distance: difference of attribute value, if it has similar distance, group the instance into cluster 2

Centroid (mean) : majority value for each attribute, if no majority choose previous value

Stop:
max distance in each cluster < 2

k-Means Implementation

Initialization:

Centroid Cluster 1: Data A

Centroid Cluster 2: Data B

Centroid Cluster 3: Data C

Iteration 1:

Assign each instance to closest centroid:

Cluster 1: A

Cluster 2: B, D, E

Cluster 3: C, F

Data	Pengalaman	Programming	Bahasa Inggris	Buta Warna	Menikah	Distance Centroid 1	Distance Centroid 2	Distance Centroid 3
A	Tidak	Ya	Ya	Tidak	Tidak	0	2	3
B	Tidak	Tidak	Ya	Tidak	Ya	2	0	5
C	Ya	Ya	Tidak	Ya	Tidak	3	5	0
D	Tidak	Tidak	Tidak	Ya	Ya	4	2	3
E	Ya	Tidak	Ya	Ya	Ya	4	2	3
F	Tidak	Ya	Tidak	Ya	Tidak	2	4	1

k-Means Implementation

Initialization:

Centroid Cluster 1: Data A

Centroid Cluster 2: Data B

Centroid Cluster 3: Data C

Iteration 1:

Adjust new centroid

Centroid Cluster 1:

{Tidak, Ya, Ya, Tidak, Tidak} □ max dist: 0

Centroid Cluster 2:

{Tidak, Tidak, Ya, Ya, Ya}

□ max dist: 1

Centroid Cluster 3:

{Ya, Ya, Tidak, Ya, Tidak}

□ max dist: 1

STOP? Yes

Data	Pengalaman	Programming	Bahasa Inggris	Buta Warna	Menikah	Distance Centroid 1	Distance Centroid 2	Distance Centroid 3
A	Tidak	Ya	Ya	Tidak	Tidak	0	2	3
B	Tidak	Tidak	Ya	Tidak	Ya	2	0	5
C	Ya	Ya	Tidak	Ya	Tidak	3	5	0
D	Tidak	Tidak	Tidak	Ya	Ya	4	2	3
E	Ya	Tidak	Ya	Ya	Ya	4	2	3
F	Tidak	Ya	Tidak	Ya	Tidak	2	4	1

Usage

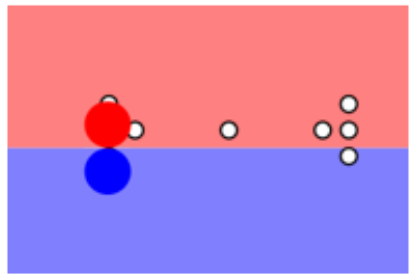
New Instance G:
{Ya, Ya, Ya, Tidak, Tidak}

Centroid Cluster 1:
{Tidak, Ya, Ya, Tidak, Tidak}
Centroid Cluster 2:
{Tidak, Tidak, Ya, Ya, Ya
Centroid Cluster 3:
{Ya, Ya, Tidak, Ya, Tidak}

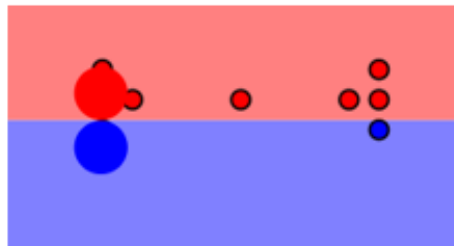
Distance Centroid 1	Distance Centroid 2	Distance Centroid 3
1	4	2

New Instance G:
Cluster 1

K-Means: Pengaruh Cluster Awal

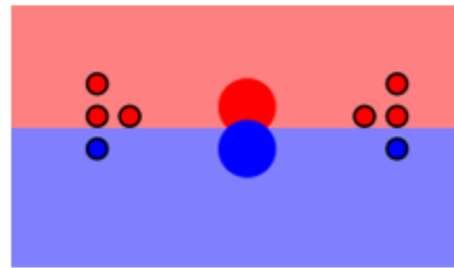


i=0

{A,B,D,E,F,G,H}, {C,I}

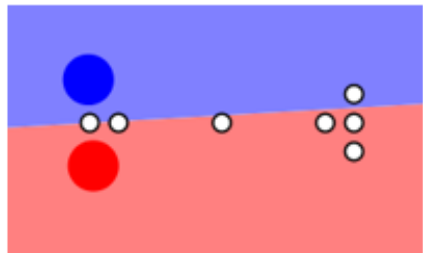
i=1

{A,B,D,E,F,G,H}, {C,I}

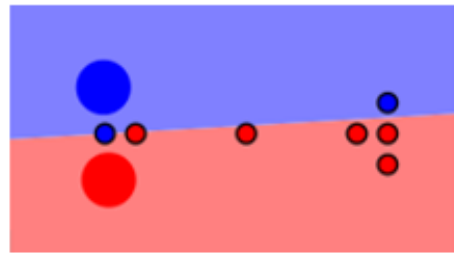


i=2

{A,B,D,E,F,G,H}, {C,I}

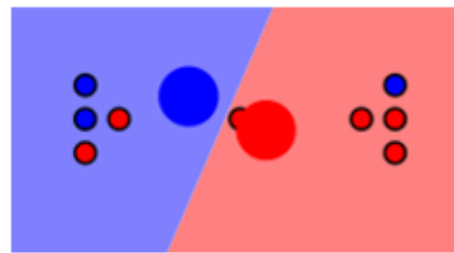


i=0

{A,G}, {B,C,D,E,F,H,I}

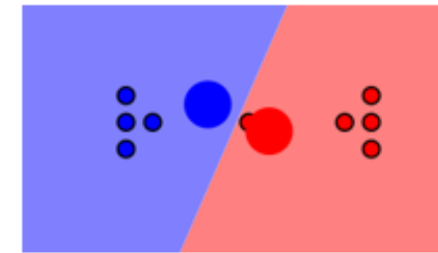
i=1

{A,B,G}, {C,D,E,F,H,I}



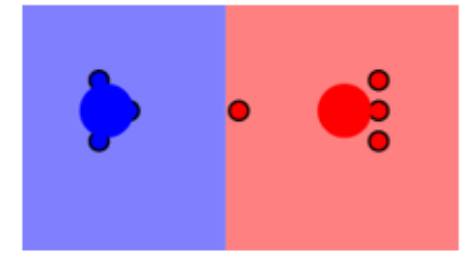
i=2

{A,B,G}, {C,D,E,F,H,I}



i=3

{A,B,C,D}, {E,F,G,H,I}



i=4

{A,B,C,D}, {E,F,G,H,I}

Kelemahan K-Means (Berkhin, 2006)

- Penentuan nilai k? ☐ Gunakan Elbow Method (slide berikutnya)
- Hasil sangat dipengaruhi oleh inisialisasi centroid
 - Sering berhenti pada optimum lokal
 - Hasil akhir tidak stabil
- Algoritma tidak *scalable*
- Mean hanya terdefinisi untuk atribut numerik
 - Atribut nominal ☐ metode k-modes
- Prosesnya sensitif terhadap outliers
 - Outlier dgn nilai yang ekstrim besar mempengaruhi mean pada centroid



Clustering with K-Means

How we know the best "K"

Within-Cluster-Sum-of-Squares: $WCSS = \sum_k^n \sum_i^m \|d_i - c_k\|^2$

Expect: Small WCSS

Elbow Method

N instances in training data,
create N cluster $\square$ $WCSS = 0$



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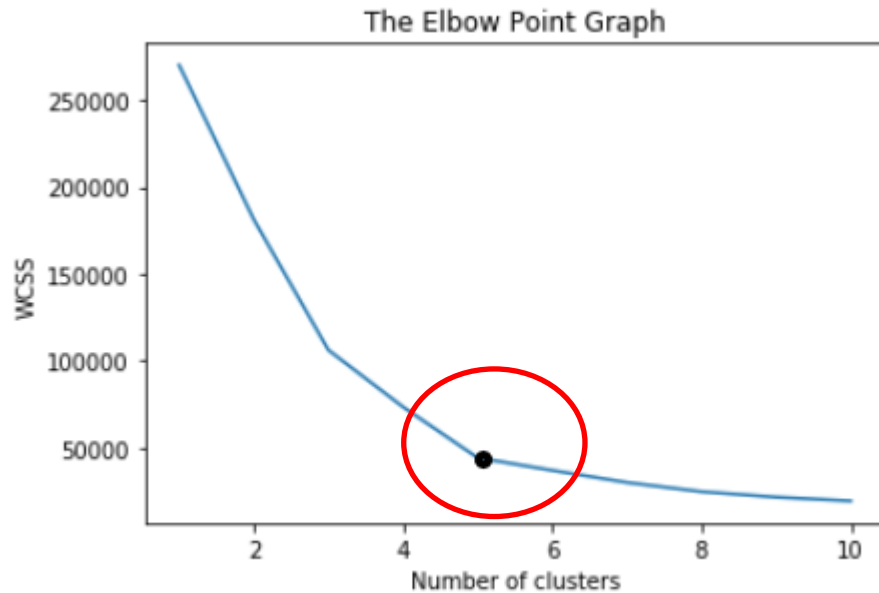


Elbow Method

Try k from the smallest number to the largest

For each k , run k-Means and plot its WCSS

Find k where the position is in the “elbow”



Q & A: K-means

- Q: Kompleksitas algoritma K-means $O(nkt)$, tetapi disebutkan tidak scalable.
- A: Jumlah iterasi untuk kasus terburuknya bisa mencapai eksponensial / superpolynomial.
 - Xu & Tian (2017): Kompleksitas algoritma K-means $O(nkt)$ dikategorikan kompleksitas rendah.
 - Dimensi diabaikan $\square$ $O(nktd)$; d dimensi data
 - Berkhin (2006): Scalability both in terms of computing time and memory requirements.
 - Arthur & Vassilvitskii (2006): The worst-case running time of k-means is superpolynomial number of iteration ($2^{\Omega(n)}$ iteration).





03 Density-based Clustering

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Outline

DBSCAN

Density-based
Clustering

Example

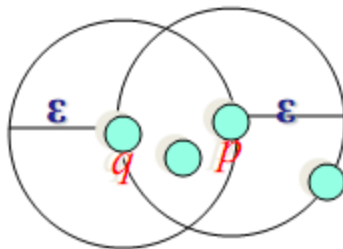
Density-based Clustering

- Clusters are dense regions in the data space, separated by regions of lower object density
- A cluster is defined as a maximal set of density-connected points

ϵ -Neighborhood – Objects within a radius of ϵ from an object.

$$N_{\epsilon}(p) : \{q \mid d(p, q) \leq \epsilon\}$$

“High density” - ϵ -Neighborhood of an object contains at least **MinPts** of objects.



ϵ -Neighborhood of p

ϵ -Neighborhood of q

Density of p is “high” (MinPts = 4)

Density of q is “low” (MinPts = 4)

DBSCAN

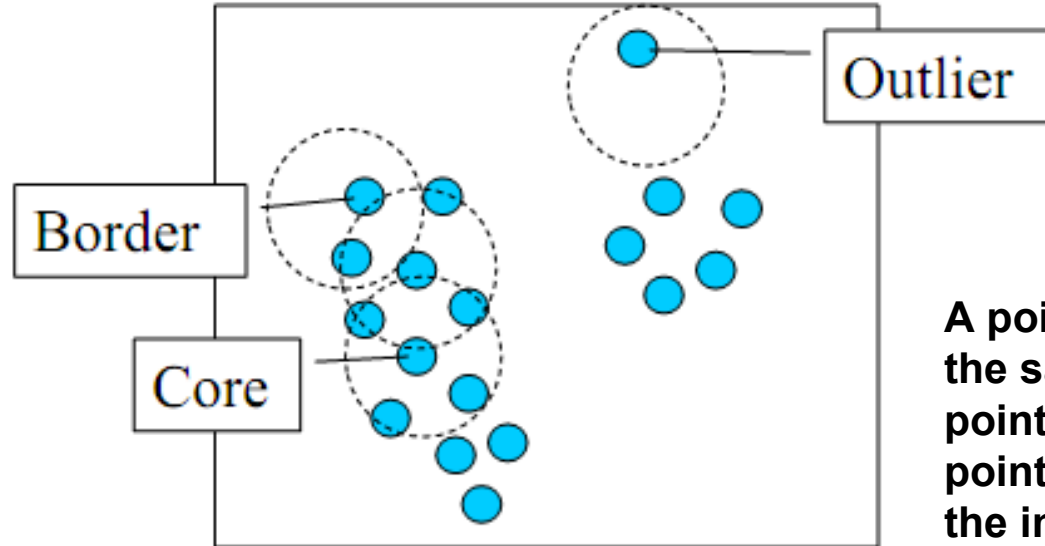
(Density Based Spatial Clustering of Applications with Noise)

- DBSCAN starts with an arbitrary point p and retrieves all points density-reachable from p wrt. Eps and $MinPts$.
 - If p is a core point, this procedure yields a cluster wrt. Eps and $MinPts$.
 - If p is a border point, no points are density-reachable from p and DBSCAN visits the next point of the database.

DBSCAN: Object Type

- The neighborhood within a radius ϵ of a given object is called the ϵ -neighborhood of the object.
- If the ϵ -neighborhood of an object contains at least a minimum number, $MinPts$, of objects, then the object is called a **core object**.
- Given a set of objects, D , we say that an object p is **directly density-reachable** from object q if p is within the ϵ -neighborhood of q , and q is a core object.
- An object p is **density-reachable** from object q with respect to ϵ and $MinPts$ in a set of objects, D , if there is a chain of objects $p_1, \dots, p_n$, where $p_1 = q$ and $p_n = p$ such that p_{i+1} is directly density-reachable from p_i with respect to ϵ and $MinPts$, for $1 \leq i \leq n$, $p_i \in D$.
- An object p is **density-connected** to object q with respect to ϵ and $MinPts$ in a set of objects, D , if there is an object $o \in D$ such that both p and q are density-reachable from o with respect to ϵ and $MinPts$.

Core, Border, Outlier



Given ε and *MinPts*, categorize the objects into three exclusive groups.

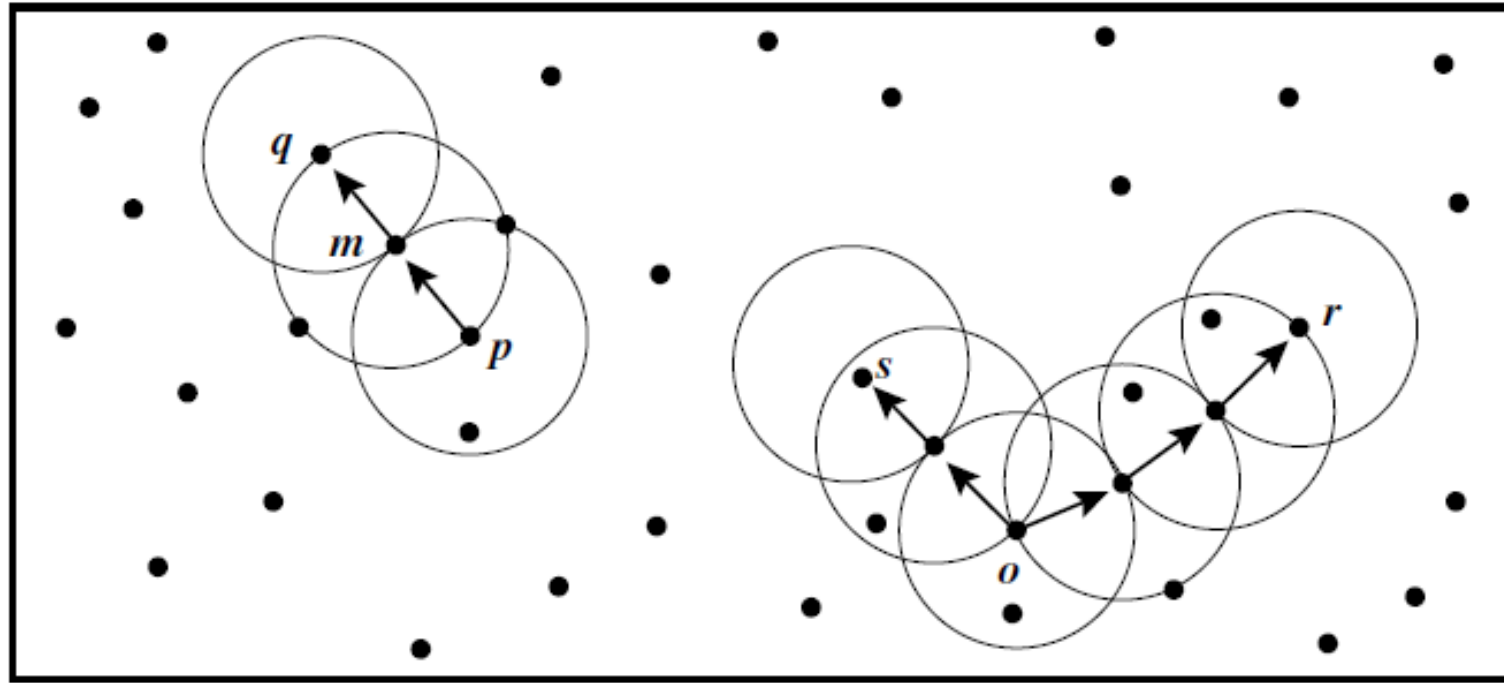
A point is a **core point** if it has at least the same as a specified number of points (*MinPts*) within *Eps* (including the point itself) – These are points that are the interior of a cluster

$\varepsilon = 1 \text{ unit}$, *MinPts* = 5

A **border point** has fewer than *MinPts* within *Eps*, but is in the neighborhood of a core point.

A **noise point** is any point that is not a core point nor a border point.

DBSCAN find clusters

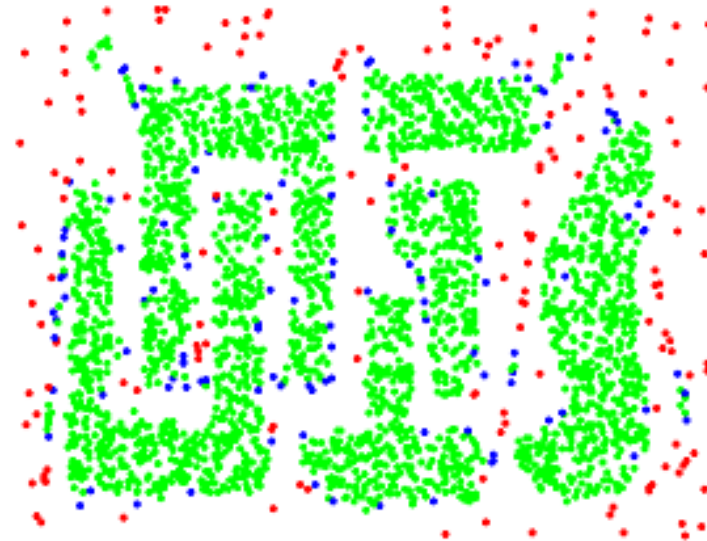


DBSCAN searches for clusters by checking the ϵ -neighborhood of each point in the database. If the ϵ -neighborhood of a point p **contains** more than $MinPts$, a new cluster with p as a core object is created. DBSCAN then iteratively collects directly density-reachable objects from these core objects, which may involve the merge of a few density-reachable clusters. The process terminates when no new point can be added to any cluster.

DBSCAN Example



Original Points



Point types: **core**,
border and **outliers**

$\epsilon = 10$, MinPts = 4

DBScan – Example (1)

- If Epsilon is 2 and minpoint is 2, what are the clusters that DBScan would discover with the following 8 examples:

$A1=(2,10)$, $A2=(2,5)$, $A3=(8,4)$, $A4=(5,8)$, $A5=(7,5)$,
 $A6=(6,4)$, $A7=(1,2)$, $A8=(4,9)$.

Matriks jarak (kuadrat):

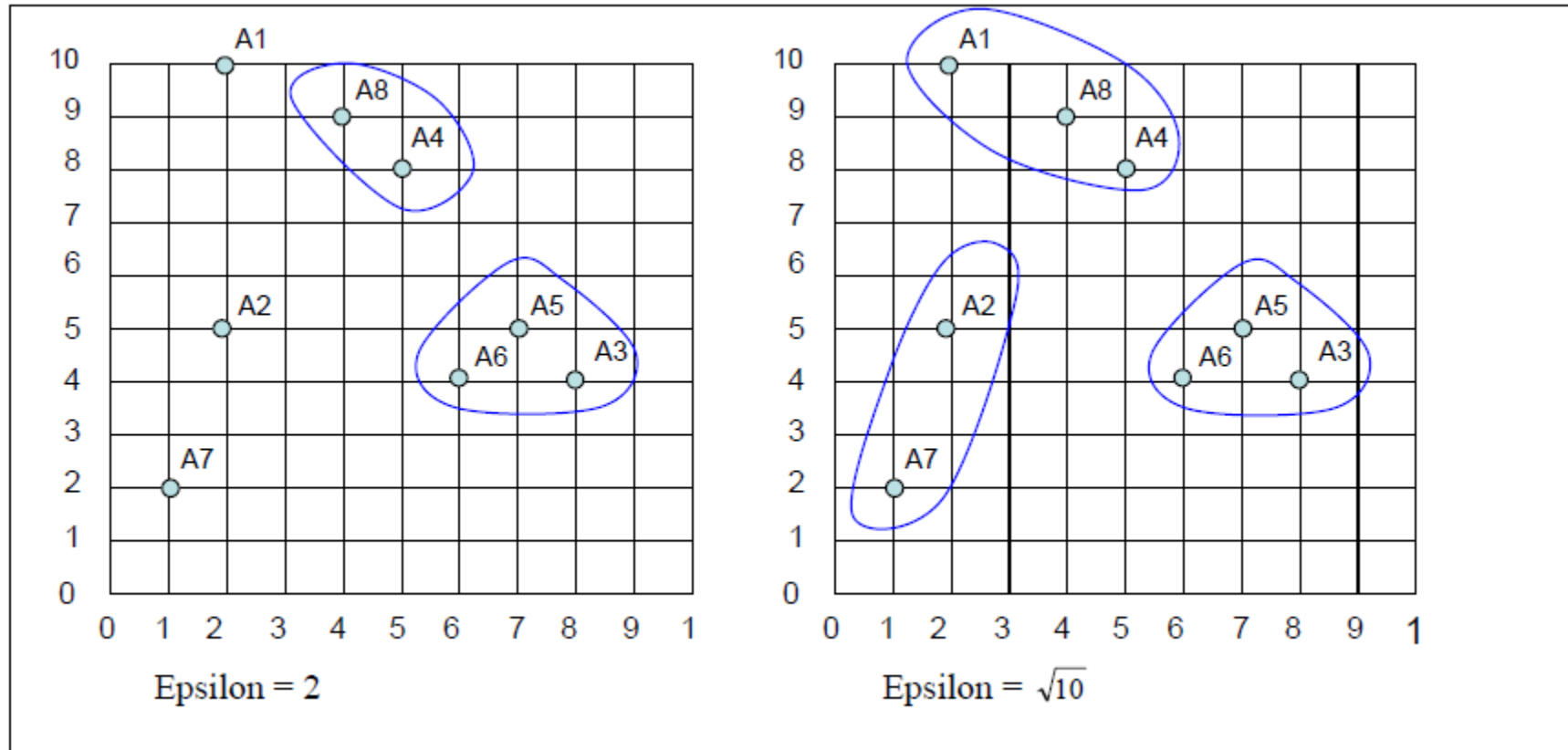
	A1	A2	A3	A4	A5	A6	A7	A8
A1	0	25	72	13	50	52	65	5
A2	25	0	37	18	25	17	10	20
A3	72	37	0	25	2	4	53	41
A4	13	18	25	0	13	17	52	2
A5	50	25	2	13	0	2	45	25
A6	52	17	4	17	2	0	29	29
A7	65	10	53	52	45	29	0	58
A8	5	20	41	2	25	29	58	0

DBScan – Example (2)

- Solutions:
- Epsilon neighborhood of each point
 - $N_2(A1)=\{\}$;
 - $N_2(A2)=\{\}$;
 - $N_2(A3)=\{A5, A6\}$;
 - $N_2(A4)=\{A8\}$;
 - $N_2(A5)=\{A3, A6\}$;
 - $N_2(A6)=\{A3, A5\}$;
 - $N_2(A7)=\{\}$;
 - $N_2(A8)=\{A4\}$
 - So A1, A2, and A7 are outliers, while we have two clusters $C1=\{A4, A8\}$ and $C2=\{A3, A5, A6\}$
- If Epsilon is $10^{1/2}$ then the neighborhood of some points will increase:
 - A1 would join the cluster C1 and A2 would joint with A7 to form cluster $C3=\{A2, A7\}$.

	A1	A2	A3	A4	A5	A6	A7	A8
A1	0	25	72	13	50	52	65	5
A2	25	0	37	18	25	17	10	20
A3	72	37	0	25	2	4	53	41
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DBScan – Example (3)





04 Cluster Evaluation

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Outline

Cluster Validity

Internal Measures

External
Measures



Cluster Validity

- For supervised classification we have a variety of measures to evaluate how good our model is
 - Accuracy, precision, recall
- For cluster analysis, the analogous question is how to evaluate the “goodness” of the resulting clusters?
- But “clusters are in the eye of the beholder”!
- Then why do we want to evaluate them?
 - To avoid finding patterns in noise
 - To compare clustering algorithms
 - To compare two sets of clusters
 - To compare two clusters

Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
 - **Internal Index:** Used to measure the goodness of a clustering structure *without* respect to external information.
 - **Cohesion and Separation, Silhouette Coefficient**, Dunn Index, Cophenetic Correlation Coefficient (CPCC), etc.
 - **External Index:** Used to measure the extent to which cluster labels match externally supplied class labels.
 - Entropy, **Purity**, Precision, Recall, F-Measure, Rand Index, etc.
 - **Relative Index:** Used to compare two different clusterings or clusters.
 - Often an external or internal index is used for this function, e.g., SSE or entropy
- Sometimes these are referred to as **criteria** instead of **indices**
 - However, sometimes criterion is the general strategy and index is the numerical measure that implements the criterion.

Internal Measures: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster; compactness
 - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters; isolation
- Example: Squared Error
 - Cohesion is measured by the within cluster sum of squares (SSE)

$$WSS = \sum_i \sum_{x \in C_i} (x - m_i)^2$$

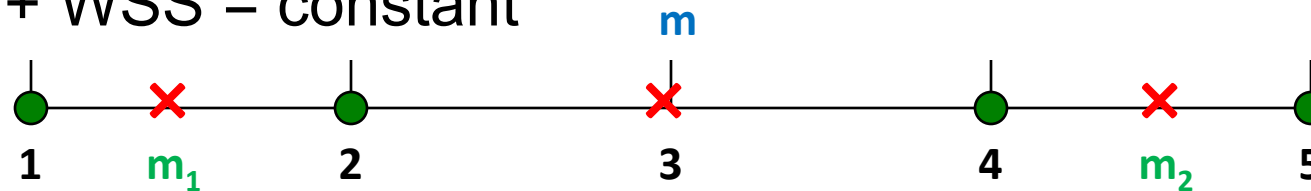
- Separation is measured by the between cluster sum of squares

$$BSS = \sum_i |C_i| (m - m_i)^2$$

- Where $|C_i|$ is the size of cluster i

Internal Measures: Cohesion and Separation

- Example: SSE
 - BSS + WSS = constant



K=1 cluster:

$$WSS = (1 - 3)^2 + (2 - 3)^2 + (4 - 3)^2 + (5 - 3)^2 = 10$$

$$BSS = 4 \times (3 - 3)^2 = 0$$

$$Total = 10 + 0 = 10$$

K=2 clusters:

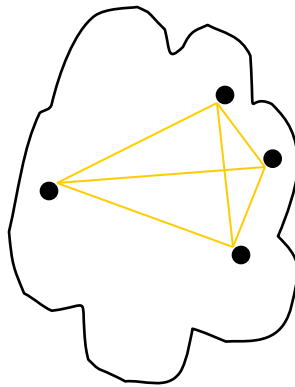
$$WSS = (1 - 1.5)^2 + (2 - 1.5)^2 + (4 - 4.5)^2 + (5 - 4.5)^2 = 1$$

$$BSS = 2 \times (3 - 1.5)^2 + 2 \times (4.5 - 3)^2 = 9$$

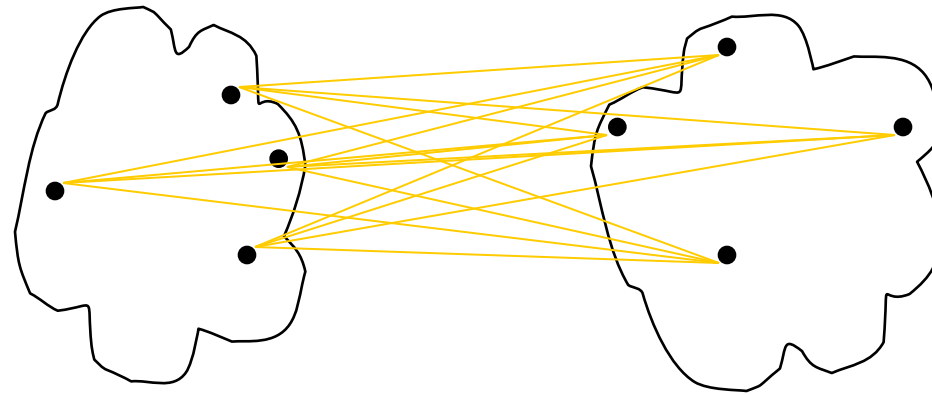
$$Total = 1 + 9 = 10$$

Internal Measures: Cohesion and Separation

- A proximity graph based approach can also be used for cohesion and separation.
 - Cluster cohesion is the sum of the weight of all links within a cluster.
 - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



cohesion

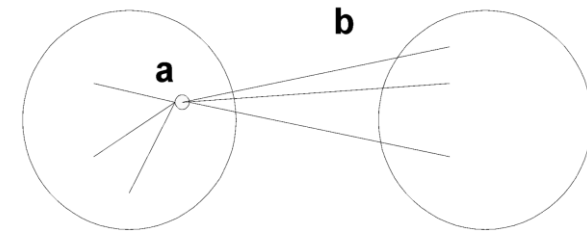


separation

Internal Measures: Silhouette Coefficient

- Silhouette Coefficient combine ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings.
- For an individual point, i
 - Calculate a = average distance of i to the points in its cluster
 - Calculate b = min (average distance of i to points in another cluster)
 - The silhouette coefficient for a point is then given by

$$s = \frac{b - a}{\max(a, b)}$$



- The score is bounded between -1 for incorrect clustering and +1 for highly dense clustering.
 - Scores around zero indicate overlapping clusters.
- Can calculate the Average Silhouette width for a cluster or a clustering

Silhouette Coefficient (SC) Example

$$s = \frac{b - a}{\max(a, b)}$$

Cluster 1:

Point A1: (2, 5)

Point A2: (3, 4)

Point A3: (4, 6)

Cluster 2:

Point B1: (8, 3)

Point B2: (9, 2)

Point B3: (10, 5)

Cluster 3:

Point C1: (6, 10)

Point C2: (7, 8)

Point C3: (8, 9)

Distance:

SC A1:
Step 1:

- Distance from Point A1 to Point A2: $\sqrt{(2-3)^2 + (5-4)^2} = \sqrt{2}$
- Distance from Point A1 to Point A3: $\sqrt{(2-4)^2 + (5-6)^2} = \sqrt{5}$
- Average cohesion for Point A1 : **$a = (\sqrt{2} + \sqrt{5}) / 2 \approx 1.825$**

Step 2a:

- Distance from Point A1 to Point B1: $\sqrt{(2-8)^2 + (5-3)^2} = \sqrt{40}$
- Distance from Point A1 to Point B2: $\sqrt{(2-7)^2 + (5-8)^2} = \sqrt{34}$
- Distance from Point A1 to Point B3: $\sqrt{(2-8)^2 + (5-9)^2} = \sqrt{52}$
- Average separation for Point A1 to cluster 2 = $(\sqrt{40} + \sqrt{34} + \sqrt{52}) / 3 \approx 6.456$

Step 2b:

- Distance from Point A1 to Point C1: $\sqrt{(2-6)^2 + (5-10)^2} = \sqrt{26}$
- Distance from Point A1 to Point C2: $\sqrt{(2-7)^2 + (5-8)^2} = \sqrt{18}$
- Distance from Point A1 to Point C3: $\sqrt{(2-8)^2 + (5-9)^2} = \sqrt{32}$
- Average separation for Point A1 to cluster 3 = $(\sqrt{26} + \sqrt{18} + \sqrt{32}) / 3 \approx 5.333$

Minimum separation A1 to other cluster: **$b = \min(6.456, 5.333) = 5.333$**

Step 3:

SC A1 = $(5.333 - 1.825) / \max(5.333, 1.825) \approx 0.657$

- Count SC for all points
- SC for a cluster is the average SC points in the cluster
- SC for all cluster is the average SC of all clusters/ all points

External Measures: Purity

- Purity: ratio between the dominant class in the cluster and the size of the cluster.

$$Purity(w_i) = \frac{1}{n_i} \max_j (n_{ij}) \quad j \in C$$

- Assume documents with C gold standard classes, while our clustering algorithm produce K clusters, $w_1, w_2, \dots, w_k$ with n_i members.

Clustering for 'Labeled Data'

Purity

Bad clustering:
Purity close to 0

Good clustering:
Purity close to 1

Purity in a
cluster:

$$P(C_r) = \frac{1}{n_r} \max_i n_r^i$$

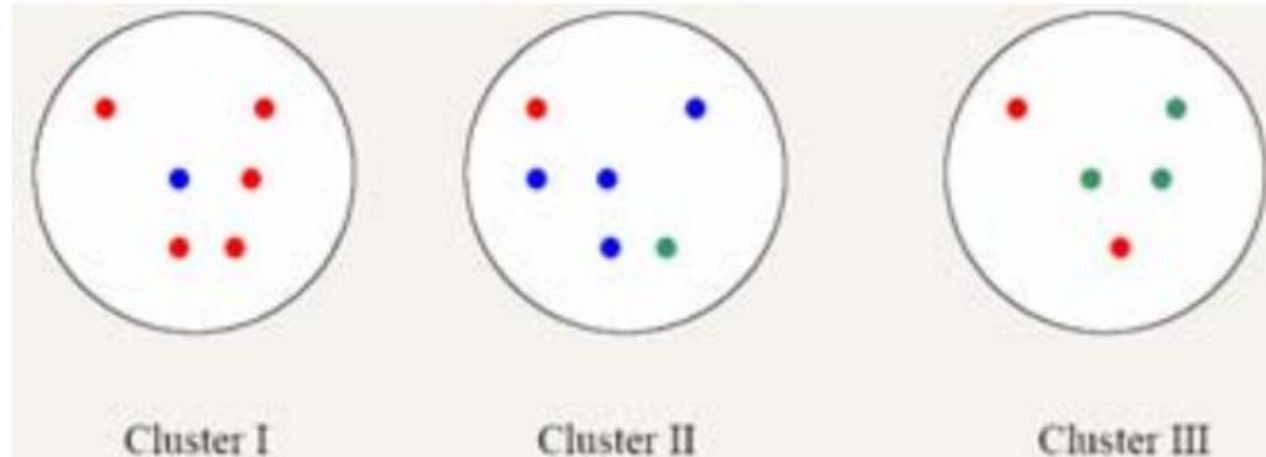
Purity of all clustering
result:

$$\begin{aligned} Purity(C) &= \sum_{r=1}^k \frac{n_r}{n} P(C_r) \\ \text{or} &= \frac{1}{n} \sum_{r=1}^k \max_i (n_r^i) \end{aligned}$$



External Measures: Purity

- Example:



- Cluster I: Purity = $1/6 (\max(5, 1, 0)) = 5/6$
- Cluster II: Purity = $1/6 (\max(1, 4, 1)) = 4/6$
- Cluster III: Purity = $1/5 (\max(2, 0, 3)) = 3/5$
- All cluster: Purity = $1/17 (5+4+3) = 12/17$

Terima Kasih